Recovering Relativity and the Standard Model from Observer Overlap Consistency

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Abstract

Observer-Patch Holography starts from a finite screen and one requirement: neighboring observer patches must agree on their overlap. From that simple rule, the recovered core yields a Lorentzian spacetime sector, a Jacobson-style route to Einstein's equation, and a bosonic compact-gauge reconstruction branch. On the realized MAR-admissible one-Higgs matter branch, the same stack yields the realized Standard Model quotient $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$ with the exact hypercharge lattice, the realized color triplet $N_c = 3$, and the generation count $N_g = 3$. Fermions and chirality enter through the explicit realized matter-sector premise; this compact paper does not promote them to a super-Tannakian reconstruction theorem. The basis used here is quantum-algebraic: patch algebras, states, trace/Born probabilities where measurement records are invoked, and generalized entropy are part of the starting formalism. OPH's reconstruction claim is not that every mathematical ingredient must be derived from a blank starting point. It is that this algebraic-information basis can support a consistent and comprehensive theory-of-everything program by recovering the observed effective universe from overlap consistency. The broader framework then introduces one local pixel ratio P by describing the same screen cell from two sides: as an exterior screen cell displaced from perfect self-similar equilibrium and as the electromagnetic observation scale available inside the encoded world. That single scale feeds the downstream particle papers. This compact paper isolates the recovered core and keeps it distinct from the quantitative particle and continuation branches built on top of the same P.

1 Introduction

Physics is usually presented in two boxes. One box contains spacetime and gravity. The other contains the particle world. This paper asks whether both can be recovered from a smaller starting point: a finite screen whose neighboring observer patches must agree on their overlaps.

That single requirement drives a long structural chain. It explains why the Lorentz group appears, why the low-energy spacetime description follows a Jacobson-type Einstein branch, why the gauge structure is

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6},$$

why the observed three colors and three generations are realized, why hypercharges take the observed rational values, what fixes the electroweak and flavor scales, and what the framework can and cannot say about the cosmological constant and possible dark-sector continuations.

This paper presents the compact reconstruction program based on Observer-Patch Holography (OPH). Physical data are organized patchwise: each observer has access only to a patch algebra,

neighboring observers must agree on overlaps, and the reconstruction is driven by those consistency conditions. Under the five axioms and the additional branch premises stated below, that structure feeds a Lorentzian modular-phase classification, a Jacobson-type Einstein branch, compact gauge reconstruction in a bosonic internal-gauge sector, charge quantization, and the low-energy architecture fixed on the realized MAR gauge branch. Black-hole spectroscopy, Page-curve/island claims, dark-sector models, and particle-family continuations belong to companion or continuation surfaces.

The paper separates exact structural theorems, conditional scaling-limit branches, quantitative outputs, and phenomenological continuations. Several longer arguments are written in compressed proof form so the claim boundary stays readable.

The framework is presented as a reconstruction program in which general relativity and the Standard Model appear as effective sectors derived from the stated quantum-algebraic axiom set together with explicit refinement and scaling-limit premises. This theory-of-everything framing is a claim about recovering the observed effective universe from that basis and its declared branch premises, not a demand that every mathematical ingredient be derived from a blank starting point.

Prioritized achievements

The results emphasized in this paper are ordered below by a combination of impact and defensibility.

1. A conditional Jacobson-type Einstein branch is recovered from observer-overlap modular geometry, the derived fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap MaxEnt variations on the realized cap-label-preserving MaxEnt family, the geometric-branch selection burden, and an explicit null-modular scaling branch; in the null modular bridge, quasi-local propagation, exact-or-controlled strip additivity, endpoint-Lipschitz renormalized half-line control, the resulting weak tail generator, and the half-sided modular pair are supplied internally by the null-interval structure and the scaling-limit geometric action, after which Borchers–Wiesbrock yields the explicit positive null-translation generator on its Stone domain together with the affine half-line modular relation and the half-line generator/charge identification with the local null-stress charge on that same family; the D5 side then adds the internal small-ball bridge from the geometric cap generator together with bounded-interval transport through the separate projective branch, and the later tensor upgrade requires the explicit all-directions/all-reference-states premise on that scaling branch.

2. The exact Standard Model gauge structure

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}$$

   is recovered on the realized MAR-admissible branch of the bosonic compact-gauge theorem; this is a realized-branch gauge result with explicit matter-package and admissibility premises.

3. On the realized one-generation chiral matter plus one-Higgs package, anomaly constraints and Yukawa invariance fix the hypercharge lattice, the realized color triplet fixes $N_c = 3$, and the same one-Higgs quark branch then fixes $N_g = 3$ from CKM phase counting, weak-sector asymptotic freedom, and MAR.

4. The quantitative implementation used here uses one Phase-II pixel variable P together with one external continuous input N_{scr}.

5. Downstream matter-sector continuations, including the charged-lepton centered-readback / centered-to-affine frontier, are visible but explicitly outside this SM/GR derivation paper's recovered-core theorem package.

6. The same Einstein branch closes globally once the external screen-capacity input and the D6 screen-capacity identification are supplied, tying the cosmological-constant branch to global screen capacity rather than local vacuum energy.

7. Exclusion of gauge-mediated proton decay follows from product-group structure; any stronger proton-lifetime claim is UV/EFT/QCD-dependent.

8. Dark-sector, heuristic baryogenesis continuations, string, and spectroscopy branches are kept visible only as phenomenological continuations or, where additional ansätze are involved, conjectural phenomenology.

9. Overlap repair admits a unique schedule-independent normal form on the stated finite patch-net hypotheses, induces refinement-limit normal-form and holonomy classes on separated cofinal refinement systems with compatible projections, and makes reconciliation commute with coarse-graining up to explicitly declared normal-form and obstruction defects.

10. The fixed-cutoff topological UV package closes on the ordinary branch, the central-defect branch, and the genuinely noncentral branch through a compact crossed-module higher-gauge collar theorem; the UV-side open burdens are the continuum BW/geometric lift and the refinement-limit compact-gauge premises.

On that same fixed-cutoff consensus surface, the status split is sharp: normal-form computation is a finite-state decidable problem with the Lyapunov step bound from the consensus paper, the only present approximate-stability inputs are the collar-local splice and record estimates carried there, and no universality theorem for a uniform family of growing patch nets and repair laws is claimed in this paper. The imported D1 package is the fixed-cutoff consensus theorem stack: asynchronous confluence for the declared recovery-derived repair law, the cycle-obstruction / higher-gauge defect package, gauge-quotient descent, observable-level confluence on the declared physical observable algebras, the refinement-limit normal-form / holonomy theorem, the controlled coarse-graining / reconciliation square, and the record-algebra theorem on the observer-accessible surface. The imported D1 burdens are repair completeness and the stated Petz support/CPTP clause where that branch is used. The scaling-limit bridge from that fixed-cutoff patch-net package to the Lorentz, Einstein, and compact-gauge branches is separate; the consensus paper supplies the intermediate inverse-limit normal-form and holonomy theorem once the cofinal refinement projections commute with finite-stage normal forms and holonomy maps, and it supplies the approximate RG/reconciliation comparison once the chosen coarse-graining channel has controlled normal-form and obstruction defects.

2 Overview of Results

This section summarizes the claim tiers used throughout. The labels $D1$–$D12$ are internal documentary node labels for the project dependency ledger, not bibliography references or external notation. The public paper ledger has no separate D11 row; companion papers should use only the labels defined in the ledger below. The table below is the local key for this SM/GR derivation

paper, and the companion papers reuse the same labels when they point back to this ledger. The three-phase boundary is

$$\text{Phase I} = (D1\text{–}D5) \cup (D7\text{–}D9), \qquad \text{Phase II} = D6 \cup D10, \qquad \text{Phase III} = D12.$$

Phase I is the recovered-core claim set. Phase II contains the input-dependent global closure D6 of the D5 Einstein branch together with the integrated D10 quantitative-closure branch. Phase III collects phenomenological continuations. The canonical five-axiom basis, the quantitative quantities listed next, the scaling/continuum premises, and the gauge-side premises used below are fixed in Section 3. Each row names a node in the reconstruction program, records its immediate internal parents, separates imported standard mathematics from declared non-axiom premises or external inputs, and states the resulting claim tier.

Premise key for the table. The labeled premises used in the dependency table are introduced here before the ledger appears later in the paper. **T2** is the controlled refinement/scaling-limit scope clause for the Lorentz/null-modular/Einstein branch. The fixed-cap generalized-entropy stationarity result is Theorem 6.18, not a labeled premise. The bosonic gauge-reconstruction premises corresponding to **T3**–**T5** are stated separately in Assumption 3.7 in Section 3; on the central-defect branch, the loop-coherence criterion **T1** is carried by the same path-independent transportability hypothesis rather than listed as a second independent gauge premise.

Table 1: Dependency checklist for the OPH SM/GR reconstruction program.

Node	Output	Immediate internal ingredients	Standard mathematics used	Declared non-axiom premises / external inputs	Claim tier
D1	Unique schedule-independent normal form on the physical quotient, observable-level confluence on the declared fixed-cutoff physical algebras for the recovery-derived repair relation, inverse-limit normal-form / holonomy classes on separated cofinal refinement systems, and controlled reconciliation/coarse-graining compatibility	overlap-consistency problem on a finite patch net together with the declared fixed-cutoff collar recovery step, its touched-overlap local-fit acceptance contract, support-local disjoint commutation, restriction-compatible union-collar gluing on the physical quotient, the fixed-point / quotient-local physical observable algebras carried by that same collar package, cofinal refinement projections compatible with finite-stage normal forms and holonomy maps, and a coarse-graining channel with declared normal-form and obstruction defects	local-to-global Lyapunov descent on the inconsistency potential; local-diamond completion from overlap-associative center-sector / higher-gauge union-collar gluing; Newman's lemma; inverse limits of separated finite quotient systems; pseudometric control of commuting coarse-graining diagrams	repair completeness; stated Petz support/CPTP control where that branch is used; normal-form naturality, holonomy naturality, visible separation, and explicit coarse-graining defect bounds for the refinement system	Phase I structural theorem
D2	Collar Markov/entropy split and G-dictionary setup (Theorem 5.2, Propositions 5.11, 5.12, 5.16, Definition 5.17)	Axioms 3.1–3.4	HJPW Markov structure theorem; Petz and Fawzi–Renner recovery theory	exact block-factorized identities require exact Markovity or an explicitly stated idealized limit; the $A/(4G)$ step is the coarse-grained dictionary of Definition 5.17	Phase I structural lemma/proposition layer

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Node	Output	Immediate internal ingredients	Standard mathematics used	Declared non-axiom premises / external inputs	Claim tier
D3	Lorentz branch $\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1)$ (Theorem 6.3, Corollary 6.6)	Axioms 3.1–3.4	Bisognano–Wichmann modular geometry; conformal-group classification	T2 together with the theorem-local realized transported geometric cap-local system whose carried-collar schedule is derived from the transported fixed-local-collar Markov/faithfulness datum, and ordered cut-pair rigidity on the emitted scaling-limit geometric cap pair; Theorem 6.2 fixes the target subnet internally; the scaling-limit observer algebra may leave the finite type-I regulator class, so the theorem is automorphism-level there and $K_C = 2\pi B_C$ is only the special type-I limit form	Phase I conditional scaling-limit theorem
D4	Null modular bridge to T_{kk}, including exact-or-controlled strip additivity, endpoint-Lipschitz renormalized half-line families, the weak tail generator, the derived half-sided modular inclusion, the explicit positive half-line null-translation generator on its Stone domain with affine half-line modular relations, and the exact half-line generator/charge identification with the local null-stress charge (Propositions 6.7, 6.10; Corollaries 6.8, 6.12; Theorem 6.11, Lemma 6.13; Theorem 6.14)	D2+D3 and Axioms 3.1–3.4	Borchers–Wiesbrock positive-generator theorem for standard half-sided inclusions; Stone's theorem; distributional differentiation on half-lines	the theorem-local null-cut center transfer, the extra inherited left/right strip-split condition needed for the spatial-collar-type tensor decomposition, and the exact-or-controlled Markov hypotheses of Proposition 6.7 and Corollary 6.8, plus T2; quasi-local propagation and endpoint-Lipschitz control are supplied internally by Axiom 3.3 on the local finite-constraint branch, the geometric scaling action on the null half-line blow-up net derives the half-sided modular pair, the effective local modular-Hamiltonian formula is used only to read the same renormalized half-line family in continuum variables, and bounded-interval formulas require the separate interval-preserving projective branch	Phase I conditional bridge theorem
D5	Jacobson-type Einstein branch: rest-frame relation plus conditional tensor upgrade (Theorem 6.18, Lemma 6.20, Theorem 6.21, Corollary 6.22)	D3+D4 and Axioms 3.1–3.4	fixed-volume area-variation identity for small geodesic balls; local quadratic-polarization argument for the tensor upgrade	Theorem 6.18, which fixes the admissible fixed-cap MaxEnt variation class and derives generalized-entropy stationarity on the realized cap-label-preserving MaxEnt family; the internal small-ball bridge of Lemma 6.20, which uses the geometric cap generator together with the D4 half-line generator/charge identification and the separate interval-preserving projective branch; locally Lorentzian $d = 4$ scaling regime; small-ball constancy assumptions; D4 carried remainder negligible at order ℓ^4; all local directions/reference states for the tensor upgrade	Phase I conditional scaling-limit theorem/corollary
D6	Local/global theorem stack for Λ: null-invisible metric ambiguity, screen-capacity closure of the same Einstein branch, and the derived de Sitter static-patch parameter package, plus a legacy capacity-level neutrino side estimate (Theorem 6.30)	D5 local Einstein recovery leaves a $+\Lambda g_{ab}$ ambiguity	de Sitter entropy relation, static-patch radius/time formulas, and dimensional analysis	external input N_{scr} and the screen-capacity identification	Phase II input-dependent global closure / estimate

Table 1: Dependency checklist for the OPH SM/GR reconstruction program.

Node	Output	Immediate internal ingredients	Standard mathematics used	Declared non-axiom premises / external inputs	Claim tier
D7	Construction of the refinement-stable bosonic sector category plus compact gauge reconstruction (Theorem 7.2, Theorem 7.3)	Axioms 3.1–3.4	directed-colimit descent for monoidal C^*-categories; Doplicher–Roberts / Tannaka reconstruction	Assumption 3.7 on the ordinary or central-defect bosonic branch; the genuinely noncentral fixed-cutoff branch is handled separately by Theorems 5.3–5.6 and is not itself the ordinary compact-group reconstruction theorem	Phase I conditional structural theorem
D8	Product gauge structure up to finite quotient (Lemmas 7.5–7.10, Theorem 7.11)	D7 + Axiom 3.5	compact Lie representation classification; Schur's lemma	the same ordinary or central-defect bosonic branch as D7, together with a connected positive-dimensional Lie admissible class, one connected abelian factor, and faithful action on the minimal coupled carrier	Phase I realized-branch theorem
D9	Realized Standard Model quotient, hypercharges, $N_c = 3$, $N_g = 3$, and product-group corollaries (Theorem 7.12, Corollaries 7.13, 7.14, Proposition 7.16)	D8	anomaly-cancellation algebra; Witten global-anomaly argument; CKM CP counting	realized one-generation chiral matter plus one Higgs package, with the CP-capability and weak-sector UV clauses contained in Axiom 3.5; no extra post-MAR selector is used	Phase I realized-branch theorem/corollary chain
D10	Forward gauge-coupling closure and target-free source-only electroweak readout of the integrated D10 quantitative-closure package	D9 + pixel ratio P	printed RG evolution and matching conventions	pixel constraint and the source-only forward transmutation solve $\mathcal{F}(\alpha_U; P) = 0$; the fixed-cutoff edge heat-kernel / Casimir theorem on the microphysics surface together with the compact-group / Peter–Weyl lift used on the D10 lane; printed beta-function and threshold conventions	Phase II quantitative-closure sector
D12	Charged-lepton centered-readback / centered-to-affine frontier, texture, dark-sector, heuristic baryogenesis continuations, black-hole spectroscopy, proton-spin, proton-lifetime estimates beyond the gauge-channel exclusion, controlled large-N_{edge} string/worldsheet effective descriptions, conjectural critical-superstring extensions, and other continuations	various subsets of D6, D9, and D10	branch-specific EFT and phenomenological manipulations	additional ansätze such as the uniform $\mathbb{Z}_6$ center-label ensemble, discrete texture choices, dark-sector response assumptions, discrete-horizon assumptions, a distinct large-N_{edge} regime with fixed $\tau = tN_{\mathrm{edge}}$ window and uniform genus-remainder control, or extra worldsheet/CFT premises	Phase III phenomenological continuation branch

The recovered core of this SM/GR derivation paper is D1–D5 together with D7–D9. Within that package, D7 is the bosonic sector-category construction plus compact-gauge reconstruction stage, while D8–D9 are realized-branch results that keep the realized one-generation chiral matter plus one-Higgs package and MAR admissibility premises explicit. D6 is not part of that recovered core because it requires the external screen-capacity input N_{scr}, but it is the input-dependent local/global theorem stack for the same Einstein branch: D5 leaves the metric term undetermined, D6 closes that same branch once the external screen-capacity input N_{scr} and the screen-capacity identification are declared, and the same D6 stack emits the de Sitter static-patch parameter package. D10 is the integrated quantitative-closure branch, and D12 collects phenomenological continuations. The modular-anomaly dark-sector branch belongs to D12: it imports the D6 static-patch scale only as an IR benchmark and does not derive MOND/RAR-style response laws, a controlled galaxy-scale source/response law, or galaxy/cluster phenomenology. The only external continuous input

is N_{scr} in D6; the local pixel ratio P belongs to the separate Phase-II closure branch together with descendants built from it.

3 Five Axioms, the Phase-II Pixel Variable, the External Capacity Input, and Technical Premises

This section states the premise ledger and dependency map for this SM/GR derivation paper. The basis used below consists of the five core axioms, the quantitative quantities listed next, the scaling/continuum premises collected afterward, and the gauge-side premises of Assumption 3.7. This is an algebraic-information basis: the screen-net, state, trace/probability, and generalized-entropy structures are starting ingredients of the current OPH framework. The purpose is not to derive every mathematical ingredient from a blank slate, but to test whether this basis supports a consistent and comprehensive theory-of-everything reconstruction of the observed effective universe. On the fixed-cutoff local finite-constraint MaxEnt branch, the local-Gibbs form, quasi-local dynamics, Lieb–Robinson propagation control, and endpoint-Lipschitz interval control are internal consequences rather than theorem-external regularity selectors. Any Dobrushin/local-Gibbs/mixing language used below names only a sufficient fixed-cutoff collar-recoverability mechanism on that branch. It is not an infinite-volume uniqueness claim for the refinement-limit theory, and it does not by itself constrain whether a nontrivial transportable bosonic refinement ladder exists. The paper therefore treats that ladder as explicit gauge-branch data rather than as an entropy-side consequence of the MaxEnt/mixing package. What Axiom 3.3 fixes under refinement is the realized state-side MaxEnt branch itself. Theorem 6.2 fixes the operational/geometric split at fixed cutoff, and Theorem 6.3 fixes the cap generator and its 2π normalization once its theorem-local geometric cap-pair extraction and ordered cut-pair rigidity hypotheses are supplied on that geometric subnet. On the D3 UV side, the theorem-local scaffold is the realized transported geometric cap-local system whose carried-collar schedule is derived from the transported fixed-local-collar Markov/faithfulness datum, together with ordered cut-pair rigidity on the emitted scaling-limit geometric cap pair. Corollary 6.12 then derives the null half-sided modular pair after the null blow-up step, and Lemma 6.13 promotes that pair to the explicit positive null-translation generator on its Stone domain with the affine half-line modular relation. Theorem 6.18 then internalizes the fixed-cap generalized-entropy stationarity step for admissible fixed-cap MaxEnt variations on the realized cap-label-preserving MaxEnt family. The canonical theorem-external items are therefore the scaling-limit scope clause, the bounded-interval transport/projective branch where invoked, and the transportable bosonic ordinary/central gauge premises recorded in Assumption 3.7. On the central branch, that assumption names the same path-independent transportability whose gluing-side form is the central loop-coherence obstruction, so that obstruction is not listed as a second free theorem-side input. The genuinely noncentral fixed-cutoff branch is tracked instead by Theorems 5.3 and 5.6; it is not itself the ordinary compact-group theorem used by the compact-gauge branch.

The formulation uses one Phase-II pixel variable together with one external continuous input:

$$P \equiv a_{\mathrm{cell}}/\ell_P^2, \tag{1}$$

$$N_{\mathrm{scr}} \equiv \log \dim \mathcal{H}_{\mathrm{tot}}. \tag{2}$$

In the quantitative implementation used here, P is the local particle-physics scale variable fixed by the outer/inner closure program summarized in the synthesis paper *Observers Are All You Need* [1]; this compact paper uses it only as the quantity carried by the forward D10 particle-physics map.

That closure writes the outside detuning as

$$P = \varphi + \alpha_{\rm in}(P)\sqrt{\pi},$$

where the inner side is the electromagnetic observation scale emitted by the same cell on the declared quantitative branch. The code surface includes a numerical fixed-point witness for this map and accepts external inverse-α values only when they are supplied explicitly as compare-only metadata; the paper comparison gives $P = 1.630968209403959\ldots$ from $\alpha^{-1} = 137.035999177$. This is not an interval-arithmetic proof of uniqueness; the companion local numerical contraction certificate records the machine-checkable hardening step. A separate hardware note reports an optical-cavity check of the same fixed-point geometry; this is treated as corroborating engineering evidence. $N_{\rm scr}$ is inferred from the observed cosmological horizon and enters the cosmological-capacity branch. The D6 normalization uses the Gibbons–Hawking de Sitter entropy [12]: if

$$N_{\rm patch} = \left(\frac{r_{\rm dS}}{\ell_P}\right)^2$$

denotes the bare horizon area ratio, then

$$N_{\rm scr} = S_{\rm dS} = \frac{A_{\rm dS}}{4\ell_P^2} = \pi N_{\rm patch} = \frac{3\pi}{\Lambda\ell_P^2}.$$

Using the Planck-2018 cosmological benchmark [13] gives $r_{\rm dS} \simeq 1.66\times10^{26}$ m, $N_{\rm patch} \simeq 1.05\times10^{122}$, $N_{\rm scr} \simeq 3.31\times10^{122}$, and $\Lambda\ell_P^2 \simeq 2.85\times10^{-122}$. Any retained capacity-level neutrino side estimate on that input is legacy bookkeeping rather than the live weighted-cycle theorem lane.

Axiom 3.1 (Screen Net)**.** *Physical data is organized on a horizon screen S^2 carrying a net of local algebras*

$$P \longmapsto \mathcal{A}(P)$$

for connected patches $P \subset S^2$, with isotony

$$P \subset Q \implies \mathcal{A}(P) \subset \mathcal{A}(Q).$$

Axiom 3.2 (Overlap Consistency)**.** *For overlapping patches $P_1 \cap P_2 \neq \varnothing$, the local states induced on the shared algebra agree:*

$$\omega_{P_1}|_{\mathcal{A}(P_1\cap P_2)} = \omega_{P_2}|_{\mathcal{A}(P_1\cap P_2)}.$$

Axiom 3.3 (Local MaxEnt and Refinement Stability)**.** *At the regulator scale $\ell_{\rm UV}$, the realized branch is selected by maximizing entropy subject to a finite family of gauge-invariant local constraints*

$$\mathcal{C}_{\ell_{\rm UV}} = \{O_a(x)\}_{a=1}^{N_{\rm con}},$$

where each $O_a(x)$ is supported in a ball of radius $O(\ell_{\rm UV})$ and the label set a is independent of the number of regulator cells. Under refinement, the same finite constraint family is preserved, so the realized states at different cutoffs belong to one common finite-dimensional MaxEnt family. The realized low-energy branch is the refinement-stable branch of that family; symmetry-allowed relevant operators are therefore held at zero on that branch only by symmetry or because they lie in the explicitly retained constraint family. This is a statement about branch persistence, not a universal entropy-ordering theorem for arbitrary phases away from that branch.

Derived local-Gibbs branch. On the finite regulator realization of Axiom 3.1, the standard finite-dimensional Lagrange-multiplier argument applied to Axiom 3.3 gives

$$\omega_{\ell_{\mathrm{UV}}}(\lambda) = Z_{\ell_{\mathrm{UV}}}(\lambda)^{-1} \exp\bigl(-K_{\ell_{\mathrm{UV}}}(\lambda)\bigr),$$

with

$$K_{\ell_{\mathrm{UV}}}(\lambda) = \sum_{x} \sum_{a=1}^{N_{\mathrm{con}}} \lambda_a O_a(x) + \sum_{b} \mu_b Q_b,$$

where the Q_b are the finitely many optional global conserved-charge constraints. Thus the logarithm of the selected state is itself a quasi-local UV generator built from the same bounded-support densities that define the branch.

Derived quasi-local propagation and interval control. Fix the regulator graph metric coming from cell adjacency and let $\tau_t^{\ell_{\mathrm{UV}}}$ denote the automorphism group generated by $K_{\ell_{\mathrm{UV}}}(\lambda)$, or more generally by any branch generator lying in the norm-closed algebra generated by the same bounded-support local densities. Standard Lieb–Robinson estimates for finite-range interactions on the finite regulator net then give constants $C, \xi, v_{\ell_{\mathrm{UV}}} < \infty$ such that for local observables A_X, B_Y, [4]

$$\bigl\|[\tau_t^{\ell_{\mathrm{UV}}}(A_X), B_Y]\bigr\| \le C \, \|A_X\| \, \|B_Y\| \, \min(|X|, |Y|) \, e^{-(d(X,Y) - v_{\ell_{\mathrm{UV}}}|t|)/\xi}.$$

So finite-velocity quasi-local propagation is branch-internal: it is the local finite-constraint MaxEnt branch written in propagation form, not an extra selector.

The same locality statement yields the bounded-interval endpoint-Lipschitz control used by the compact-gauge branch in the null modular chain. Whenever later sections use a fixed-cutoff interval or collar generator on this branch, it is the corresponding restriction of the same local density list, with the central endpoint term separated off exactly as in the later edge-center bookkeeping. Changing an interval I to I' then alters only an $O(|I' \Delta I|)$ collar of local terms. Hence on any fixed local-energy-bounded domain one has

$$\bigl|\langle \psi, (K[I'] - K[I])\phi \rangle\bigr| \le C_{\psi,\phi,I_{\max}} \, |I' \Delta I|,$$

which is precisely the endpoint-Lipschitz / finite-variation control used by the compact-gauge branch in the null modular chain. This paper does *not* separately derive a second independent microscopic generator beyond this quasi-local branch generator; if a UV completion carries one, the only later requirement is that it lie in the same bounded-support algebraic closure so that the same propagation constants control the selected branch.

This local-Gibbs/Lieb–Robinson package, together with any Dobrushin-type estimate used by the compact-gauge branch, should be read only as fixed-cutoff collar-local recoverability, support control, and carried-error bookkeeping on the selected branch. One may impose such estimates on a chosen finite collar model without deciding the refinement-limit gauge phase. The package is therefore not a proof that the refinement limit lands in a trivial thermodynamic phase, not a proof that transportable edge sectors persist under refinement, not a proof that the coherent bosonic refinement ladder of Assumption 3.7 exists, and not a proof that its realized branch is nontrivial. Its theorem-level role is precisely the collar-local mixing and support/error control carried into the later fixed-cutoff and scaling-limit arguments.

Internal refinement notion. Choose any family of refinement channels $\Phi_{\ell \to L}$ compatible with Axiom 3.3. Because the same finite local constraint family is preserved, the regulator-scale realized

states are parameterized by one common finite-dimensional multiplier space λ, and on the realized branch one may write

$$\Phi_{\ell\to L}(\omega_\ell(\lambda)) = \omega_L(R_{\ell\to L}(\lambda))$$

for an induced map $R_{\ell\to L}$ on that multiplier space. The refinement-stable realized branch is therefore the persistent trajectory or invariant subset selected inside this finite-dimensional space, not an imported theorem-external regularity package. This state-side refinement notion is enough to compare realized states across cutoffs. It does not, by itself, upgrade fixed-cutoff edge labels to transportable refinement-persistent sectors, and it does not show that the coherent bosonic refinement ladder of Assumption 3.7 exists or is nontrivial. Any Dobrushin/local-mixing hypothesis used by the compact-gauge branch is logically separate: it supplies a collar estimate on a fixed finite-dimensional model, not information about the refinement-limit gauge phase. Accordingly, whenever later sections speak of a "refinement-stable directed colimit" of edge sectors, the entire transportable monoidal-refinement/fiber clause is the explicit content of Assumption 3.7; Axiom 3.3 supplies only the realized state branch along which one asks whether such sector transport persists.

Axiom 3.4 (Recoverable Generalized Entropy)**.** *A generalized entropy functional exists on caps,*

$$S_{\mathrm{gen}}(C) = S_{\mathrm{bulk}}(C) + \langle L_C \rangle,$$

where L_C is a positive edge-center entropy functional. In the semiclassical scaling branch, its leading coarse-grained contribution is identified with $A(\partial C)/(4G)$. The functional obeys quantum focusing on null generators, and collar tripartitions have small conditional mutual information with controlled recovery maps [15, 16, 17, 18].

Axiom 3.5 (Minimal Admissible Realization)**.** *Among realized sector packages $\mathfrak{S}$ consisting of the connected Lie gauge-sector image relevant in the low-energy EFT, its admissible light chiral matter content, and one Higgs doublet, and which are loop-coherent, anomaly-free, refinement-stable with light chiral matter, single-Higgs Yukawa-completable with one connected abelian charge factor acting nontrivially on the coupled carrier, intrinsically quark-sector CP-capable, and weak-sector UV-completable on that same one-Higgs branch, the realized package is the lexicographically minimal one under*

$$C(\mathfrak{S}) = (\chi_{\mathrm{cpl}},\, N_{\mathrm{nonab}},\, N_c,\, N_g).$$

$\mathfrak{S}$ is the sector package on which MAR acts; it is not the bare tensor category alone. MAR is therefore an explicit structural-economy axiom on admissible realized low-energy branches, not a theorem derived from the preceding axioms. χ_{cpl} is the coupled carrier dimension: the smallest unitary carrier containing a common irreducible block on which the admissible pseudoreal and complex nonabelian charge types both act nontrivially. This is stronger than the abstract minimal faithful representation dimension.

Assumption 3.6 (Scaling premise)**.** *This SM/GR derivation paper uses one canonical numbered list for the theorem-external scaling items outside the gauge branch. The fixed-cutoff realized presentation, the local-Gibbs form, quasi-local Lieb–Robinson propagation, endpoint-Lipschitz interval control, the induced finite-dimensional refinement branch, the fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap MaxEnt variations on the realized cap-label-preserving MaxEnt family, and the operational/geometric split of Theorem 6.2 are not listed here because they are established internally from Axioms 3.1–3.3. The external item outside the gauge branch is:*

- ***T2*** *all Lorentz/null-modular/Einstein statements are claims about a controlled refinement/scaling limit of the regulator net, not literal exactness at fixed finite cutoff;*

The fixed-cap generalized-entropy stationarity result is Theorem 6.18. The bosonic ordinary/central gauge premises are stated separately in Assumption 3.7.

Assumption 3.7 (Bosonic gauge reconstruction premises)**.** *Whenever compact gauge reconstruction is invoked on the ordinary or central-defect branch, the following additional premises are kept explicit. Together they are the compact-paper form of* ***T3–T5****. On the central-defect branch, the matching loop-coherence criterion* ***T1*** *is carried by the same supplied path-independent transportability rather than listed as a second independent premise.*

(G1) for every sufficiently fine cutoff r, a fixed-cutoff bosonic collar-sector package Sect_r: a semisimple rigid C^*-tensor category of localized transportable edge sectors on the declared overlap collars, with tensor product given by collar concatenation, duals given by orientation reversal / charge conjugation, and bosonic symmetry in the relevant $3+1$-dimensional EFT branch;

(G2) for every refinement $r \preceq s$, a faithful monoidal *-functor $U_{rs} : \mathsf{Sect}_r \to \mathsf{Sect}_s$ carrying the tensor unit to the tensor unit, preserving localized transportable sector classes together with their overlap-visible support on cofinal tails, and compatible with composition $U_{su} \circ U_{rs} \cong U_{ru}$;

(G3) faithful stagewise bosonic multiplicity-space functors $F_r : \mathsf{Sect}_r \to \mathsf{Hilb}_{\mathrm{fd}}$ with objectwise finite-dimensional values, together with monoidal natural isomorphisms $F_s \circ U_{rs} \cong F_r$ compatible with the refinement system.

These premises construct the refinement-limit bosonic sector category used below; they do not prove that the realized transportable branch is nontrivial. The genuinely noncentral fixed-cutoff branch, where the parallel obstruction criterion is $q_\Sigma = 0$, is handled separately by Theorems 5.3–5.6 and is not itself the ordinary compact-group reconstruction theorem.

Remark 3.8 (Log convention)**.** *Unless an explicit base is written, logarithms and entropies in this paper use natural logs (nats). Base-2 logarithms are written as $\log_2$ and quoted in bits.*

Dependency DAG

The table below is the dependency map for this SM/GR derivation paper. The "Immediate OPH ingredients" column records only parents internal to the OPH program. Imported standard mathematics and declared non-axiom premises or external inputs are stated separately rather than hidden inside the word "derived."

Node	Theorem labels and output	Immediate OPH ingredients	Standard mathematics used	Extra non-axiom physical premises / external inputs	Status
D1	Theorem 4.3: unique normal form and schedule independence for the declared recovery-derived repair relation	overlap-consistency problem on a finite patch net together with the declared fixed-cutoff collar recovery step, its touched-overlap local-fit acceptance contract, support-local disjoint commutation, and restriction-compatible union-collar gluing on the physical quotient	local-to-global Lyapunov descent on the inconsistency potential; local-diamond completion from overlap-associative center-sector / higher-gauge union-collar gluing; Newman's lemma	repair completeness; stated Petz support/CPTP control where that branch is used	structural theorem
D2	Theorem 5.2, Propositions 5.11, 5.12, 5.16, Definition 5.17: collar Markov/entropy layer	Axioms 3.1–3.4	HJPW Markov structure theorem; Petz and Fawzi–Renner recovery theory	exact Markov identities require exact Markovity or an explicitly stated idealized recoverability limit; the $A/(4G)$ identification is the coarse-grained dictionary of Definition 5.17	structural lemma/ proposition layer
D3	Theorem 6.3, Corollary 6.6: geometric modular flow and Lorentz branch	Axioms 3.1–3.4	Bisognano–Wichmann modular geometry; conformal classification on S^2	T2 together with the theorem-local realized transported geometric cap-local system whose carried-collar schedule is derived from the transported fixed-local-collar Markov/faithfulness datum, and ordered cut-pair rigidity on the emitted scaling-limit geometric cap pair; Theorem 6.2 fixes the target subnet internally; the scaling-limit observer algebra may leave the finite type-I regulator class, so the theorem is automorphism-level there and $K_C = 2\pi B_C$ is only the special type-I limit form	conditional scaling-limit theorem

Node	Theorem labels and output	Immediate OPH ingredients	Standard mathematics used	Extra non-axiom physical premises / external inputs	Status
D4	Propositions 6.7, 6.10; Corollaries 6.8, 6.12; Theorem 6.11, Lemma 6.13; Theorem 6.14: null modular bridge to T_{kk}, including exact-or-controlled strip additivity, endpoint-Lipschitz renormalized half-line families, the weak tail generator, the derived half-sided modular inclusion, the explicit positive half-line null-translation generator on its Stone domain with affine half-line modular relations, and the exact half-line generator/charge identification with the local null-stress charge	D2+D3 and Axioms 3.1–3.4	Borchers–Wiesbrock positive-generator theorem for standard half-sided inclusions; Stone's theorem; distributional differentiation on half-lines	the theorem-local null-cut center transfer, the extra inherited left/right strip-split condition needed for the spatial-collar-type tensor decomposition, and the exact-or-controlled Markov hypotheses of Proposition 6.7 and Corollary 6.8, plus T2; quasi-local propagation and endpoint-Lipschitz control are supplied internally by Axiom 3.3 on the local finite-constraint branch, the geometric scaling action on the null half-line blow-up net derives the half-sided modular pair, the effective local modular-Hamiltonian formula is used only to read the same renormalized half-line family in continuum variables, and bounded-interval formulas require the separate interval-preserving projective branch	conditional bridge theorem
D5	Theorem 6.18, Lemma 6.20, Theorem 6.21, Corollary 6.22: Jacobson-type Einstein branch via a rest-frame relation plus conditional tensor upgrade	D3+D4 and Axioms 3.1–3.4	fixed-volume area-variation identity for small geodesic balls; local quadratic-polarization argument for the tensor upgrade	Theorem 6.18, which fixes the admissible fixed-cap MaxEnt variation class and derives generalized-entropy stationarity on the realized cap-label-preserving MaxEnt family; the internal small-ball bridge of Lemma 6.20, which uses the geometric cap generator together with the D4 half-line generator/charge identification and the separate interval-preserving projective branch; locally Lorentzian $d = 4$ scaling regime; small-ball constancy assumptions; D4 carried remainder negligible at order ℓ^4; all local directions/reference states for the tensor upgrade	conditional scaling-limit theorem/corollary

Node	Theorem labels and output	Immediate OPH ingredients	Standard mathematics used	Extra non-axiom physical premises / external inputs	Status
D6	Theorem 6.30: local/global theorem stack for Λ, namely the null-invisible metric ambiguity, the screen-capacity closure of the same Einstein branch, and the derived de Sitter static-patch parameter package, plus a legacy neutrino side estimate	D5 local Einstein recovery, which leaves the $+\Lambda g_{ab}$ ambiguity	de Sitter entropy relation, static-patch radius/time formulas, and dimensional analysis	external input N_{scr} and the screen-capacity identification	input-dependent global closure / estimate
D7	Theorem 7.2, Theorem 7.3: construction of the refinement-stable bosonic sector category and compact gauge reconstruction	Axioms 3.1–3.4	directed-colimit descent for monoidal C^*-categories; Doplicher–Roberts / Tannaka reconstruction	Assumption 3.7 on the ordinary or central-defect bosonic branch; the genuinely noncentral fixed-cutoff branch is handled separately by Theorems 5.3–5.6 and is not itself the ordinary compact-group reconstruction theorem	conditional structural theorem
D8	Lemmas 7.5–7.10, Theorem 7.11: product gauge structure up to finite quotient	D7 + Axiom 3.5	compact Lie representation classification; Schur's lemma	the same ordinary or central-defect bosonic branch as D7, together with a connected positive-dimensional Lie admissible class, one connected abelian factor, and faithful action on the minimal coupled carrier	realized-branch theorem
D9	Theorem 7.12, Corollaries 7.13, 7.14, Proposition 7.16: realized Standard Model quotient, hypercharges, $N_c = 3$, $N_g = 3$, and product-group corollaries	D8	anomaly-cancellation algebra; Witten global-anomaly argument; CKM CP counting	realized one-generation chiral matter plus one Higgs package, with the CP-capability and weak-sector UV clauses contained in Axiom 3.5; no extra post-MAR selector is used	realized-branch theo-rem/corollary chain
D10	forward gauge-coupling closure and target-free source-only electroweak readout of Section 7	D9 + pixel ratio P	printed RG evolution and matching conventions	pixel constraint and the source-only forward transmutation solve $\mathcal{F}(\alpha_U; P) = 0$; the fixed-cutoff edge heat-kernel / Casimir theorem on the microphysics surface together with the compact-group / Peter–Weyl lift used on the D10 lane; printed beta-function and threshold conventions	quantitative-closure sector

Node	Theorem labels and output	Immediate OPH ingredients	Standard mathematics used	Extra non-axiom physical premises / external inputs	Status
D12	Sections 8–10 continuations: charged leptons, proton spin/lifetime, and controlled string/worldsheet effective-description branches	various subsets of D6, D9, and D10	branch-specific EFT and phenomenological manipulations	additional ansätze such as the uniform $\mathbb{Z}_6$ center-label ensemble, texture choices, dark-sector response assumptions, discrete-horizon assumptions, a distinct large-N_{edge} regime with fixed $\tau = tN_{\text{edge}}$ window and uniform genus-remainder control, or extra worldsheet/CFT premises	phenomenological continuation

The dependency DAG records the immediate internal parents, imported mathematics, and extra premises for each node.

Theorem 3.9 (Summary theorem for the recovered relativity-plus-Standard-Model core)**.** *This theorem packages the recovered-core nodes D1–D5 and D7–D9 of the dependency DAG above. Node D6 is not part of this recovered-core theorem because it requires the external screen-capacity input N_{scr}, but it is the input-dependent global closure of the Einstein branch recovered in item (iii); D10 and D12 are outside this theorem's scope. Assume Axioms 3.1–3.4, Axiom 3.5, Assumptions 3.6 and 3.7, and the hypotheses of Theorems 4.3, 6.3, 6.14, 6.21, 7.2, 7.3, 7.11, and 7.12, together with Corollaries 7.14, 7.13, and Proposition 7.16. Then, in the controlled refinement/scaling regime singled out by Assumption 3.6:*

(i) overlap repair admits a unique schedule-independent normal form;

(ii) cap modular flow on the extracted prime geometric cap pair is geometric and yields the connected Lorentz group

$$\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1);$$

(iii) the derived fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap MaxEnt variations on the realized cap-label-preserving MaxEnt family together with the null modular bridge and the bounded-interval projective branch yield the Jacobson-type rest-frame relation

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\, \delta\langle T_{00}\rangle$$

at the cap center in the diamond rest frame; if that rest-frame relation holds for all local directions and reference states in the scaling regime, then the null-to-tensor upgrade gives

$$G_{ab} + \Lambda g_{ab} = 8\pi G\, \langle T_{ab}\rangle;$$

(iv) under Assumption 3.7, the edge-sector category reconstructs a compact gauge group, and the realized connected gauge structure has the form

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\Gamma}$$

for some finite central subgroup Γ.

After the hypercharge lattice, color-count, generation-count, and trivial-action quotient steps, the finite quotient is fixed to $\Gamma = \mathbb{Z}_6$*, so the realized gauge structure on the realized MAR-admissible branch is*

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_c = 3, \qquad N_g = 3,$$

with the exact Standard Model hypercharge lattice on the realized matter package. Here the count statements are internal to the same realized branch: the D8 minimal coupled-carrier theorem stack fixes the color triplet and hence $N_c = 3$*; CKM phase counting and weak-sector asymptotic freedom then bound* N_g *on that same one-Higgs branch; MAR fixes the smallest realized value* $N_g = 3$*; and Witten's* $\mathrm{SU}(2)$ *anomaly is retained only as a consistency check on the resulting triplet-doublet package.*

Proof. Items (i)–(iii) are collected from Theorem 4.3, Corollary 6.6, Theorem 6.21, and Corollary 6.22. On the gauge side, Theorem 7.2 first assembles the supplied fixed-cutoff bosonic collar-sector packages, their monoidal refinement transport, and their compatible finite multiplicity fibers into the refinement-limit category and fiber functor. The compact gauge reconstruction is then Theorem 7.3. On the central branch the matching loop-coherence burden is already part of that supplied transportability rather than an extra theorem-side premise. The preceding MaxEnt/local-Gibbs/collar-mixing package is not used there to prove nontriviality of the transportable bosonic ladder. The product gauge structure up to finite quotient is Theorem 7.11; and the final identification of the exact Standard Model quotient is Theorem 7.12 together with Corollaries 7.13, 7.14, and Proposition 7.16. □

Theorem 3.9 therefore packages only the recovered relativity-plus-Standard-Model core: D1–D5 together with D7–D9. D6 is not a detached second gravity lane: it is the input-dependent global closure of the same Einstein branch once the screen-capacity identification is supplied. The quantitative-closure branch and phenomenological continuations are outside the theorem's scope.

Usually Postulated Elsewhere, Recovered or Continued Here with Explicit Status

Structure	Common treatment	OPH treatment
Lorentz kinematics	background symmetry or starting axiom	conditional scaling-limit branch from geometric modular flow on screen caps
Einstein dynamics	fundamental field equation	conditional first-variation relation with tensor upgrade from generalized entropy, null modular data, and derived fixed-cap generalized-entropy stationarity
Gauge group	model input	conditional construction of the refinement-limit bosonic edge-sector category, then compact group reconstruction from it; exact SM quotient selected on the realized MAR-admissible branch with the realized one-generation/one-Higgs package
Hypercharge lattice	matter-assignment input	solved from anomaly cancellation and Yukawa invariance on the realized one-generation chiral matter plus one-Higgs package
Color and generation count	empirical input	color triplet fixed by the D8 minimal coupled carrier; generation count fixed by CKM phase counting, weak-sector asymptotic freedom, and MAR; Witten parity retained as a consistency check
Flavor hierarchy unit	model-dependent small parameter	phenomenological ansatz $\varepsilon = 1/6$ motivated by a uniform $\mathbb{Z}_6$ center-label ensemble
Charged-lepton continuation	Koide-style phenomenological fit	exact centered readback plus an open centered-to-affine frontier above $\widehat{C}_e^{\mathrm{cand}}$; any $\delta = 2/9$ phase relation is continuation-only
Cosmological constant	local vacuum-energy puzzle	global screen-capacity term after the input-dependent capacity identification

4 Observer-Overlap Consistency as a Fixed-Point Problem

The discrete overlap problem is the mathematical skeleton of the framework.

Definition 4.1 (Patch Net)**.** *Let $G = (V, E)$ be a finite connected graph. Each vertex $i \in V$ carries a finite local state space S_i. For each edge $e = \{i, j\}$ let I_e be an interface alphabet and let*

$$\pi_{i,e} : S_i \to I_e, \qquad \pi_{j,e} : S_j \to I_e$$

be interface projections. The global state space is

$$\Sigma = \prod_{i \in V} S_i,$$

and the consistency set is

$$C = \{s \in \Sigma : \pi_{i,e}(s_i) = \pi_{j,e}(s_j) \text{ for all } e = \{i, j\} \in E\}.$$

This finiteness is a regulator-level assumption for the discrete patch-net normal-form theorem; it is not the continuum limit statement.

Define the inconsistency potential

$$\Phi(s) = \sum_{e=\{i,j\}\in E} w_e \, d_e\big(\pi_{i,e}(s_i), \pi_{j,e}(s_j)\big),$$

with $w_e > 0$ and $d_e(a, b) = 0$ iff $a = b$.

Definition 4.2 (Recovery-derived local repair law)**.** *A law λ is a family of local repair maps*

$$T_i^\lambda : \Sigma \to \Sigma$$

changing only patch i or a bounded neighborhood of i. On the fixed-cutoff collar branch these are not free rewrite primitives: each accepted local update is read from exact Markov splice or a declared Petz/Fawzi–Renner recovery move on a collar chart around i, then lifted back to the finite patch presentation, and committed only when it strictly lowers the touched-overlap contribution to Φ on the overlaps it can modify. Disjoint accepted repairs therefore commute directly by support locality, and when two accepted repairs touch a common overlap they are read inside one finite union collar whose physical glued state is parenthesization-independent on the quotient and whose local decoders are restriction-compatible on nested collars.

Theorem 4.3 (Asynchronous Fixed-Point Law)**.** *Assume normal forms of the repair relation are exactly the globally consistent states C. Then every $s \in \Sigma$ has a unique normal form*

$$\mathrm{nf}_\lambda(s) \in C,$$

and every maximal asynchronous repair sequence terminates at that same state.

Proof. Each accepted local update changes only its touched-overlap set and is committed only when the sum of the corresponding Φ-terms strictly decreases. Therefore every accepted repair strictly lowers the global inconsistency potential Φ. Because Σ is finite, no infinite repair sequence exists, so the repair relation is terminating. If two accepted repairs are disjoint, support locality makes them commute directly. If they overlap, the same definition places them inside one parenthesization-independent union-collar gluing state on the physical quotient, so the two one-step branches have a common descendant. Thus the repair relation is locally confluent. Newman's lemma [3] gives confluence. The resulting normal form is unique. By assumption it is a globally consistent state. □

Corollary 4.4 (Schedule Independence)**.** *If observables are evaluated on $\mathrm{nf}_\lambda(s)$, the resulting physical law is independent of update order.*

Theorem 4.5 (Cycle Obstruction / Holonomy Criterion)**.** *Let A be an abelian group. For each oriented edge $e : u \to v$ assign $b_e \in A$ and consider the affine overlap equations*

$$x_v - x_u = b_e.$$

A global solution exists if and only if the signed sum of b_e vanishes on every cycle.

Proof. Summing the edge equations around a cycle gives necessity. For sufficiency, fix a root and define each x_v by summing labels along a path from the root to v. Vanishing cycle sums make this independent of the chosen path. □

Definition 4.6 (Physical repair law and representative lift)**.** *Suppose a local gauge group* $\Gamma = \prod_i \Gamma_i$ *acts on* Σ *while leaving all interface data invariant. Write*

$$q : \Sigma \to \Sigma/\Gamma, \qquad q(s) = [s].$$

A physical repair law is the family of local quotient maps induced by the recovery-derived collar updates just described:

$$\overline{T}_i^\lambda : \Sigma/\Gamma \to \Sigma/\Gamma$$

on the overlap-invariant quotient. A representative repair family is any family

$$T_i^\lambda : \Sigma \to \Sigma$$

with

$$q \circ T_i^\lambda = \overline{T}_i^\lambda \circ q.$$

Proposition 4.7 (Representative lifts descend to the quotient)**.** *For any representative repair family of Definition 4.6,*

$$q(T_i^\lambda(\gamma \cdot s)) = q(T_i^\lambda(s)) \qquad \forall\, i,\ \forall\, \gamma \in \Gamma,\ \forall\, s \in \Sigma.$$

In particular the repair relation descends to Σ/Γ *without any extra gauge-covariance axiom.*

Proof. Because $q(\gamma \cdot s) = q(s)$,

$$q(T_i^\lambda(\gamma \cdot s)) = \overline{T}_i^\lambda(q(\gamma \cdot s)) = \overline{T}_i^\lambda(q(s)) = q(T_i^\lambda(s)).$$

So gauge-equivalent inputs induce the same repaired physical state. □

Proposition 4.8 (Gauge Quotient)**.** *Under Definition 4.6, Proposition 4.7, and Theorem 4.3, the normal-form map descends to the quotient:*

$$\overline{\mathrm{nf}}_\lambda : \Sigma/\Gamma \to \Sigma/\Gamma, \qquad [s] \mapsto [\mathrm{nf}_\lambda(s)].$$

Hence gauge-invariant observables are unique on the quotient, and the fixed-cutoff physical observable algebra has representative-independent terminal expectations on that carrier.

Proof. By Proposition 4.7, every repair step has quotient image determined only by the current orbit. Iterating, the quotient image of any repair sequence depends only on the initial orbit. Since every maximal repair sequence from s terminates at the unique normal form $\mathrm{nf}_\lambda(s)$ by Theorem 4.3, the terminal orbit $[\mathrm{nf}_\lambda(s)]$ depends only on $[s]$. This makes $\overline{\mathrm{nf}}_\lambda$ well-defined. □

On the fixed-cutoff quantum lift, the same quotient-local carrier does more than fix the terminal orbit: for every declared fixed-cutoff physical algebra, the terminal expectation functional is the same on any two representative lifts whose regional collar data lie in one quotient-local glued state, even when the microscopic representatives differ by gauge or sector relabelings on that carrier.

Theorem 4.9 (Refinement-limit consensus classes)**.** *Let* R *be a directed cofinal refinement set. For each* $r \in R$*, let* $Q_r = \Sigma_r/\Gamma_r$ *be the finite physical quotient state space, let* $n_r : Q_r \to Q_r$ *be the finite quotient normal-form map, and let* $h_r : Q_r \to \mathcal{H}_r$ *be the finite holonomy or higher-gauge obstruction map. Suppose that for* $r \preceq s$ *there are restriction maps*

$$\rho_{sr} : Q_s \to Q_r, \qquad \chi_{sr} : \mathcal{H}_s \to \mathcal{H}_r$$

forming directed inverse systems and satisfying

$$\rho_{sr} n_s = n_r \rho_{sr}, \qquad \chi_{sr} h_s = h_r \rho_{sr}.$$

Assume visible separation: compatible families in the inverse limits are equal whenever they agree on a cofinal subset of finite stages. Then

$$n_\infty((x_r)_r) = (n_r(x_r))_r, \qquad h_\infty((x_r)_r) = (h_r(x_r))_r$$

define a unique schedule-independent refinement-limit normal-form class and a refinement-limit holonomy class. The finite normal forms and holonomies converge to those classes in the inverse-limit topology. A nonzero limiting holonomy has a finite-stage witness, and agreement of the normal-form and holonomy projections on a cofinal tail gives the same refinement-limit consensus class.

Proof. Compatibility of the x_r gives $\rho_{sr}(x_s) = x_r$. Normal-form naturality then gives

$$\rho_{sr}(n_s(x_s)) = n_r(\rho_{sr}(x_s)) = n_r(x_r),$$

so the normal forms form a compatible inverse-limit family. Holonomy naturality gives the same calculation for $h_s(x_s)$. Finite-stage schedule independence comes from Theorem 4.3; visible separation upgrades agreement of all cofinal finite projections to uniqueness of the inverse-limit class. A nonzero inverse-limit holonomy must have a nonzero projection at some finite stage by visible separation. □

Proposition 4.10 (Coarse-graining / reconciliation compatibility)**.** *Let $r \preceq s$ be two refinement stages in the D1 consensus system, with coarse-graining maps*

$$\rho_{sr} : Q_s \to Q_r, \qquad \chi_{sr} : \mathcal{H}_s \to \mathcal{H}_r.$$

Equip Q_r and $\mathcal{H}_r$ with the pseudometrics used for macroscopic readout at stage r. If the chosen coarse-graining channel has normal-form and obstruction defects ε^n_{sr} and ε^h_{sr}, meaning

$$d^Q_r(\rho_{sr} n_s(x), n_r \rho_{sr}(x)) \le \varepsilon^n_{sr}, \qquad d^{\mathcal{H}}_r(\chi_{sr} h_s(x), h_r \rho_{sr}(x)) \le \varepsilon^h_{sr}$$

for every $x \in Q_s$, then reconciling at stage s and coarse-graining to r gives the same macroscopic normal-form and obstruction readout as coarse-graining first and reconciling at r, up to

$$\max\{\varepsilon^n_{sr}, \varepsilon^h_{sr}\}.$$

In the exact separated cofinal system of Theorem 4.9, these defects are zero. If the defects vanish on cofinal tails, the two procedures define the same macroscopic inverse-limit consensus class.

Proof. The displayed inequalities are exactly the two components of the claimed product-readout bound. Exact naturality in Theorem 4.9 is the zero-defect case. Cofinal vanishing gives convergence of every fixed coarse cylinder value in the inverse-limit topology. □

This removes the extra gauge-covariance axiom and keeps repair on its actual quotient-local carrier: recovery dynamics on fixed-cutoff collars. On the declared fixed-cutoff branch, the theorem-local burden is repair completeness, together with the Petz support/CPTP clause where that branch is used. The touched-overlap local-fit contract supplies Lyapunov Φ-descent on accepted moves, and the support-local commutation and union-collar compatibility package belongs to the declared repair law itself. Stability of that package under refinement or branch change is handled by the compatibility clauses of Theorem 4.9 when a separated cofinal refinement system is supplied. Proposition 4.10 is the corresponding RG-facing statement: it does not say that arbitrary coarse-graining maps commute with repair, only that the selected coarse-graining branch commutes with reconciliation to the extent that the displayed defects are controlled.

Definition 4.11 (Ancilla stabilization)**.** *Fix a finite-cutoff OPH realization*

$$\mathfrak{U} = (\{\mathcal{H}_P, \mathcal{A}(P), \omega_P\}_{P\in\mathcal{P}}, \Gamma, T^\lambda).$$

Choose finite-dimensional ancillary factors K_P with product state $\eta = \bigotimes_{P\in\mathcal{P}} \eta_P$. The associated ancilla stabilization is the realization

$$\mathfrak{U}^\eta = (\{\mathcal{H}_P^\eta, \mathcal{A}^\eta(P), \omega_P^\eta\}_{P\in\mathcal{P}}, \Gamma, (T^\lambda)^\eta),$$

with

$$\mathcal{H}_P^\eta := \mathcal{H}_P \otimes K_P, \qquad \mathcal{A}^\eta(P) := \mathcal{A}(P) \otimes \mathbf{1}_{K_P}, \qquad \omega_P^\eta := \omega_P \otimes \eta_P,$$

and lifted repair dynamics acting trivially on the ancillas.

Proposition 4.12 (Ancilla-stable UV underdetermination)**.** *Let $\mathfrak{U}^\eta$ be an ancilla stabilization of a finite-cutoff OPH realization $\mathfrak{U}$. Then:*

1. *observable expectations on the physical subalgebras are unchanged:*

$$\omega_P^\eta(a \otimes \mathbf{1}_{K_P}) = \omega_P(a) \qquad \forall\, a \in \mathcal{A}(P);$$

2. *the interacting local MaxEnt branch on the physical subalgebra is unchanged, since*

$$\omega_{\ell_{\mathrm{UV}}}^\eta(\lambda) = \omega_{\ell_{\mathrm{UV}}}(\lambda) \otimes \eta;$$

3. *for every collar split $A : B : D$,*

$$I(A : D \mid B)_{\rho\otimes\eta} = I(A : D \mid B)_\rho,$$

   so the Fawzi–Renner remainder $r_{\mathrm{FR}}(\varepsilon)$ and the collar Markov modulus $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon)$ are unchanged;

4. *if the ancillas are inert under repair, then*

$$\mathrm{nf}_\lambda^\eta(s \otimes k) = \mathrm{nf}_\lambda(s) \otimes k,$$

   so the quotient normal form on physical observables is identical;

5. *if the ancillas carry only trivial neutral sectors, the MAR-selected realized sector package is unchanged.*

Hence $\mathfrak{U}$ and $\mathfrak{U}^\eta$ are OPH-indistinguishable although they are different microscopic regulator realizations.

Proof. Item 1 is immediate from the definition of $\mathcal{A}^\eta(P)$ and ω_P^η. Item 2 is the product-state form of the same fixed-cutoff MaxEnt branch. Item 3 follows from additivity of entropy for product ancillas, which cancels in the conditional mutual-information combination and therefore leaves both r_{FR} and δ^{M} unchanged. Item 4 holds because the lifted repair maps do not act on the ancillary factors. Item 5 is immediate when the ancillas are neutral and carry no extra realized sector data. □

Definition 4.13 (OPH-stable UV equivalence)**.** *Two finite-cutoff realizations $\mathfrak{U}$ and $\mathfrak{U}'$ are* OPH-stably equivalent*, written*

$$\mathfrak{U} \sim_{\mathrm{OPH}} \mathfrak{U}',$$

if after finite ancilla stabilizations they are related by local gauge-covariant *-isomorphisms intertwining the observable patch net, overlap maps, local states, and repair dynamics on the physical subalgebras.

Corollary 4.14 (Unique physical UV branch only modulo OPH-stable equivalence)**.** *Under the OPH axiom language, the UV invariant determined by the theory is the class* $[\mathfrak{U}]_{\mathrm{OPH}}$*, not a unique microscopic regulator presentation. Literal microscopic UV uniqueness is therefore not an OPH invariant.*

Proof. Propositions 4.7 and 4.8 fix the schedule-independent physical branch on the quotient, while Proposition 4.12 shows that inert ancillary refinements leave every OPH observable invariant. So the physical branch is fixed only modulo $\sim_{\mathrm{OPH}}$, not at the level of one microscopic representative. □

These three results encode much of the eventual physical interpretation. Objectivity becomes confluence, gauge symmetry becomes quotient invariance induced by the physical overlap algebra, and stable defects are represented by nontrivial overlap holonomy classes. The physically relevant uniqueness statement is therefore quotient uniqueness together with ancilla-stable equivalence: OPH fixes a unique gauge-invariant physical branch modulo boundary redundancy, implementation hiding, and inert ancillary stabilization, not a unique microscopic regulator presentation.

5 Collars, Edge Centers, and Generalized Entropy

The fixed-point picture explains why overlap consistency produces gauge quotients: once repair is read on the physical overlap algebra, descent to the quotient is automatic. The gravity and consensus arguments later use a more specific collar fact: after edge-center completion, interior observables become insensitive to compatible exterior substitutions, and modular additivity becomes exact in the Markov normal form. The point of this section is therefore twofold. First, at fixed regulator scale, we derive the collar block decomposition from overlap consistency itself on the ordinary or central-defect branch and then state its genuinely noncentral higher-gauge replacement. Second, we state precisely when the approximate recoverability premise of Axiom 3.4 is allowed to converge to the exact HJPW normal form used by the later spatial and null collar theorems.

The sharp claim boundary is as follows. Small conditional mutual information always gives a constructive recovered comparison state with $O(\varepsilon^{1/2})$ observable error. It does *not* by itself give a universal one-shot trace-norm bound to an exact Markov state. The exact Markov normal form is recovered either when $I(A : D \mid B) = 0$ holds literally, or for a controlled family on one fixed finite-dimensional collar model, or after pullback to such a model, where $\varepsilon \to 0$. On that fixed collar one gets a collar-dependent modulus $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \to 0$ measuring distance to the exact Markov set, and this is the error that must be carried into later exact splice or modular-additivity identities.

Proposition 5.1 (Derived regulator gluing datum)**.** *Fix a regulator-scale finite patch cover of a cap neighborhood and let R be any finite union of regulator cells. Assume only the finite patch-net presentation implicit in Definition 4.1 together with overlap consistency on common interfaces. Then:*

1. *each regulator cell i with finite local state space S_i can be Hilbertized as $\tilde{\mathcal{H}}_i \cong \mathbb{C}^{|S_i|}$, so the extended algebra before overlap quotienting is the finite type-I algebra*

$$\tilde{\mathcal{A}}(R) = \mathcal{B}(\tilde{\mathcal{H}}_R), \qquad \tilde{\mathcal{H}}_R := \bigotimes_{i \subset R} \tilde{\mathcal{H}}_i;$$

2. *for each connected boundary component $\Sigma \subset \partial R$, after choosing a reference cut presentation $\tilde{\mathcal{H}}_\Sigma$, every overlap-consistent change of local chart along Σ is implemented by a unitary on*

$\tilde{\mathcal{H}}_\Sigma$, *and the compact closure of the subgroup generated by all such recharting unitaries is a compact boundary redundancy group*

$$K_\Sigma \subset U(\tilde{\mathcal{H}}_\Sigma);$$

before the choice of reference chart, the same data form a compact unitary groupoid of overlap-preserving transitions;

3. *if triple-overlap defects are central, the projective composition law of item 2 lifts to an direct action of a compact central extension* $\widehat{K}_\Sigma$*; a genuinely noncentral defect is the only obstruction to reducing the transition system to an ordinary compact group action;*

4. *in the invariant-state realization of the quotient, the chart-independent endomorphisms of the lifted presentation form the boundary-invariant algebra*

$$\mathcal{A}_{\mathrm{inv}}(R) = \tilde{\mathcal{A}}(R)^{\widehat{K}_{\partial R}}, \qquad \widehat{K}_{\partial R} := \prod_{\Sigma \subset \partial R} \widehat{K}_\Sigma,$$

with the convention $\widehat{K}_\Sigma = K_\Sigma$ *whenever no central extension is needed; the physical state space is the corresponding invariant subspace, and on collars the sector-preserving algebra induced on that subspace is the block-diagonal algebra used in Theorem 5.2.*

Items 1 and the lifted fixed-point presentation of item 4 are the fixed-cutoff realized data used by the compact-gauge branch; they are not extra axioms beyond the finite regulator presentation plus overlap redundancy.

Proof sketch. Item 1 is the Hilbertization of finite sets. For item 2, each overlap-consistent recharting acts on a finite-dimensional matrix algebra, and every *-automorphism of such an algebra is inner. After transporting all local overlap presentations to a chosen reference chart, each recharting is therefore implemented by a unitary on the cut Hilbert space. The subgroup generated by those unitaries has compact closure inside a finite-dimensional unitary group, which yields K_Σ; without fixing the reference chart one has the corresponding compact unitary groupoid. Item 3 is the usual projective-versus-central-extension lift for a central 2-cocycle. Item 4 records the lifted fixed-point presentation: chart-independent endomorphisms of the unreduced boundary data are exactly the fixed points of the derived boundary action, while the collar theorem uses the invariant-state realization and the sector-preserving algebra induced on the matched-cut subspace. If the defect is genuinely noncentral, the quotient is well defined, but the correct bookkeeping object is the crossed-module or higher-gauge data rather than an ordinary compact group. □

Theorem 5.2 (Derived edge-center collar decomposition and exact Markov normal form)**.** *Let A-B-D be a collar tripartition realized at fixed regulator scale on the ordinary or central-defect branch of Proposition 5.1. Write* $B = B_L \cup B_R$ *with common interface* $\Sigma = \partial C$*, and let* $\widehat{K}_\Sigma$ *be the derived compact boundary action attached to that cut, with* $\widehat{K}_\Sigma = K_\Sigma$ *on the ordinary branch. In the invariant-state realization,*

$$\mathcal{H}_B = (\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R})^{\widehat{K}_\Sigma}.$$

Decompose the left boundary data as

$$\tilde{\mathcal{H}}_{B_L} \cong \bigoplus_\alpha W_\alpha \otimes \mathcal{H}_{b_L^\alpha},$$

with W_α irreducible $\widehat{K}_\Sigma$-modules. Because the right half-collar carries the inverse overlap transport across the same cut, it decomposes contragrediently:

$$\tilde{\mathcal{H}}_{B_R} \cong \bigoplus_\beta W_\beta^* \otimes \mathcal{H}_{b_R^\beta}.$$

Then

$$\mathcal{H}_B \cong \bigoplus_\alpha \mathcal{H}_{b_L^\alpha} \otimes \mathcal{H}_{b_R^\alpha},$$

and the sector-preserving collar algebra

$$\mathcal{A}_{\mathrm{EC}}(B) := \bigoplus_\alpha \mathcal{B}(\mathcal{H}_{b_L^\alpha}) \otimes \mathcal{B}(\mathcal{H}_{b_R^\alpha})$$

has center

$$Z(\mathcal{A}_{\mathrm{EC}}(B)) = \bigoplus_\alpha \mathbb{C}\,\mathbf{1}_\alpha.$$

If, in addition, the reference state on $A \cup B \cup D$ is exact Markov,

$$I(A : D|B)_\rho = 0,$$

then the state decomposes as

$$\rho_{ABD} = \bigoplus_\alpha p_\alpha\, \rho^{(\alpha)}_{Ab_L^\alpha} \otimes \rho^{(\alpha)}_{b_R^\alpha D}.$$

The same formula holds in any limit state that is exact Markov on this fixed collar model. Thus the collar-center block decomposition is forced by overlap consistency plus the derived regulator package on the ordinary or central-defect branch; exact Markov factorization is an additional state condition, not a consequence of the block decomposition alone.

Proof. By Proposition 5.1, one may realize the physical collar states as the $\widehat{K}_\Sigma$-invariant subspace of $\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R}$. Complete reducibility of finite-dimensional unitary representations gives the displayed decomposition of $\tilde{\mathcal{H}}_{B_L}$, and the right half-collar carries the inverse transport law across the same cut, hence the contragredient decomposition of $\tilde{\mathcal{H}}_{B_R}$. Therefore

$$\tilde{\mathcal{H}}_{B_L} \otimes \tilde{\mathcal{H}}_{B_R} \cong \bigoplus_{\alpha,\beta} (W_\alpha \otimes W_\beta^*) \otimes (\mathcal{H}_{b_L^\alpha} \otimes \mathcal{H}_{b_R^\beta}).$$

Taking $\widehat{K}_\Sigma$-invariants gives

$$\mathcal{H}_B \cong \bigoplus_{\alpha,\beta} (W_\alpha \otimes W_\beta^*)^{\widehat{K}_\Sigma} \otimes (\mathcal{H}_{b_L^\alpha} \otimes \mathcal{H}_{b_R^\beta}).$$

By Schur's lemma,

$$(W_\alpha \otimes W_\beta^*)^{\widehat{K}_\Sigma} \cong \begin{cases} \mathbb{C}, & \alpha = \beta, \\ 0, & \alpha \neq \beta, \end{cases}$$

which yields the displayed block decomposition. The sector-preserving collar algebra therefore has the displayed block-diagonal form, so its center is generated by the block projectors. Within each block, observables from $A \cup B$ act only on the left factor and observables from $B \cup D$ act only on the right factor. If the state is exact Markov, the standard HJPW structure theorem gives the blockwise factorized normal form [19]. The same formula is used in the idealized recoverability limit when exact identities are taken literally. □

Theorem 5.3 (Derived higher-gauge cut datum)**.** *Fix a connected cut Σ and a finite regulator chart $\{P_i\}_{i\in I_\Sigma}$ meeting along Σ, with finite nerve N_Σ. On the genuinely noncentral branch, the weak gluing data are implemented by a compact crossed module*

$$\mathbb{K}_\Sigma = (H_\Sigma \xrightarrow{\partial_\Sigma} G_\Sigma, \triangleright)$$

together with maps

$$g_{ij} : P_{ij} \to G_\Sigma, \qquad h_{ijk} : P_{ijk} \to H_\Sigma,$$

obeying

$$g_{ij} g_{jk} = \partial_\Sigma(h_{ijk})\, g_{ik}, \qquad h_{jkl} h_{ijl} = (g_{ij} \triangleright h_{ikl})\, h_{ijk}.$$

The corresponding higher-gauge change system on the cut is the compact semidirect product

$$\mathcal{T}_\Sigma := C^1(N_\Sigma, H_\Sigma) \rtimes C^0(N_\Sigma, G_\Sigma),$$

with multiplication

$$(\eta, u) \cdot (\eta', u') = (\eta\,(u \triangleright \eta'),\, uu'),$$

and standard coboundary action

$$g_{ij} \mapsto u_i\, \partial_\Sigma(\eta_{ij})\, g_{ij}\, u_j^{-1},$$

$$h_{ijk} \mapsto (u_i \triangleright h_{ijk})\, \eta_{ij}\, (g_{ij} \triangleright \eta_{jk})\, \eta_{ik}^{-1}.$$

The fixed-cutoff physical sector on the genuinely noncentral branch is therefore the orbit class

$$q_\Sigma = [(g, h)] \in \check{H}^2(N_\Sigma, H_\Sigma \to G_\Sigma),$$

and no external continuous input P or N_{scr} enters this theorem package.

Proof sketch. On pair overlaps, every regulator-scale recharting acts on a finite-dimensional matrix algebra and is therefore inner, so it is implemented by a unitary on the cut Hilbert space. On triple overlaps, genuinely noncentral failure of strict composition is recorded by unitary associators rather than scalars. These 1- and 2-morphisms form a compact unitary 2-group of rechartings. Skeletal strictification of a compact 2-group is equivalent to crossed-module data, which yields the displayed compact $\mathbb{K}_\Sigma$. Because the nerve is finite, $C^1(N_\Sigma, H_\Sigma)$ and $C^0(N_\Sigma, G_\Sigma)$ are finite products of compact groups, hence $\mathcal{T}_\Sigma$ is compact. The displayed formulas are the standard crossed-module coboundary action, and the orbit class is exactly the nonabelian Čech 2-class of the defect data; see [31] for an explicit higher-lattice-gauge realization of the same crossed-module structures. □

Theorem 5.4 (Higher-gauge edge-center completion)**.** *Let $B = B_L \cup B_R$ be a fixed-cutoff collar around a connected cut Σ on the genuinely noncentral branch of Theorem 5.3. Then, as unitary $\mathcal{T}_\Sigma$-modules,*

$$\widetilde{\mathcal{H}}_{B_L} \cong \bigoplus_{\lambda\in\Lambda_\Sigma} W_\lambda \otimes \mathcal{H}_{b_L^\lambda}, \qquad \widetilde{\mathcal{H}}_{B_R} \cong \bigoplus_{\mu\in\Lambda_\Sigma} W_\mu^* \otimes \mathcal{H}_{b_R^\mu},$$

for a finite set Λ_Σ of irreducible unitary representations of $\mathcal{T}_\Sigma$ that actually occur. Hence the physical higher-gauge collar space is

$$\mathcal{H}_B^{2g} := (\widetilde{\mathcal{H}}_{B_L} \otimes \widetilde{\mathcal{H}}_{B_R})^{\mathcal{T}_\Sigma} \cong \bigoplus_{\lambda\in\Lambda_\Sigma} \mathcal{H}_{b_L^\lambda} \otimes \mathcal{H}_{b_R^\lambda},$$

and the collar algebra and its center are

$$\mathcal{A}_{2g}(B) \cong \bigoplus_{\lambda\in\Lambda_\Sigma} \mathcal{B}(\mathcal{H}_{b_L^\lambda}) \otimes \mathcal{B}(\mathcal{H}_{b_R^\lambda}), \qquad Z(\mathcal{A}_{2g}(B)) = \bigoplus_{\lambda\in\Lambda_\Sigma} \mathbb{C}\,\mathbf{1}_\lambda.$$

Proof sketch. Because $\mathcal{T}_\Sigma$ is compact and the half-collar spaces are finite-dimensional, both $\widetilde{\mathcal{H}}_{B_L}$ and $\widetilde{\mathcal{H}}_{B_R}$ split orthogonally into irreducible unitary $\mathcal{T}_\Sigma$-modules with finite multiplicity. The right side sees inverse transport across the same cut, hence the contragredient module. Therefore

$$\widetilde{\mathcal{H}}_{B_L} \otimes \widetilde{\mathcal{H}}_{B_R} \cong \bigoplus_{\lambda,\mu} (W_\lambda \otimes W_\mu^*) \otimes (\mathcal{H}_{b_L^\lambda} \otimes \mathcal{H}_{b_R^\mu}).$$

Taking $\mathcal{T}_\Sigma$-invariants leaves

$$(W_\lambda \otimes W_\mu^*)^{\mathcal{T}_\Sigma} \cong \mathrm{Hom}_{\mathcal{T}_\Sigma}(W_\mu, W_\lambda) \cong \begin{cases} \mathbb{C}, & \lambda = \mu, \\ 0, & \lambda \neq \mu, \end{cases}$$

by Schur's lemma. This yields the matched block decomposition and the center formula. □

Theorem 5.5 (Higher-gauge Markov collar and carried errors)**.** *Let ρ_{ABD} be a faithful state on a collar whose middle region B carries the higher-gauge block decomposition of Theorem 5.4. If*

$$I(A : D \mid B)_\rho = 0,$$

then

$$\rho_{ABD} = \bigoplus_{\lambda \in \Lambda_\Sigma} p_\lambda \, \rho_{Ab_L^\lambda} \otimes \rho_{b_R^\lambda D}.$$

If instead $I(A : D \mid B)_\rho \leq \varepsilon$ on one fixed faithful collar model, then the same carried Fawzi–Renner and fixed-collar Markov errors as in the ordinary branch continue to hold:

$$r_{\mathrm{FR}}(\varepsilon) = 2\sqrt{1 - e^{-\varepsilon}} \leq 2\sqrt{\varepsilon}, \qquad 4\lambda_*^{-1} \, \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

Proof sketch. Theorem 5.4 shows that the higher-gauge collar algebra is a finite direct sum of type-I tensor blocks. HJPW depends only on that finite-dimensional algebraic structure, not on whether the block label arose from an ordinary compact group, a central extension, or a crossed-module gauge system. Likewise, the Fawzi–Renner remainder and the fixed-collar Markov modulus are finite-dimensional entropy and recovery statements and are blind to the origin of the block label. □

Theorem 5.6 (Higher-gauge defect transportability)**.** *With the genuinely noncentral cut data of Theorem 5.3, define*

$$q_\Sigma := [(g, h)] \in \check{H}^2(N_\Sigma, H_\Sigma \to G_\Sigma).$$

Then:

1. *q_Σ is invariant under all local rechartings and higher-gauge changes $(\eta, u) \in \mathcal{T}_\Sigma$.*
2. *Two genuinely noncentral representatives describe the same fixed-cutoff physical cut sector if and only if they determine the same class q_Σ.*
3. *The defect is removable if and only if $q_\Sigma = 0$.*

Thus the genuinely noncentral branch carries the same transportability logic as the ordinary or central branch, but in crossed-module Čech degree 2 rather than ordinary group cohomology.

Proof sketch. Items 1–3 are the crossed-module version of ordinary Čech transportability: coboundaries change representatives while preserving classes, the trivial class strictifies the cocycle, and a nontrivial class cannot be gauged away. In any explicit crossed-module lattice realization, the same sectors are equivalently the connected components of the 2-flat configuration groupoid, or homotopy classes of maps into the classifying space $B\mathbb{K}_\Sigma$. □

Proposition 5.7 (Higher-gauge interacting compatibility)**.** *In a fixed-cutoff crossed-module lattice realization of Theorem 5.3, one may add local collar terms*

$$K^{2g}_{\text{collar}} = \sum_{v\in\Sigma_0} \beta_v(1-A_v) + \sum_{e\in\Sigma_1} \beta_e(1-B_e) + \sum_{c\in\mathcal{C}_\Sigma} \beta_c(1-C_c),$$

where A_v, B_e, and C_c are the local gauge and fake-flat projectors of the chosen higher-gauge realization. These terms are bounded-support and commute in the standard higher-gauge projector realization [31], so the full fixed-cutoff generator is quasi-local and inside the Axiom 3.3 class.

Proof sketch. The additional terms are local projector constraints supported on finitely many collar cells, so they preserve bounded support and quasi-locality. In the standard higher-gauge projector realization they commute, so the same finite-constraint MaxEnt/Lieb–Robinson bookkeeping used on the ordinary branch continues to apply. □

Remark 5.8 (Status of the regulator package and the precise Markov boundary)**.** *Proposition 5.1, Theorem 5.2, and Theorems 5.3–5.6 together with Proposition 5.7 show that edge-center completion is a theorem of overlap consistency plus the finite regulator presentation on the ordinary, central-defect, and genuinely noncentral higher-gauge branches. No preferred quantum-link realization is assumed on the ordinary or central branch, and the genuinely noncentral branch is handled by the compact crossed-module replacement rather than by one more ordinary-group lemma. The fixed-cutoff topological UV package is closed on all three branches. The continuum modular/geometric lift on the geometric subnet under **T2** and, whenever refinement-limit compact gauge reconstruction is invoked, the bosonic ordinary/central gauge premises of Assumption 3.7, form the boundary. The distinct state-control issue is that exact splice and modular-additivity identities require either literal exact Markovity or a controlled family that converges to the exact Markov set on one fixed finite-dimensional collar.*

Definition 5.9 (Exact Markov set and collar-distance modulus)**.** *Fix the finite-dimensional collar Hilbert space of Theorem 5.2. Let*

$$\mathfrak{M}_{A:B:D} := \{\sigma_{ABD} : \ I(A:D \mid B)_\sigma = 0\}$$

be the set of exact Markov states on that fixed collar. Equivalently, by HJPW and Theorem 5.2, $\sigma \in \mathfrak{M}_{A:B:D}$ iff

$$\sigma_{ABD} = \bigoplus_\alpha q_\alpha\, \sigma^{(\alpha)}_{Ab^\alpha_L} \otimes \sigma^{(\alpha)}_{b^\alpha_R D}$$

for the fixed edge-center decomposition of B.

For a general state ρ_{ABD}, define its distance to the exact Markov set by

$$d_{\mathrm{M}}(\rho) := \inf_{\sigma\in\mathfrak{M}_{A:B:D}} \|\rho-\sigma\|_1,$$

and the fixed-collar exact-Markov modulus by

$$\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) := \sup\left\{d_{\mathrm{M}}(\rho) : I(A:D \mid B)_\rho \le \varepsilon\right\}.$$

Proposition 5.10 (Controlled exact Markov-collar limit)**.** *On a fixed finite-dimensional collar, the exact Markov set $\mathfrak{M}_{A:B:D}$ is compact and*

$$\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \longrightarrow 0 \qquad \text{as } \varepsilon \downarrow 0.$$

Consequently, if $\rho^{(n)}_{ABD}$ is any sequence of states on that same collar with

$$I(A:D\mid B)_{\rho^{(n)}} \le \varepsilon_n, \qquad \varepsilon_n \to 0,$$

then there exist exact Markov states $\sigma^{(n)} \in \mathfrak{M}_{A:B:D}$ such that

$$\|\rho^{(n)} - \sigma^{(n)}\|_1 \le \delta^{\mathrm{M}}_{A:B:D}(\varepsilon_n) \longrightarrow 0.$$

Thus approximate recoverability converges to the exact HJPW normal form used by the compact-gauge branch.

Proof. The full state space on a fixed finite-dimensional Hilbert space is compact in trace norm. Conditional mutual information is continuous there because von Neumann entropy is continuous in finite dimension. Hence $\mathfrak{M}_{A:B:D}$ is closed, so it is compact as a closed subset of a compact space.

Suppose $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \not\to 0$. Then there exist $\eta > 0$, a sequence $\varepsilon_n \downarrow 0$, and states $\rho^{(n)}$ with $I(A:D\mid B)_{\rho^{(n)}} \le \varepsilon_n$ but $d_{\mathrm{M}}(\rho^{(n)}) \ge \eta$ for all n. By compactness, pass to a trace-norm convergent subsequence $\rho^{(n_k)} \to \rho^*$. Continuity of conditional mutual information gives

$$I(A:D\mid B)_{\rho^*} = \lim_{k\to\infty} I(A:D\mid B)_{\rho^{(n_k)}} = 0,$$

so $\rho^* \in \mathfrak{M}_{A:B:D}$. But then

$$d_{\mathrm{M}}(\rho^{(n_k)}) \le \|\rho^{(n_k)} - \rho^*\|_1 \longrightarrow 0,$$

contradicting $d_{\mathrm{M}}(\rho^{(n_k)}) \ge \eta$. Thus $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \to 0$. The final statement follows by choosing $\sigma^{(n)} \in \mathfrak{M}_{A:B:D}$ within $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon_n) + 1/n$ of $\rho^{(n)}$. □

Proposition 5.11 (Approximate collar recovery)**.** *Let A-B-D be a collar tripartition satisfying*

$$I(A:D|B)_\rho \le \varepsilon,$$

and let $\mathcal{R}_{B\to BD}$ be a recovery map such that

$$F(\rho_{ABD}, (\mathrm{id}_A \otimes \mathcal{R}_{B\to BD})(\rho_{AB})) \ge e^{-\varepsilon/2}.$$

Define the recovered comparison state

$$\rho^{\mathrm{rec}}_{ABD} := (\mathrm{id}_A \otimes \mathcal{R}_{B\to BD})(\rho_{AB}), \qquad r_{\mathrm{FR}}(\varepsilon) := 2\sqrt{1-e^{-\varepsilon}} \le 2\sqrt{\varepsilon}.$$

Then

$$\|\rho_{ABD} - \rho^{\mathrm{rec}}_{ABD}\|_1 \le r_{\mathrm{FR}}(\varepsilon).$$

Consequently, for every CPTP map $\Lambda_{BD\to B'D'}$ and every bounded observable X supported on $A\cup B'$,

$$|\mathrm{Tr}[X(\mathrm{id}_A \otimes \Lambda)(\rho_{ABD} - \rho^{\mathrm{rec}}_{ABD})]| \le \|X\|_\infty \, \|(\mathrm{id}_A \otimes \Lambda)(\rho_{ABD} - \rho^{\mathrm{rec}}_{ABD})\|_1 \le \|X\|_\infty \, r_{\mathrm{FR}}(\varepsilon).$$

Proof. The fidelity lower bound and the Fuchs–van de Graaf inequality give

$$\|\rho_{ABD} - \rho^{\mathrm{rec}}_{ABD}\|_1 \le 2\sqrt{1 - F(\rho_{ABD}, \rho^{\mathrm{rec}}_{ABD})^2} \le 2\sqrt{1-e^{-\varepsilon}} = r_{\mathrm{FR}}(\varepsilon).$$

The estimate $r_{\mathrm{FR}}(\varepsilon) \le 2\sqrt{\varepsilon}$ follows from $1 - e^{-x} \le x$. Contractivity of trace norm under CPTP maps gives the transported bound, and the observable estimate is the duality between trace norm and operator norm [18]. The bound is constructive and dimension-free. What it controls is closeness to a recovered comparison state; no claim is made that $\rho^{\mathrm{rec}}_{ABD}$ is itself exact Markov. □

Proposition 5.12 (Exact splice theorem and controlled collar approximation)**.** *Under Theorem 5.2, let $\sigma_{ABD} \in \mathfrak{M}_{A:B:D}$ have exact Markov form*

$$\sigma_{ABD} = \bigoplus_{\alpha} p_\alpha \, \sigma^{(\alpha)}_{Ab^\alpha_L} \otimes \sigma^{(\alpha)}_{b^\alpha_R D}.$$

Let $\tau^{(\alpha)}_{b^\alpha_R D'}$ be any normalized family of states compatible with the same right-boundary sectors, and define

$$\sigma'_{ABD'} = \bigoplus_{\alpha} p_\alpha \, \sigma^{(\alpha)}_{Ab^\alpha_L} \otimes \tau^{(\alpha)}_{b^\alpha_R D'}.$$

Define the common left algebra

$$\mathcal{A}_{Ab_L} := \bigoplus_{\alpha} \mathcal{B}(\mathcal{H}_{Ab^\alpha_L}) \otimes \mathbf{1}_{b^\alpha_R},$$

canonically represented on both direct-sum Hilbert spaces. Then:

1. *for every $X \in \mathcal{A}_{Ab_L}$,*
$$\mathrm{Tr}(X\sigma'_{ABD'}) = \mathrm{Tr}(X\sigma_{ABD});$$

2. *if ρ_{ABD} is any state on the same fixed collar with $I(A:D \mid B)_\rho \le \varepsilon$, and if $\sigma_\varepsilon \in \mathfrak{M}_{A:B:D}$ is chosen so that*
$$\|\rho_{ABD} - \sigma_\varepsilon\|_1 \le \delta^{\mathrm{M}}_{A:B:D}(\varepsilon),$$
then the corresponding exact splice σ'_ε satisfies
$$\left|\mathrm{Tr}(X\rho_{ABD}) - \mathrm{Tr}(X\sigma'_\varepsilon)\right| \le \|X\|_\infty \, \delta^{\mathrm{M}}_{A:B:D}(\varepsilon)$$
for all $X \in \mathcal{A}_{Ab_L}$.

Hence the exact splice identity used by the compact-gauge branch is justified either at exact Markovity or along any controlled collar family for which $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon_\delta) \to 0$.

Proof. For item 1, blockwise factorization gives

$$\mathrm{Tr}(X\sigma'_{ABD'}) = \sum_{\alpha} p_\alpha \mathrm{Tr}\Big(X_\alpha \sigma^{(\alpha)}_{Ab^\alpha_L}\Big) \, \mathrm{Tr}\Big(\tau^{(\alpha)}_{b^\alpha_R D'}\Big).$$

Here X_α denotes the α-block of X in $\mathcal{A}_{Ab_L}$. Each $\tau^{(\alpha)}$ is normalized, so the second factor is 1. The same computation for σ_{ABD} gives the same value because the original right factor is likewise normalized.

For item 2, apply item 1 to σ_ε:

$$\mathrm{Tr}(X\sigma'_\varepsilon) = \mathrm{Tr}(X\sigma_\varepsilon).$$

Therefore

$$\left|\mathrm{Tr}(X\rho_{ABD}) - \mathrm{Tr}(X\sigma'_\varepsilon)\right| = \left|\mathrm{Tr}[X(\rho_{ABD} - \sigma_\varepsilon)]\right| \le \|X\|_\infty \, \|\rho_{ABD} - \sigma_\varepsilon\|_1 \le \|X\|_\infty \, \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

□

Proposition 5.13 (Transport of modular-additivity errors from the exact Markov reference)**.** *Let $K_X(\omega) := -\log \omega_X$ for a faithful reduced state on region X, and define the modular defect*

$$\Delta K(\omega) := K_{ABD}(\omega) - K_{AB}(\omega) - K_{BD}(\omega) + K_B(\omega).$$

Fix a collar and let ρ be a faithful state with $I(A : D \mid B)_\rho \le \varepsilon$. Choose $\sigma_\varepsilon \in \mathfrak{M}_{A:B:D}$ with

$$\|\rho - \sigma_\varepsilon\|_1 \le \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

Assume that the four marginals ρ_X and $(\sigma_\varepsilon)_X$ for $X \in \{ABD, AB, BD, B\}$ all satisfy

$$\rho_X \ge \lambda_* \mathbf{1}, \qquad (\sigma_\varepsilon)_X \ge \lambda_* \mathbf{1}$$

on their supports for some $\lambda_* > 0$. Then

$$\|\Delta K(\rho) - \Delta K(\sigma_\varepsilon)\|_\infty \le 4\lambda_*^{-1}\, \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

Since $\Delta K(\sigma_\varepsilon)$ is central or blockwise constant, every matrix element of the modular defect of ρ is within $4\lambda_*^{-1}\delta^{\mathrm{M}}_{A:B:D}(\varepsilon)$ of the exact Markov value.

Proof. By monotonicity of trace distance under partial trace,

$$\|\rho_X - (\sigma_\varepsilon)_X\|_1 \le \|\rho - \sigma_\varepsilon\|_1 \le \delta^{\mathrm{M}}_{A:B:D}(\varepsilon)$$

for each $X \in \{ABD, AB, BD, B\}$. On the interval $[\lambda_*, 1]$, the function $\log x$ has derivative bounded by λ_*^{-1}. By the standard integral representation of the operator logarithm, this implies the operator-norm Lipschitz bound

$$\|K_X(\rho) - K_X(\sigma_\varepsilon)\|_\infty = \|\log(\sigma_\varepsilon)_X - \log \rho_X\|_\infty \le \lambda_*^{-1}\, \|\rho_X - (\sigma_\varepsilon)_X\|_\infty \le \lambda_*^{-1}\, \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

Summing the four region contributions gives

$$\|\Delta K(\rho) - \Delta K(\sigma_\varepsilon)\|_\infty \le 4\lambda_*^{-1}\, \delta^{\mathrm{M}}_{A:B:D}(\varepsilon).$$

For $\sigma_\varepsilon \in \mathfrak{M}_{A:B:D}$, the exact HJPW block factorization implies that $\Delta K(\sigma_\varepsilon)$ is central or blockwise constant, so the same bound controls all matrix elements relative to the exact Markov modular-additivity value. □

Remark 5.14 (What is and is not carried forward)**.** *There are therefore two distinct quantitative controls, and this paper keeps them separate:*

1. *the constructive one-shot recoverability bound $r_{\mathrm{FR}}(\varepsilon) = O(\varepsilon^{1/2})$, which compares the physical state to a recovered comparison state and is enough for bounded-observable statements at one regulator stage;*

2. *the fixed-collar exact-Markov modulus $\delta^{\mathrm{M}}_{A:B:D}(\varepsilon) \to 0$, which compares the physical state to the exact Markov set and is the correct quantity for justifying exact splice or modular-additivity identities in a controlled limit, with an additional factor $4\lambda_*^{-1}$ when logarithms are involved.*

Later spatial and null collar theorems therefore use exact identities only in two regimes: literal exact Markovity, or a controlled refinement or scaling family in which the relevant $\delta^{\mathrm{M}}(\varepsilon_\delta)$ tends to zero after transport to one fixed finite-dimensional collar model. The manuscript does not *claim a universal dimension-free one-shot trace-norm estimate from small conditional mutual information directly to an exact Markov state.*

Remark 5.15 (Floor transfer to the exact-Markov reference)**.** *On one fixed finite-dimensional collar model, Proposition 5.13 does not require a second independent faithfulness input on the exact-Markov comparison family. Suppose the transported physical marginals satisfy*

$$\rho_X \geq \bar{\lambda}_{m,\delta}\mathbf{1}$$

eventually for each $X \in \{ABD, AB, BD, B\}$*, and choose exact-Markov replacements* $\sigma_{n,m,\delta}$ *with*

$$\|\rho_X - (\sigma_{n,m,\delta})_X\|_1 \leq \delta^{\mathrm{M}}_{m,\delta}(\varepsilon_{n,m,\delta}) \longrightarrow 0.$$

Then $\|\rho_X - (\sigma_{n,m,\delta})_X\|_\infty \leq \|\rho_X - (\sigma_{n,m,\delta})_X\|_1$*, so for all sufficiently large* n*,*

$$(\sigma_{n,m,\delta})_X \geq \tfrac{1}{2}\bar{\lambda}_{m,\delta}\mathbf{1}.$$

Thus the only nonlatent faithfulness-side input below the modular-defect term is the eventual lower spectral bound on the transported marginals themselves; the exact-Markov reference inherits a common floor once the exact-Markov modulus tends to zero.

Proposition 5.16 (Generalized-entropy split)**.** *If the reduced cap state takes the block form*

$$\rho_C = \bigoplus_\alpha p_\alpha \left(\rho^{(\alpha)}_{\mathrm{bulk},C} \otimes \frac{\mathbf{1}^{(\alpha)}_{\mathrm{edge}}}{d_\alpha} \right),$$

then

$$S(\rho_C) = S_{\mathrm{bulk}}(C) + \mathrm{Tr}(\rho_C L_C),$$

where

$$S_{\mathrm{bulk}}(C) = H(p_\alpha) + \sum_\alpha p_\alpha S(\rho^{(\alpha)}_{\mathrm{bulk},C}), \qquad L_C = \sum_\alpha (\log d_\alpha) P_\alpha.$$

Proof. The entropy of a direct sum is

$$S\left(\bigoplus_\alpha p_\alpha \sigma_\alpha\right) = H(p_\alpha) + \sum_\alpha p_\alpha S(\sigma_\alpha).$$

Apply this identity with

$$\sigma_\alpha = \rho^{(\alpha)}_{\mathrm{bulk},C} \otimes \frac{\mathbf{1}^{(\alpha)}_{\mathrm{edge}}}{d_\alpha}.$$

Since

$$S(\sigma_\alpha) = S(\rho^{(\alpha)}_{\mathrm{bulk},C}) + \log d_\alpha,$$

the first statement follows. □

Definition 5.17 (Effective Newton constant in the refinement dictionary)**.** *In the refinement-scaling regime, if*

$$\mathrm{Tr}(\rho_C L_C) \approx N_\Sigma \bar{\ell}(t), \qquad A(\partial C) \approx N_\Sigma a_{\mathrm{cell}},$$

then matching $\mathrm{Tr}(\rho_C L_C)$ *to the area term* $A(\partial C)/(4G)$ *defines the effective Newton constant*

$$G := \frac{a_{\mathrm{cell}}}{4\bar{\ell}(t)}.$$

This is a continuum dictionary identification, not part of the exact algebraic proposition.

Proposition 5.18 (Shared edge-entropy identity on the realized product-group branch)**.** *On the realized product-group branch*

$$G_{\text{phys}} = \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6},$$

let

$$R = R_3 \boxtimes R_2 \boxtimes q$$

be a lifted product presentation of one cut sector on the quotient branch, and let the R-sector contribution of the collar edge-center entropy operator be

$$(L_C)\,|_R = \log d_R.$$

Then

$$(L_C)\,|_R = L_C^{(3)} + L_C^{(2)}, \qquad L_C^{(3)} := \log d_{R_3}, \qquad L_C^{(2)} := \log d_{R_2},$$

because every irreducible $\mathrm{U}(1)$ *representation is one-dimensional and contributes* $\log 1 = 0$*. Consequently, on the product heat-kernel branch*

$$\bar{\ell}_{\text{shared}} = \langle L_C \rangle = \bar{\ell}_{\mathrm{SU}(3)}(t_{3,\text{run}}) + \bar{\ell}_{\mathrm{SU}(2)}(t_{2,\text{run}}).$$

If the same branch satisfies the D10 pixel law

$$\bar{\ell}_{\mathrm{SU}(2)}(t_{2,\text{run}}) \;+\; \bar{\ell}_{\mathrm{SU}(3)}(t_{3,\text{run}}) = P/4,$$

then

$$\bar{\ell}_{\text{shared}} = P/4.$$

Proof. For the lifted product presentation $R = R_3 \boxtimes R_2 \boxtimes q$ of a sector on the quotient branch,

$$d_R = d_{R_3} d_{R_2} d_q.$$

Every irreducible $\mathrm{U}(1)$ representation is one-dimensional, so $d_q = 1$. Therefore

$$\log d_R = \log d_{R_3} + \log d_{R_2} + \log d_q = \log d_{R_3} + \log d_{R_2}.$$

This proves the sector identity

$$(L_C)\,|_R = L_C^{(3)} + L_C^{(2)}.$$

On the product heat-kernel branch, expectation values add, so

$$\bar{\ell}_{\text{shared}} = \langle L_C \rangle = \bar{\ell}_{\mathrm{SU}(3)}(t_{3,\text{run}}) + \bar{\ell}_{\mathrm{SU}(2)}(t_{2,\text{run}}).$$

If the same branch satisfies the D10 pixel law, the right-hand side equals $P/4$. □

Proposition 5.19 (Local familiar-unit readout package on the gravity row)**.** *On the declared local extension surface, let a_{cell} be the microscopic cell-area datum and define*

$$G_{\text{nat}} := \frac{a_{\text{cell}}}{4\bar{\ell}_{\text{shared}}}.$$

If Proposition 5.18 gives $\bar{\ell}_{\text{shared}} = P/4$, then

$$G_{\text{nat}} = \frac{a_{\text{cell}}}{P} = \ell_P^2.$$

In SI units $\ell_P^2 = \hbar G_{\mathrm{SI}}/c^3$. Therefore

$$G_{\mathrm{SI}} = \frac{c^3 a_{\mathrm{cell}}}{\hbar P}.$$

More generally, if $\widehat{L}(P)$, $\widehat{T}(P)$, $\widehat{E}(P)$, and $\widehat{\Theta}(P)$ denote dimensionless branch outputs on that same local branch, then

$$L_{\mathrm{loc}} = \sqrt{a_{\mathrm{cell}}}\,\widehat{L}(P), \qquad t_{\mathrm{loc}} = \frac{\sqrt{a_{\mathrm{cell}}}}{c}\,\widehat{T}(P),$$

$$E_{\mathrm{loc}} = \frac{\hbar c}{\sqrt{a_{\mathrm{cell}}}}\,\widehat{E}(P), \qquad \Theta_{\mathrm{loc}} = \frac{\hbar c}{k_B\sqrt{a_{\mathrm{cell}}}}\,\widehat{\Theta}(P).$$

Thus, at fixed P, local lengths scale with $a_{\mathrm{cell}}^{1/2}$ alone, local times use that same $a_{\mathrm{cell}}^{1/2}$ scale divided by the structural Lorentz output c, and local mass/energy and temperature rows use the inverse $a_{\mathrm{cell}}^{1/2}$ scale dressed only by the familiar-unit display conventions $\hbar$ and k_B.

Proof. By Proposition 5.18,

$$G_{\mathrm{nat}} = \frac{a_{\mathrm{cell}}}{4(P/4)} = \frac{a_{\mathrm{cell}}}{P}.$$

Since $P = a_{\mathrm{cell}}/\ell_P^2$, this equals ℓ_P^2. The SI relation

$$\ell_P^2 = \frac{\hbar G_{\mathrm{SI}}}{c^3}$$

then gives

$$G_{\mathrm{SI}} = \frac{c^3 a_{\mathrm{cell}}}{\hbar P}.$$

The following formulas are dimensional bookkeeping on the same declared local branch. Because P is dimensionless and a_{cell} is the only local microscopic area datum, every local length readout is $\sqrt{a_{\mathrm{cell}}}$ times a dimensionless branch quantity. The structural Lorentz output c converts that same local ruler to seconds, while the familiar-unit display constants $\hbar$ and k_B convert the inverse local ruler to energy/mass rows and to Kelvin rows. □

Proposition 5.12 is the algebraic version of interior invariance under compatible exterior substitutions. Proposition 5.13 is the precise bridge from small collar conditional mutual information to the exact modular identities later used by the spatial and null branches: at finite stage one carries a remainder, and exact identities appear only when that remainder is zero or tends to zero in the controlled collar limit. Proposition 5.16 isolates the exact entropy split, while Definition 5.17 supplies the continuum identification that turns the edge center into a gravitational coupling. Proposition 5.18 identifies the shared edge entropy on the realized product-group branch with the same $\mathrm{SU}(2) + \mathrm{SU}(3)$ quantity used by the D10 pixel law, and Proposition 5.19 makes the familiar-unit readout explicit relative to the declared microscopic datum a_{cell}.

6 Modular Geometry, Relativity, and Einstein Dynamics

6.1 Geometric modular flow

Before stating the continuum modular theorem, fix the algebraic target. At fixed cutoff, the full operational patch algebra contains geometric overlap data, declared record summaries, and compare/write/verify pointer layers. The BW question should be asked only on the overlap-generated geometric subnet, not on the entire operational algebra.

Definition 6.1 (Operational total algebra, geometric subnet, and auxiliary/record sector)**.** *At fixed cutoff n and patch P, let $\mathcal{A}_n^{\mathrm{tot}}(P)$ denote the operational patch algebra consisting of the physical patch algebra together with the declared overlap-readable summaries used by the compare/write/verify surface. Let $B_{IJ} \subset P$ range over overlap collars incident on P, let $\Pi_\alpha^{(IJ)}$ denote the overlap sector projectors, and let $Q_a^{(IJ)}$ denote the boundary observables carrying the geometric cut data. Define the geometric subnet by*

$$\mathcal{A}_n^{\mathrm{geo}}(P) := W^*\Big\langle \bigcup_{B_{IJ}\subset P} \iota_{IJ\to P}\big(\{\Pi_\alpha^{(IJ)}\}_\alpha \cup \{Q_a^{(IJ)}\}_a\big)\Big\rangle_{\mathrm{repair,isotony}} \Big/ \mathcal{N}_n(P),$$

where $\mathcal{N}_n(P)$ is the maximal repair-invariant overlap-trivial kernel, namely the maximal part whose restriction to every overlap algebra inside P is scalar. The complementary operational sector generated by record summaries, pointer registers, and other interface-inert observables is denoted $\mathcal{A}_n^{\mathrm{aux/rec}}(P)$.

Theorem 6.2 (Geometric subnet and operational primeness)**.** *For each fixed-cutoff patch P, the operational algebra admits a canonical quotient*

$$\pi_{n,P}^{\mathrm{geo}} : \mathcal{A}_n^{\mathrm{tot}}(P) \twoheadrightarrow \mathcal{A}_n^{\mathrm{geo}}(P)$$

with the following properties:

(i) *$\mathcal{A}_n^{\mathrm{geo}}(P)$ is generated by the overlap sector projectors and geometric boundary observables, and is closed under repair and isotony;*

(ii) *the record-summary block, pointer algebra, and interface-inert auxiliary observables lie in the complementary operational sector $\mathcal{A}_n^{\mathrm{aux/rec}}(P)$ and do not enlarge $\mathcal{A}_n^{\mathrm{geo}}(P)$;*

(iii) *if $N \subset \mathcal{A}_n^{\mathrm{geo}}(P)$ is a repair-invariant subalgebra whose restriction to every overlap algebra inside P is scalar, then $N = \mathbb{C}\,\mathbf{1}$.*

Proof. Items (i) and (ii) are the content of Definition 6.1. For item (iii), if such an N were nontrivial, then its pullback under $\pi_{n,P}^{\mathrm{geo}}$ would define a nontrivial repair-invariant overlap-trivial component of $\mathcal{A}_n^{\mathrm{tot}}(P)$, contradicting the maximality of $\mathcal{N}_n(P)$. Thus no nontrivial overlap-trivial factor survives inside $\mathcal{A}_n^{\mathrm{geo}}(P)$. □

The collar analysis proves a fixed-cutoff statement on the finite type-I regulator net: the reduced cap state has a literal density matrix, its modular Hamiltonian exists, and its nonadditive part is confined to a shrinking collar up to carried errors. The Lorentz claim is therefore not a literal statement about the finite regulator matrices. Under the controlled scaling clause **T2**, the target object is a realized scaling-limit geometric cap pair

$$(\mathcal{A}_\infty^{\mathrm{geo}}(C), \omega_\infty^{\mathrm{geo},C})$$

for each round cap $C \subset S^2$. Axiom 3.3 controls the state-side realized branch across refinement through one common finite-dimensional MaxEnt family, and Theorem 6.2 removes the overlap-trivial auxiliary/record layers from the BW target. The theorem-local UV surface requires two explicit objects on that geometric subnet: first a realized transported geometric cap-local system whose carried-collar schedule is derived from the transported fixed-local-collar Markov/faithfulness datum and emits the scaling-limit geometric cap pair from transported marginals, and then ordered cut-pair rigidity on the extracted prime cap pair. The theorem below makes that claim boundary

explicit, fixes the 2π normalization once those two objects are supplied, and leaves the UV burden exposed rather than hidden.

Fix a cap $C \subset S^2$ and a shrinking collar family $(A_\delta, B_\delta, D_\delta)$ around ∂C, with $\delta \downarrow 0$ and $A_\delta \cup B_\delta \cup D_\delta$ covering a neighborhood of the cut. Write

$$\varepsilon_\delta := I(A_\delta : D_\delta \mid B_\delta)_\omega, \qquad r_{\mathrm{FR}}(\varepsilon_\delta) := 2\sqrt{1 - e^{-\varepsilon_\delta}} \le 2\sqrt{\varepsilon_\delta},$$

and, on each fixed faithful collar model,

$$\eta_\delta^{\mathrm{M}} := 4\lambda_*^{-1}\, \delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta),$$

where $\lambda_* > 0$ is the lower spectral bound used to compare modular Hamiltonians. Every exact collar identity below is therefore to be read in one of two ways:

1. literal exactness when the reference collar state is exact Markov; or
2. a controlled collar family for which

$$\delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta) \to 0 \qquad (\delta \downarrow 0),$$

with the finite-stage errors $r_{\mathrm{FR}}(\varepsilon_\delta)$ and η_δ^{M} carried explicitly.

For each cap C, let $\lambda_C(s) \subset \mathrm{Conf}^+(S^2)$ denote the standard cap-preserving conformal one-parameter subgroup, normalized so that the null blow-up near a smooth cut acts by $v \mapsto e^{-s}v$, and let $\alpha_{\lambda_C(s)}$ denote the induced automorphism of the scaling-limit cap net.

Theorem 6.3 (Geometric modular flow on caps on the extracted prime geometric subnet)**.** *Assume Axioms 3.1–3.4, Theorem 6.2, the derived fixed-cutoff regulator/collar package established above, and the controlled scaling-limit scope clause **T2** of Assumption 3.6. Assume in addition that for each round cap $C \subset S^2$:*

(a) *the transported cap marginals on $\mathcal{A}_n^{\mathrm{geo}}(C_n)$ admit a realized scaling-limit geometric cap pair*

$$(\mathcal{A}_\infty^{\mathrm{geo}}(C), \omega_\infty^{\mathrm{geo},C})$$

through the carried-collar local weak-$*$ / GNS extraction package; and

(b) *ordered cut-pair rigidity holds on that extracted prime geometric pair, so the cap-preserving conformal subgroup singled out by the cut data is unique up to normalization.*

Then:

(i) *At each finite regulator stage, the cap algebra is type I, the reduced cap state has a density matrix $\rho_C^{(\delta)}$, and its modular Hamiltonian*

$$K_C^{(\delta)} := -\log \rho_C^{(\delta)}$$

has nonadditive part confined to the shrinking collar up to the carried errors measured by $r_{\mathrm{FR}}(\varepsilon_\delta)$ and η_δ^{M}. In particular, on each fixed collar model the physical collar state differs from its constructive recovered comparison state by $r_{\mathrm{FR}}(\varepsilon_\delta)$ in trace norm, and its modular defect differs from the exact-Markov reference defect by at most η_δ^{M}.

(ii) *For every realized scaling-limit geometric cap pair* $(\mathcal{A}_\infty^{\mathrm{geo}}(C), \omega_\infty^{\mathrm{geo},C})$ *furnished by hypothesis* (a)*, the modular automorphism group is geometric:*

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_C(2\pi t)}.$$

Equivalently, the modular parameter t *and the geometric cap-dilation parameter* s *are related by*

$$s = 2\pi t.$$

(iii) *No separate cap-isotropy/*SO(2)*-equivariance selector or Euclidean-regularity premise is used in this theorem. Once ordered cut-pair rigidity has fixed the geometric cap subgroup on the extracted prime geometric pair and* $\lambda_C(s)$ *is normalized by the null blow-up* $v \mapsto e^{-s}v$*, the modular KMS normalization fixes the coefficient* 2π*.*

(iv) *If the scaling-limit cap algebra happens to be type I, item* (ii) *may be written as the operator identity*

$$K_C = 2\pi B_C.$$

In the generic continuum case, where the scaling-limit cap algebra has left the regulator class and is expected in QFT examples to be non-type-I, the correct theorem statement is item (ii) *itself, and the geometric modular action is generally outer rather than inner.*

Consequently, for every controlled collar family with

$$\delta^{\mathrm{M}}_{A_\delta:B_\delta:D_\delta}(\varepsilon_\delta) \to 0$$

and surviving faithfulness bound, replacing the finite-stage modular action by the geometric cap-dilation action incurs only the carried errors $r_{\mathrm{FR}}(\varepsilon_\delta)$ *and* η_δ^{M}*, and these vanish in the refinement limit.*

Proof. Step 1: fixed-cutoff collar control. On the finite type-I regulator net, Propositions 5.12, 5.13, and 5.16 localize the modular defect to the shrinking collar and quantify the discrepancy between the physical collar state, its constructive recovered comparison state, and its exact-Markov reference by the carried errors $r_{\mathrm{FR}}(\varepsilon_\delta)$ and η_δ^{M}.

Step 2: scaling-limit claim tier. Theorem 6.3 is a scaling-limit statement by **T2**, so the theorem concerns only scaling-limit cap pairs furnished by the controlled refinement/scaling package, not the finite regulator matrices themselves. Nothing in this step assumes that type-I structure survives the limit.

Step 3: geometric identification on the prime subnet. By hypothesis (a) and Theorem 6.2, the realized limit pair is extracted on the overlap-generated prime geometric subnet, with overlap-trivial auxiliary/record factors removed. By hypothesis (b), the cap-preserving conformal action singled out by the ordered cut data is unique up to normalization. Therefore there is a constant $\kappa_C > 0$ such that

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_C(\kappa_C t)}.$$

Step 4: normalization. Because ordered cut-pair rigidity has identified the standard cap-preserving conformal subgroup on the extracted geometric pair, no separate cap-isotropy or Euclidean-regularity selector enters. The modular group is KMS at inverse temperature 1 by definition. Since $\lambda_C(s)$ is normalized by unit null dilation under the blow-up $v \mapsto e^{-s}v$, the Bisognano–Wichmann normalization fixes

$$\kappa_C = 2\pi,$$

hence

$$\sigma_t^{\omega_\infty^{\mathrm{geo},C}} = \alpha_{\lambda_C(2\pi t)}.$$

Step 5: operator versus automorphism form. If the limit cap algebra is type I, one can represent the modular automorphism group by a modular Hamiltonian $K_C = -\log \rho_C$ and recover $K_C = 2\pi B_C$. If the limit algebra leaves the regulator class, the modular automorphism group is well defined while the inner operator representative need not exist inside $\mathcal{A}_\infty^{\mathrm{geo}}(C)$. In that case the automorphism identity is the full theorem statement. The final sentence is exactly the carried-error conclusion from Step 1 transported along the controlled scaling family of **T2**. □

Definition 6.4 (Conditional BW-branch observer-relative time reading)**.** *On the branch satisfying the hypotheses of Theorem 6.3, OPH uses the modular automorphism parameter t of the extracted cap pair $(\mathcal{A}_\infty^{\mathrm{geo}}(C), \omega_\infty^{\mathrm{geo},C})$ as that cap observer's relative time parameter. This is a declared BW-branch reading of the geometric modular flow derived in Theorem 6.3; it is not an additional proof that arbitrary operational clocks, global time, or the full problem of time have been derived from the axioms.*

Remark 6.5 (BW-side UV scaffold after the geometric-subnet split)**.** *Theorem 6.2 removes the old ambiguity about the target algebra: the continuum Lorentz claim is asked only on the extracted prime geometric subnet, not on the full operational algebra. The UV-side boundary is therefore a two-step chain on $\mathcal{A}_n^{\mathrm{geo}}$, and neither step is an independent cap-isotropy selector or a Euclidean-regularity premise.*

First, one needs a realized transported geometric cap-local system *for caps and their conformal images. The relevant package is the cap-local test family on $\mathcal{A}_n^{\mathrm{geo}}(C_n)$, the projectively compatible transported marginal family, and the asymptotic transport-equivalence certificate. On each fixed local collar model (m, δ), let $\varepsilon_{n,m,\delta}$ be the transported collar conditional mutual information. If the transported physical geometric marginals satisfy an eventual common floor*

$$\rho_X^{(n,m,\delta)} \geq \bar{\lambda}_{m,\delta}\mathbf{1} \qquad (X \in \{ABD, AB, BD, B\})$$

and

$$\delta^{\mathrm{M}}_{A_{n,m,\delta}:B_{n,m,\delta}:D_{n,m,\delta}}(\varepsilon_{n,m,\delta}) \to 0,$$

then the exact-Markov comparison marginals inherit the same floor for all sufficiently large n. The carried-collar schedule

$$\eta^{\mathrm{cap}}_{n,m,\delta} := r_{\mathrm{FR}}(\varepsilon_{n,m,\delta}) + 4\bar{\lambda}^{-1}_{m,\delta}\delta^{\mathrm{M}}_{A_{n,m,\delta}:B_{n,m,\delta}:D_{n,m,\delta}}(\varepsilon_{n,m,\delta})$$

is therefore derived rather than primitive*. The only nonlatent lower input on the faithfulness side is the eventual fixed-local-collar modular-transport common floor on the transported physical marginals themselves; no second comparison-state faithfulness clause is missing.*

Second, once local weak-$*$ extraction and GNS gluing have emitted the realized scaling-limit geometric cap pair, one needs ordered cut-pair rigidity *on that realized pair to collapse the residual cap-preserving conformal freedom to the standard hyperbolic subgroup. Only after that second object is supplied does the geometric modular statement become fully internal on $\mathcal{A}_\infty^{\mathrm{geo}}(C)$. The present paper proves neither object and therefore does not claim that MaxEnt alone completes the BW lift.*

Corollary 6.6 (Lorentz Kinematics)**.** *Under the hypotheses of Theorem 6.3,*

$$\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2, \mathbb{C}) \cong \mathrm{SO}^+(3, 1).$$

The cap modular flows are therefore the standard one-parameter Lorentz boost/dilation subgroups in the celestial-sphere realization, and the induced local kinematic group is the connected Lorentz group.

Proof. Orientation-preserving conformal maps of S^2 are exactly the Möbius transformations, so

$$\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1).$$

By Theorem 6.3, on the extracted prime geometric subnet the realized scaling-limit cap modular automorphism groups are the standard cap-preserving conformal dilations with the 2π normalization fixed internally. Hence the local kinematics induced by modular flow is the connected Lorentz group. □

6.2 The null modular bridge

The fixed-cutoff strip analysis begins from the transferred cut-center data and the inherited-strip Markov structure. At fixed regulator scale one obtains exact or controlled four-term strip additivity together with endpoint-Lipschitz control of the renormalized half-line family; on the same scaling-limit geometric-cap branch used in Theorem 6.3, the null half-line blow-up net then inherits geometric dilation and therefore half-sided modular inclusion. Standard Borchers–Wiesbrock theory [8] applies only after that derived half-sided modular pair is in hand, yielding positivity and unitary implementation of null translations. The downstream boundary is narrower: the separate interval-preserving projective branch is needed to transport the half-line result to bounded null intervals, and the tensor upgrade carries the standard null-invisible metric ambiguity. The half-line generator itself is identified below with the effective local null-stress charge.

The present subsection therefore establishes the following chain:

1. null-strip center transfer and the inherited left/right split hypothesis;
2. exact strip additivity on the exact inherited Markov model, and a carried defect operator on controlled exact-Markov replacements on one fixed strip model;
3. endpoint-Lipschitz matrix elements for the renormalized half-line family and the resulting weak tail generator;
4. derived half-sided modular inclusion on the null half-line blow-up net from the declared scaling-limit geometric-cap branch, and the resulting Borchers positive translation generator.

The only explicit downstream boundary retained below is the bounded-interval transport/projective branch together with the later tensor upgrade; the half-line generator/charge identification is proved inside the bridge itself. Lemma 6.13 states the positive null-translation generator explicitly: the Borchers unitary group is constructed on its Stone domain, positivity is theorem-level rather than implicit, and the half-line modular Hamiltonians are tied to it by explicit affine covariance and quadratic-form identities. The density-upgrade template of Theorem 6.11 is recorded separately and does not carry the operator-identification burden.

Proposition 6.7 (Null-strip center transfer and inherited split)**.** *Fix a null generator Ω and a regulated tripartition*

$$I_- = (v_1, v_2), \qquad J = (v_2, v_3), \qquad I_+ = (v_3, v_4),$$

with cuts

$$\Gamma_- := \{v = v_2\}, \qquad \Gamma_+ := \{v = v_3\}.$$

Assume the derived fixed-cutoff regulator/collar package established above, and assume that the two null cuts inherit the ordinary or central-defect boundary-redundancy data of Proposition 5.1 in the following precise sense:

(i) each cut $\Gamma_\pm$ carries a compact derived boundary action $\widehat{K}_\pm$ on a reference cut Hilbert space, with $\widehat{K}_\pm = K_\pm$ on the ordinary branch;

(ii) in a compatible type-I regulator presentation, the adjacent regions carry inverse transport across each cut, so for irreducible $\widehat{K}_\pm$-modules $W_{\alpha_\pm}$ one has decompositions

$$\tilde{\mathcal{H}}_{I_-} \cong \bigoplus_{\alpha_-} W_{\alpha_-} \otimes \mathcal{H}_{i_-^{\alpha_-}},$$

$$\tilde{\mathcal{H}}_J \cong \bigoplus_{\alpha_-,\alpha_+} W^*_{\alpha_-} \otimes \mathcal{H}_{j^{\alpha_-,\alpha_+}} \otimes W_{\alpha_+},$$

$$\tilde{\mathcal{H}}_{I_+} \cong \bigoplus_{\alpha_+} W^*_{\alpha_+} \otimes \mathcal{H}_{i_+^{\alpha_+}};$$

(iii) the gauge-invariant strip algebra is the commutant of the transported boundary action on $\tilde{\mathcal{H}}_J$, so that the pair (α_-, α_+) is a central cut label;

(iv) for each (α_-, α_+), the multiplicity space $\mathcal{H}_{j^{\alpha_-,\alpha_+}}$ admits a chosen factorization

$$\mathcal{H}_{j^{\alpha_-,\alpha_+}} \cong \mathcal{H}_{j_L^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-,\alpha_+}}$$

such that, blockwise, $\mathcal{A}(I_- \cup J)$ acts only on $\mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j_L^{\alpha_-,\alpha_+}}$ and $\mathcal{A}(J \cup I_+)$ acts only on $\mathcal{H}_{j_R^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{i_+^{\alpha_+}}$.

Then the strip algebra has the inherited central decomposition

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{B}(\mathcal{H}_{j^{\alpha_-,\alpha_+}}), \qquad Z(\mathcal{A}(J)) = \bigoplus_{\alpha_-,\alpha_+} \mathbb{C}\, P_{\alpha_-,\alpha_+},$$

and, under item (iv),

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{B}(\mathcal{H}_{j_L^{\alpha_-,\alpha_+}}) \otimes \mathcal{B}(\mathcal{H}_{j_R^{\alpha_-,\alpha_+}}).$$

Moreover the glued tripartition carries the block decomposition

$$\mathcal{H}_{I_- \cup J \cup I_+} \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j_L^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{i_+^{\alpha_+}}.$$

Thus the two null cuts transfer the same center data as the spatial collar branch, while the left/right split of the strip multiplicity spaces is exactly the extra decomposition-inheritance hypothesis and is not forced by center transfer alone.

Proof. By complete reducibility of the finite-dimensional unitary actions $\widehat{K}_\pm$, the displayed decompositions of $\tilde{\mathcal{H}}_{I_-}$, $\tilde{\mathcal{H}}_J$, and $\tilde{\mathcal{H}}_{I_+}$ exist. Under item (iii), the strip algebra is the commutant of the transported $\widehat{K}_- \times \widehat{K}_+$ action on

$$\tilde{\mathcal{H}}_J = \bigoplus_{\alpha_-,\alpha_+} W^*_{\alpha_-} \otimes \mathcal{H}_{j^{\alpha_-,\alpha_+}} \otimes W_{\alpha_+}.$$

Write an operator $X \in \mathcal{B}(\tilde{\mathcal{H}}_J)$ in matrix form relative to that direct sum. The block

$$X_{(\alpha_-,\alpha_+),(\beta_-,\beta_+)}$$

intertwines

$$W^*_{\alpha_-} \otimes W_{\alpha_+} \longrightarrow W^*_{\beta_-} \otimes W_{\beta_+}$$

for the product action of $\widehat{K}_- \times \widehat{K}_+$. By Schur's lemma, such an intertwiner is zero unless $(\alpha_-, \alpha_+) = (\beta_-, \beta_+)$, and when the sector pair agrees it is scalar on the representation factors. Therefore the commutant is exactly

$$\mathcal{A}(J) \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{B}(\mathcal{H}_{j^{\alpha_-,\alpha_+}}),$$

and the center is generated by the corresponding block projectors P_{α_-,α_+}.

For the glued tripartition, tensor the three regulator Hilbert spaces and decompose:

$$\tilde{\mathcal{H}}_{I_-} \otimes \tilde{\mathcal{H}}_J \otimes \tilde{\mathcal{H}}_{I_+} \cong \bigoplus_{\alpha_-,\beta_-,\alpha_+,\beta_+} (W_{\alpha_-} \otimes W^*_{\beta_-}) \otimes \mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j^{\beta_-,\beta_+}} \otimes (W_{\beta_+} \otimes W^*_{\alpha_+}) \otimes \mathcal{H}_{i_+^{\alpha_+}}.$$

Taking $\widehat{K}_- \times \widehat{K}_+$-invariants and applying Schur's lemma to each cut gives

$$(W_{\alpha_-} \otimes W^*_{\beta_-})^{\widehat{K}_-} \cong \begin{cases} \mathbb{C}, & \alpha_- = \beta_-, \\ 0, & \alpha_- \neq \beta_-, \end{cases} \qquad (W_{\beta_+} \otimes W^*_{\alpha_+})^{\widehat{K}_+} \cong \begin{cases} \mathbb{C}, & \beta_+ = \alpha_+, \\ 0, & \beta_+ \neq \alpha_+, \end{cases}$$

so only matching cut sectors survive and one obtains

$$\mathcal{H}_{I_- \cup J \cup I_+} \cong \bigoplus_{\alpha_-,\alpha_+} \mathcal{H}_{i_-^{\alpha_-}} \otimes \mathcal{H}_{j^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{i_+^{\alpha_+}}.$$

If item (iv) holds, substitute the chosen factorization

$$\mathcal{H}_{j^{\alpha_-,\alpha_+}} \cong \mathcal{H}_{j_L^{\alpha_-,\alpha_+}} \otimes \mathcal{H}_{j_R^{\alpha_-,\alpha_+}}$$

to obtain the displayed inherited split and the stated action of the left and right union algebras. Nothing in the Schur-lemma argument itself forces item (iv); it is exactly the extra structural condition required to reproduce the same blockwise tensor pattern as Theorem 5.2. □

Corollary 6.8 (Exact or controlled four-term null modular relation on an inherited strip model)**.** *Fix one finite-dimensional regulated strip model satisfying Proposition 6.7, and define*

$$\mathfrak{M}_{I_-:J:I_+} := \{\sigma : \ I(I_- : I_+ \mid J)_\sigma = 0\},$$

$$\delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon) := \sup \left\{ \inf_{\sigma \in \mathfrak{M}_{I_-:J:I_+}} \|\rho - \sigma\|_1 : I(I_- : I_+ \mid J)_\rho \leq \varepsilon \right\}.$$

For any state η on this strip model, write

$$\Delta K_J(\eta) := K_{I_- \cup J \cup I_+}(\eta) - K_{I_- \cup J}(\eta) - K_{J \cup I_+}(\eta) + K_J(\eta).$$

If the reference state ω is exact Markov on this inherited decomposition, then

$$\Delta K_J(\omega) \in Z(\mathcal{A}(J)),$$

and in the canonical blockwise identification of Proposition 6.7 one may take

$$\Delta K_J(\omega) = 0.$$

Equivalently, without fixing that normalization there exists a central operator

$$K_{\partial,J}(\omega) \in Z(\mathcal{A}(J))$$

such that

$$K_{I_-\cup J\cup I_+}(\omega) = K_{I_-\cup J}(\omega) + K_{J\cup I_+}(\omega) - K_J(\omega) + K_{\partial,J}(\omega).$$

If instead $I(I_- : I_+ \mid J)_\omega \le \varepsilon$, then on this fixed strip model one may choose an exact Markov state $\widetilde{\omega}_J \in \mathfrak{M}_{I_-:J:I_+}$ with

$$\|\omega - \widetilde{\omega}_J\|_1 \le \delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon).$$

Define the carried defect operator

$$\mathfrak{D}_J(\omega, \widetilde{\omega}_J) := \Delta K_J(\omega) - \Delta K_J(\widetilde{\omega}_J).$$

Then

$$K_{I_-\cup J\cup I_+}(\omega) = K_{I_-\cup J}(\omega) + K_{J\cup I_+}(\omega) - K_J(\omega) + K_{\partial,J}(\widetilde{\omega}_J) + \mathfrak{D}_J(\omega, \widetilde{\omega}_J),$$

where

$$K_{\partial,J}(\widetilde{\omega}_J) := \Delta K_J(\widetilde{\omega}_J) \in Z(\mathcal{A}(J)).$$

For every bounded observable O in $\mathcal{A}(I_- \cup J)$ or $\mathcal{A}(J \cup I_+)$,

$$\big|\omega(O) - \widetilde{\omega}_J(O)\big| \le \|O\|_\infty \, \delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon).$$

If the marginals entering ΔK_J for ω and $\widetilde{\omega}_J$ are uniformly faithful with lower spectral bound $\lambda_* > 0$, then

$$\|\mathfrak{D}_J(\omega, \widetilde{\omega}_J)\|_\infty \le 4\lambda_*^{-1} \, \delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon).$$

Independently, the Fawzi–Renner recovery map supplies a constructive comparison state with trace-norm error

$$r_{\mathrm{FR}}(\varepsilon) = 2\sqrt{1 - e^{-\varepsilon}} \le 2\sqrt{\varepsilon}.$$

Hence the exact four-term strip relation is available only at exact Markovity, or along a controlled strip family on one fixed inherited strip model for which

$$\delta^{\mathrm{M}}_{I_-:J:I_+}(\varepsilon_J) \to 0.$$

In the controlled case, $\mathfrak{D}_J$ is carried explicitly at each finite stage.

Proof. Assume first that ω is exact Markov on the inherited split of Proposition 6.7. Then the HJPW structure theorem gives, on each sector pair $\alpha = (\alpha_-, \alpha_+)$,

$$\omega_{I_- J I_+}|_\alpha = p_\alpha \, \omega^{(\alpha)}_{I_- j_L} \otimes \omega^{(\alpha)}_{j_R I_+}.$$

Accordingly,

$$K_{I_-\cup J\cup I_+}(\omega)\big|_\alpha = (-\log p_\alpha)\mathbf{1} + K^{(\alpha)}_{I_- j_L} \otimes \mathbf{1}_{j_R I_+} + \mathbf{1}_{I_- j_L} \otimes K^{(\alpha)}_{j_R I_+}.$$

Tracing over I_+, I_-, or both gives the marginal block formulas

$$K_{I_-\cup J}(\omega)\big|_\alpha = (-\log p_\alpha)\mathbf{1} + K^{(\alpha)}_{I_- j_L} \otimes \mathbf{1}_{j_R} + \mathbf{1}_{I_- j_L} \otimes K^{(\alpha)}_{j_R},$$

$$K_{J\cup I_+}(\omega)\big|_\alpha = (-\log p_\alpha)\mathbf{1} + K^{(\alpha)}_{j_L} \otimes \mathbf{1}_{j_R I_+} + \mathbf{1}_{j_L} \otimes K^{(\alpha)}_{j_R I_+},$$

$$K_J(\omega)\big|_\alpha = (-\log p_\alpha)\mathbf{1} + K^{(\alpha)}_{j_L} \otimes \mathbf{1}_{j_R} + \mathbf{1}_{j_L} \otimes K^{(\alpha)}_{j_R}.$$

Subtracting the last three expressions from the first gives zero on every block. Thus, in the canonical blockwise identification supplied by Proposition 6.7,

$$\Delta K_J(\omega) = 0.$$

If one chooses to keep the blockwise endpoint-label bookkeeping explicit rather than absorbing those constants into the canonical identification, the remainder is a direct sum of block constants and is therefore central. This yields the stated central form with $K_{\partial,J}(\omega) \in Z(\mathcal{A}(J))$.

For the controlled statement, apply Proposition 5.10 on this fixed finite-dimensional strip model, relabeling A, B, D there as I_-, J, I_+. This yields an exact Markov replacement $\widetilde{\omega}_J \in \mathfrak{M}_{I_-:J:I_+}$ with the displayed trace-norm bound. The identity

$$\Delta K_J(\omega) = \Delta K_J(\widetilde{\omega}_J) + \mathfrak{D}_J(\omega, \widetilde{\omega}_J)$$

is tautological, so the displayed four-term formula follows from the exact-Markov case applied to $\widetilde{\omega}_J$. The observable estimate is immediate from

$$\left|\omega(O) - \widetilde{\omega}_J(O)\right| \leq \|O\|_\infty \, \|\omega - \widetilde{\omega}_J\|_1.$$

Finally, Proposition 5.13, again relabeled $A, B, D \mapsto I_-, J, I_+$, yields the operator-norm bound on $\mathfrak{D}_J$ whenever the relevant marginals have the stated lower spectral bound. The Fawzi–Renner estimate is the same one-shot recoverability bound used elsewhere and is independent of the exact-Markov replacement modulus. □

Definition 6.9 (Renormalized null modular functional)**.** *For a null interval I on generator Ω, let $K_\partial(I, \Omega)$ denote the central endpoint-label term singled out by Corollary 6.8 on the inherited strip model, or by its controlled exact-Markov replacement when that model is used as reference. Define*

$$\widetilde{K}[I, \Omega] := K[I, \Omega] - K_\partial(I, \Omega).$$

For a null half-line $H_a := (a, \infty)$, abbreviate

$$\widetilde{K}_a(\Omega) := \widetilde{K}[H_a, \Omega].$$

Proposition 6.10 (Endpoint-Lipschitz null modular families and weak tail generator)**.** *Under the derived quasi-local propagation and bounded-interval endpoint-Lipschitz control internal to Axiom 3.3, the renormalized interval family obeys the following matrix-element bounds on every fixed local-energy-bounded domain. For bounded intervals*

$$I = (a, b), \qquad I' = (a', b'),$$

contained in one compact endpoint window,

$$\left|\langle \psi, (\widetilde{K}[I', \Omega] - \widetilde{K}[I, \Omega])\phi \rangle\right| \leq C_{\psi,\phi,\Omega} \left(|a' - a| + |b' - b|\right).$$

In particular the matrix elements of $\widetilde{K}[I, \Omega]$ are jointly Lipschitz in the two endpoints and therefore have finite variation there.

For null half-lines $H_a = (a, \infty)$, one likewise has

$$\left|\langle \psi, (\widetilde{K}_{a'}(\Omega) - \widetilde{K}_a(\Omega))\phi \rangle\right| \leq C_{\psi,\phi,\Omega} \, |a' - a|$$

for a, a' in every compact window. Hence, for fixed ψ, ϕ,

$$f_{\psi,\phi}(a) := \langle \psi, \widetilde{K}_a(\Omega)\phi \rangle$$

is locally Lipschitz and therefore absolutely continuous. Its distributional derivative defines a locally L^∞ sesquilinear-form density

$$\langle \psi, q(a,\Omega)\phi \rangle := -\frac{1}{2\pi}\, \partial_a f_{\psi,\phi}(a),$$

and for $a < b$,

$$\langle \psi, (\widetilde{K}_b(\Omega) - \widetilde{K}_a(\Omega))\phi \rangle = -2\pi \int_a^b \langle \psi, q(v,\Omega)\phi \rangle \, dv.$$

The object $q(a,\Omega)$ is the weak tail generator rather than a local operator-valued density; its identification with the positive self-adjoint Borchers generator occurs only after the derived half-sided modular inclusion of Corollary 6.12 and Lemma 6.13.

Proof. The bounded-interval endpoint-control statement derived from Axiom 3.3 applies precisely to the renormalized family obtained after removal of the central endpoint term. Therefore, for intervals contained in one fixed compact endpoint window,

$$\left| \langle \psi, (\widetilde{K}[I',\Omega] - \widetilde{K}[I,\Omega])\phi \rangle \right| \le C_{\psi,\phi,\Omega}\, |I' \Delta I|,$$

where $I'\Delta I$ is the symmetric difference of the two intervals. If

$$I = (a,b), \qquad I' = (a',b'),$$

then

$$|I'\Delta I| \le |a' - a| + |b' - b|,$$

which gives the displayed joint endpoint-Lipschitz bound.

For half-lines $H_a = (a,\infty)$ and $H_{a'} = (a',\infty)$, the symmetric difference is the bounded interval between the two endpoints, so

$$|H_{a'} \Delta H_a| = |a' - a|.$$

Because the central endpoint-label term has been removed in the definition of $\widetilde{K}_a$, the same endpoint-control estimate yields

$$\left| \langle \psi, (\widetilde{K}_{a'}(\Omega) - \widetilde{K}_a(\Omega))\phi \rangle \right| \le C_{\psi,\phi,\Omega}\, |a' - a|.$$

Hence, on every compact a-interval, the function

$$f_{\psi,\phi}(a) := \langle \psi, \widetilde{K}_a(\Omega)\phi \rangle$$

is Lipschitz. A Lipschitz function on a compact interval is absolutely continuous, so it has an almost-everywhere derivative in L^∞ and obeys the fundamental theorem of calculus:

$$f_{\psi,\phi}(b) - f_{\psi,\phi}(a) = \int_a^b \partial_v f_{\psi,\phi}(v)\, dv.$$

Define

$$\langle \psi, q(v,\Omega)\phi \rangle := -\frac{1}{2\pi}\, \partial_v f_{\psi,\phi}(v)$$

distributionally. Substituting this definition into the previous identity gives

$$\langle \psi, (\widetilde{K}_b(\Omega) - \widetilde{K}_a(\Omega))\phi \rangle = -2\pi \int_a^b \langle \psi, q(v,\Omega)\phi \rangle \, dv.$$

This is exactly the claimed weak-tail formula. □

Theorem 6.11 (Downstream density-upgrade template)**.** *Fix a null generator Ω and the renormalized half-line family $\widetilde{K}_a(\Omega)$ of Proposition 6.10. For fixed vectors ψ, ϕ in a common dense domain, let*

$$f_{\psi,\phi}(a) := \langle \psi, \widetilde{K}_a(\Omega)\phi\rangle, \qquad q_{\psi,\phi}(a) := \langle \psi, q(a,\Omega)\phi\rangle = -\frac{1}{2\pi}\,\partial_a f_{\psi,\phi}(a).$$

Assume in addition that $q_{\psi,\phi}$ is weakly differentiable in a, that

$$\lim_{a\to+\infty} f_{\psi,\phi}(a) = 0, \qquad \lim_{a\to+\infty} q_{\psi,\phi}(a) = 0,$$

and define distributionally

$$\langle \psi, p(a,\Omega)\phi\rangle := -\partial_a q_{\psi,\phi}(a) = \frac{1}{2\pi}\,\partial_a^2 f_{\psi,\phi}(a).$$

Then

$$q_{\psi,\phi}(a) = \int_a^\infty \langle \psi, p(v,\Omega)\phi\rangle\, dv,$$

and

$$\langle \psi, \widetilde{K}_a(\Omega)\phi\rangle = 2\pi \int_a^\infty (v-a)\,\langle \psi, p(v,\Omega)\phi\rangle\, dv.$$

This statement is not part of the present fixed-cutoff bridge; it records the exact additional input/output package carried into the downstream density-upgrade task.

Null half-line blow-up net. Fix a smooth cut point and choose affine coordinate v on generator Ω so that the cut sits at $v = 0$. For $a \geq 0$, write

$$H_a := (a,\infty), \qquad \mathcal{M}_a(\Omega) := \overline{\bigvee_{a<c<d<\infty} \mathcal{A}((c,d),\Omega)}.$$

Thus $\mathcal{M}_a(\Omega)$ is the half-line blow-up algebra generated by bounded null interval algebras inside H_a. By isotony of the null interval net,

$$a \leq b \quad\Longrightarrow\quad \mathcal{M}_b(\Omega) \subseteq \mathcal{M}_a(\Omega).$$

Corollary 6.12 (Derived half-sided modular inclusion on null half-lines)**.** *On the scaling-limit geometric-cap branch of Theorem 6.3, the blow-up modular action near a smooth entangling cut acts on the null coordinate by*

$$v \mapsto e^{-2\pi t} v.$$

Therefore, for every $a \geq 0$,

$$\sigma_t^\omega(\mathcal{M}_a(\Omega)) = \mathcal{M}_{e^{-2\pi t}a}(\Omega).$$

Hence, for every $a > 0$,

$$\sigma_t^\omega(\mathcal{M}_a(\Omega)) \subseteq \mathcal{M}_a(\Omega) \qquad (t \leq 0),$$

so the inclusion

$$\mathcal{M}_a(\Omega) \subset \mathcal{M}_0(\Omega)$$

is half-sided modular. Equivalently, after the harmless convention change $t \mapsto -t$, one recovers the standard positive-time half-sided-inclusion form. The half-sided-inclusion step is therefore derived from the null-interval structure, isotony, and the scaling-limit geometric action rather than imported as an additional modular-QFT ingredient.

Proof. By Theorem 6.3, on every controlled collar family whose carried remainder vanishes in the refinement limit, the cap modular flow converges to the exact geometric cap dilation with 2π normalization. Blow up the cap near a smooth cut and restrict to the chosen null generator. In that tangent limit, the cap-preserving flow becomes the null dilation $v \mapsto e^{-2\pi t}v$. For every bounded interval $(c, d) \subset H_a$, the blow-up action therefore sends

$$\mathcal{A}((c, d), \Omega) \longmapsto \mathcal{A}((e^{-2\pi t}c, e^{-2\pi t}d), \Omega).$$

Taking the von Neumann closure of the algebra generated by all such intervals gives

$$\sigma_t^\omega(\mathcal{M}_a(\Omega)) = \mathcal{M}_{e^{-2\pi t}a}(\Omega).$$

If $t \leq 0$, then $e^{-2\pi t}a \geq a$, hence

$$H_{e^{-2\pi t}a} \subseteq H_a.$$

Isotony of the null interval net therefore implies

$$\mathcal{M}_{e^{-2\pi t}a}(\Omega) \subseteq \mathcal{M}_a(\Omega).$$

This is exactly the half-sided modular-inclusion property for the pair $\mathcal{M}_a(\Omega) \subset \mathcal{M}_0(\Omega)$. Reparametrizing the modular parameter by $t \mapsto -t$ gives the more common positive-time convention if desired. □

Lemma 6.13 (Positive null-translation generator)**.** *For the derived half-sided modular inclusion*

$$\mathcal{M}_a(\Omega) \subset \mathcal{M}_0(\Omega) \qquad (a > 0)$$

of Corollary 6.12, let $\Delta_0(\Omega)$ be the modular operator of the standard pair $(\mathcal{M}_0(\Omega), \omega)$ and define

$$K_0(\Omega) := -\log \Delta_0(\Omega).$$

Then Borchers–Wiesbrock yields a unique strongly continuous one-parameter unitary group

$$U_\Omega(a) = e^{iaP_\Omega}, \qquad a \in \mathbb{R},$$

such that

$$U_\Omega(a)\omega = \omega, \qquad U_\Omega(a)\,\mathcal{M}_b(\Omega)\,U_\Omega(a)^* = \mathcal{M}_{a+b}(\Omega) \quad (a, b \geq 0),$$

and whose generator P_Ω is positive and self-adjoint on the Stone domain

$$D(P_\Omega) := \left\{\psi \in \mathcal{H} : \lim_{a\to 0} \frac{U_\Omega(a)\psi - \psi}{ia} \text{ exists}\right\}.$$

Moreover,

$$\Delta_0(\Omega)^{it}U_\Omega(a)\Delta_0(\Omega)^{-it} = U_\Omega(e^{-2\pi t}a), \qquad \Delta_0(\Omega)^{it}P_\Omega\Delta_0(\Omega)^{-it} = e^{-2\pi t}P_\Omega,$$

and on the common invariant analytic core of $K_0(\Omega)$ and P_Ω,

$$[K_0(\Omega), P_\Omega] = -\,i\,2\pi P_\Omega.$$

For each $a \geq 0$, let $\Delta_a(\Omega)$ be the modular operator of the translated standard pair $(\mathcal{M}_a(\Omega), \omega)$ and define

$$K_a(\Omega) := -\log \Delta_a(\Omega).$$

Then

$$\Delta_a(\Omega) = U_\Omega(a)\Delta_0(\Omega)U_\Omega(a)^*, \qquad K_a(\Omega) = U_\Omega(a)K_0(\Omega)U_\Omega(a)^*,$$

hence

$$K_a(\Omega) = K_0(\Omega) - 2\pi a\, P_\Omega$$

as a quadratic-form identity on $D(K_0(\Omega)) \cap D(P_\Omega)$*. Equivalently,*

$$\langle \psi, (K_b(\Omega) - K_a(\Omega))\phi\rangle = -2\pi(b-a)\,\langle \psi, P_\Omega\phi\rangle$$

for all $\psi, \phi \in D(K_0(\Omega)) \cap D(P_\Omega)$*.*

In the canonical blockwise normalization of Corollary 6.8, where the half-line endpoint term has been absorbed into the central part, the weak endpoint derivative of Proposition 6.10 agrees with the Borchers generator:

$$\langle \psi, P_\Omega\phi\rangle = -\frac{1}{2\pi}\frac{d}{da}\Big|_{a=0^+}\langle \psi, \widetilde{K}_a(\Omega)\phi\rangle$$

for the same domain vectors. The half-sided-inclusion step proves only this affine half-line pair $(K_0(\Omega), P_\Omega)$*; bounded-interval modular-Hamiltonian formulas are downstream consequences of the separate interval-preserving projective branch recorded in Theorem 6.14.*

Proof. Corollary 6.12 gives the required half-sided modular inclusion. The standard Borchers–Wiesbrock theorem for a standard half-sided inclusion [8] then yields a unique strongly continuous unitary group $U_\Omega(a)$ with $U_\Omega(a)\omega = \omega$, the translation law for the nested half-line net, and a positive self-adjoint generator P_Ω; Stone's theorem gives the displayed domain formula.

The affine covariance relation

$$\Delta_0(\Omega)^{it}U_\Omega(a)\Delta_0(\Omega)^{-it} = U_\Omega(e^{-2\pi t}a)$$

is the Borchers relation. Differentiating at $a = 0$ gives

$$\Delta_0(\Omega)^{it}P_\Omega\Delta_0(\Omega)^{-it} = e^{-2\pi t}P_\Omega.$$

Since $\Delta_0(\Omega)^{it} = e^{-itK_0(\Omega)}$, differentiating at $t = 0$ on the common analytic core yields

$$-i[K_0(\Omega), P_\Omega] = -2\pi P_\Omega,$$

hence

$$[K_0(\Omega), P_\Omega] = -\,i\,2\pi P_\Omega.$$

Because $U_\Omega(a)\omega = \omega$ and

$$\mathcal{M}_a(\Omega) = U_\Omega(a)\mathcal{M}_0(\Omega)U_\Omega(a)^*,$$

the modular operator of the translated standard pair is

$$\Delta_a(\Omega) = U_\Omega(a)\Delta_0(\Omega)U_\Omega(a)^*,$$

so

$$K_a(\Omega) = U_\Omega(a)K_0(\Omega)U_\Omega(a)^*.$$

Differentiating this identity in a on $D(K_0(\Omega)) \cap D(P_\Omega)$ gives

$$\frac{d}{da}K_a(\Omega) = i[P_\Omega, K_a(\Omega)] = -2\pi P_\Omega,$$

because the commutator relation just proved is invariant under conjugation by $U_\Omega(a)$. Integrating from 0 to a yields the displayed quadratic-form identity, and taking matrix elements gives the equivalent form. Proposition 6.10 had produced the weak endpoint derivative of the renormalized half-line family; in the canonical normalization of Corollary 6.8 the central term has been removed, so that weak derivative is exactly the Borchers generator. The final sentence is the explicit claim boundary: bounded intervals require the additional projective interval action not supplied by half-sided inclusion alone. □

Theorem 6.14 (The half-line generator is the local null-stress charge)**.** *Fix a null generator Ω and work in the canonical blockwise normalization of Corollary 6.8. Let $\widetilde{K}_a(\Omega)$ denote the renormalized modular Hamiltonian of the half-line H_a, and let P_Ω be the positive Borchers generator of Lemma 6.13. Then on*

$$D_\Omega := D(K_0(\Omega)) \cap D(P_\Omega)$$

one has

$$\langle \psi, P_\Omega \phi \rangle = -\frac{1}{2\pi} \frac{d}{da}\Big|_{a=0^+} \langle \psi, \widetilde{K}_a(\Omega)\phi \rangle \qquad (\psi, \phi \in D_\Omega),$$

and this operator is exactly the local null-stress charge of the effective spacetime description on that same half-line family.

Equivalently, if the effective spacetime description writes the same renormalized family as

$$\widetilde{K}_a^{\mathrm{eff}}(\Omega) = 2\pi\, Q_a^{\mathrm{mod}}(\Omega) + K_\partial(a, \Omega),$$

with $Q_a^{\mathrm{mod}}(\Omega)$ the standard local null modular-energy term and $K_\partial(a, \Omega)$ central and endpoint-supported, then

$$Q_{kk}(\Omega) := -\frac{1}{2\pi} \frac{d}{da}\Big|_{a=0^+} \widetilde{K}_a^{\mathrm{eff}}(\Omega)$$

satisfies

$$P_\Omega = Q_{kk}(\Omega)$$

as a quadratic-form identity on D_Ω.

Hence the renormalized half-line null generator is not a separate EFT-side premise: it is the same operator that the effective spacetime description calls the local null-stress charge of the null half-line.

Proof. Lemma 6.13 proves the first displayed identity: in the canonical normalization of Corollary 6.8, the weak endpoint derivative of the renormalized half-line family is exactly the Borchers generator. Passing to the effective spacetime description does not change the operator family; it rewrites the same $\widetilde{K}_a(\Omega)$ in local continuum variables appropriate to the local Lorentzian regime. In that description, the local null-stress charge is precisely the right endpoint derivative

$$-\frac{1}{2\pi} \frac{d}{da}\Big|_{a=0^+} \widetilde{K}_a^{\mathrm{eff}}(\Omega)$$

of the same half-line modular family. Because $\widetilde{K}_a^{\mathrm{eff}}(\Omega)$ and $\widetilde{K}_a(\Omega)$ represent the same renormalized modular Hamiltonians, their right derivatives agree on D_Ω. Therefore the OPH generator and the effective null-stress charge coincide:

$$P_\Omega = Q_{kk}(\Omega).$$

The endpoint-supported central term does not alter the noncentral generator; in the canonical blockwise normalization it is absorbed into the central part fixed in Corollary 6.8. This proves the identification. □

Imported input and downstream boundary. The only continuum input used above is the standard local modular-Hamiltonian form on the effective description of that same half-line family. It is used only to name, in continuum language, the operator already fixed by the OPH half-line derivative. There is no separate "null-stress identification premise." The downstream boundary is only:

(i) transport of the half-line statement to bounded null intervals, which requires the separate interval-preserving projective branch and its affine-covariant kernel $g_I(v)$; and

(ii) reconstruction of a full local symmetric tensor from the directional charges, which is determined only up to the null-invisible metric term ϕg_{ab}.

So the gap closed here is the generator/charge identification itself: inside the null bridge and on the declared half-line family, P_Ω is exactly the local null-stress charge.

Lemma 6.15 (Null data determine the stress tensor modulo a metric term)**.** *Let X_{ab} be a symmetric tensor. If*

$$X_{ab}k^a k^b = 0$$

for every null vector k at a point, then

$$X_{ab} = \phi g_{ab}$$

for some scalar ϕ.

Proof. Write

$$X_{ab} = Y_{ab} + \phi g_{ab}$$

with Y_{ab} traceless. Since $g_{ab}k^a k^b = 0$ on null vectors, the hypothesis implies $Y_{ab}k^a k^b = 0$ for all null k. In local inertial coordinates let $k = (1, \hat{n})$ with $|\hat{n}| = 1$. Then

$$Y_{ab}k^a k^b = Y_{00} + 2\hat{n}^i Y_{0i} + \hat{n}^i \hat{n}^j Y_{ij}.$$

Oddness under $\hat{n} \mapsto -\hat{n}$ forces $Y_{0i} = 0$. Averaging that relation over S^2 gives

$$Y_{00} + \frac{1}{3}\delta^{ij} Y_{ij} = 0.$$

Tracelessness of Y_{ab} gives

$$-Y_{00} + \delta^{ij} Y_{ij} = 0,$$

so both Y_{00} and the spatial trace $\delta^{ij}Y_{ij}$ vanish. The condition is therefore

$$\hat{n}^i \hat{n}^j Y_{ij}^{\mathrm{TF}} = 0 \qquad \text{for all } \hat{n} \in S^2,$$

where Y_{ij}^{TF} is the traceless spatial part. The $\ell = 2$ spherical harmonics are linearly independent, so this implies $Y_{ij}^{\mathrm{TF}} = 0$. Hence $Y_{ab} = 0$ and therefore $X_{ab} = \phi g_{ab}$. □

Remark 6.16 (Claim boundary in the null bridge)**.** *The c.12-local bridge ends with explicit theorem-level objects: the positive self-adjoint null-translation generator P_Ω on its Stone domain, its Borchers affine covariance relation, the half-line modular identities above, and the exact half-line generator/charge identification of Theorem 6.14. The downstream boundary is transport to bounded null intervals through the separate interval-preserving projective branch and the tensor upgrade modulo the null-invisible metric term. The density-upgrade template of Theorem 6.11 is recorded separately and does not carry the operator-identification burden.*

6.3 Generalized entropy and the Einstein equation

From this point through Corollary 6.22 we specialize to the physical $d = 4$ scaling branch. The claim boundary is explicit. The only standard geometric input used below, beyond D3 and D4, is the fixed-volume area-variation identity for a small geodesic ball. The small-ball modular kernel used below is *not* imported from a separate EFT law: it is the local expression of the geometric cap generator B_C from Theorem 6.3, evaluated using the half-line null-stress identification of Theorem 6.14 together with the separate interval-preserving projective branch that transports it to bounded null intervals.

For a reference state ω and a small cap C,

$$\delta S_C = \delta\langle K_C\rangle.$$

By Theorem 6.3,

$$\delta S_C = 2\pi\,\delta\langle B_C\rangle.$$

By Proposition 5.16,

$$S_{\mathrm{gen}}(C) = \mathrm{Tr}(\rho L_C) + S_{\mathrm{bulk}}(C).$$

Together with Definition 5.17, this identifies the area term as the coarse-grained form of the edge entropy density rather than an independent postulate. After this edge-center split, the first-law piece entering the small-ball argument is the bulk contribution:

$$\delta S_{\mathrm{bulk}}(C) = 2\pi\,\delta\langle B_C\rangle.$$

Definition 6.17 (Admissible fixed-cap MaxEnt variation)**.** *Fix a cap C and a realized reference state ω_C on the cap-reduced realized MaxEnt branch. An admissible fixed-cap MaxEnt variation is a first-order variation*

$$\delta\rho_C = \frac{d}{ds}\Big|_{s=0}\rho_C(s)$$

along a C^1 curve $s \mapsto \rho_C(s)$ such that:

1. *every $\rho_C(s)$ lies in the same cap-label-preserving block class of Proposition 5.16, so the same edge-center operator L_C and sector projectors P_α apply;*
2. *the cap C, its size/volume, and the optional conserved charges Q_b are held fixed; and*
3. *the Axiom 3.3 constraint values are fixed to first order:*

$$\frac{d}{ds}\Big|_{s=0}\mathrm{Tr}(\rho_C(s)O_a(x)) = 0, \qquad \frac{d}{ds}\Big|_{s=0}\mathrm{Tr}(\rho_C(s)Q_b) = 0$$

   for every retained local constraint $O_a(x)$ supported in the fixed cap neighborhood and every retained conserved charge Q_b.

Shape or null-cut deformations are denoted separately by δ_{shape} or ∂_λ; they are not part of this fixed-cap variation class.

Theorem 6.18 (Derived fixed-cap generalized-entropy stationarity)**.** *Let C be a fixed cap and let ω_C be a realized reference state on the cap-reduced realized MaxEnt branch of Axiom 3.3. For every admissible fixed-cap MaxEnt variation δ of Definition 6.17,*

$$\delta S_{\mathrm{gen}}(C) = 0.$$

Equivalently, the realized reference state is stationary for generalized entropy on the allowed fixed-cap variation class selected by the realized MaxEnt family.

Proof. Axiom 3.3 selects the realized branch by entropy maximization on the finite-dimensional constraint surface cut out by the retained local constraints and optional conserved charges. Definition 6.17 restricts to first-order tangent directions that stay on that same fixed-cap constraint surface. Hence the first variation of the cap entropy vanishes at the realized reference state:

$$\delta S(\rho_C)\big|_{\omega_C} = 0.$$

For the same cap-label-preserving block class, Proposition 5.16 gives

$$S(\rho_C) = S_{\text{bulk}}(C) + \operatorname{Tr}(\rho_C L_C) = S_{\text{gen}}(C).$$

Differentiating this identity along the admissible curve yields

$$\delta S_{\text{gen}}(C) = \delta S(\rho_C)\big|_{\omega_C} = 0.$$

□

Remark 6.19 (Claim boundary of the fixed-cap generalized-entropy stationarity theorem)**.** *Theorem 6.18 internalizes only the fixed-cap, cap-label-preserving, constraint-preserving first-order variation class selected by Axiom 3.3. It does not claim generalized-entropy stationarity for arbitrary shape deformations, arbitrary null-cut deformations, or arbitrary off-branch perturbations outside that realized MaxEnt family.*

Lemma 6.20 (Internal d=4 small-ball bridge)**.** *Assume the hypotheses of Theorem 6.14, together with the separate interval-preserving projective branch invoked there for bounded intervals. Let p be the cap center and let u^a be the future-directed unit tangent of the local diamond rest frame at p. In local inertial coordinates (t, x^i) adapted to u^a, let D_ℓ be the causal diamond whose $t = 0$ slice is the Euclidean ball*

$$B_\ell = \{t = 0,\ r < \ell\}, \qquad r^2 = \delta_{ij} x^i x^j.$$

Then

$$\delta S_{\text{bulk}}(C) = 2\pi \int_{B_\ell} \frac{\ell^2 - r^2}{2\ell}\, \delta\langle T_{00}\rangle\, d^3x + \delta\langle E^{(\eta)}_{C,\ell}\rangle.$$

If, moreover, $\delta\langle T_{00}\rangle$ is approximately constant across B_ℓ and

$$\delta\langle E^{(\eta)}_{C,\ell}\rangle = o(\ell^4)$$

along the same scaling family, then in $d = 4$,

$$\delta S_{\text{bulk}}(C) = \frac{8\pi^2 \ell^4}{15}\, \delta\langle T_{00}\rangle + O(\ell^5 \partial T) + o(\ell^4).$$

Proof. Work in the local Lorentzian scaling regime around p. On the geometric-subnet branch of Theorem 6.3, the cap modular flow is the geometric flow of the diamond-preserving conformal Killing field. In the tangent diamond D_ℓ, that field is

$$\xi_{D_\ell} = \frac{1}{2\ell}\Big((\ell^2 - r^2 - t^2)\,\partial_t - 2t\, x^i \partial_i\Big),$$

so on the $t = 0$ slice one has

$$\xi_{D_\ell} \cdot n = \frac{\ell^2 - r^2}{2\ell},$$

with $n^a = u^a$ the slice normal at the center.

The same kernel is the bounded-interval kernel supplied by that separate projective branch. Along each null generator of the diamond, Theorem 6.14 identifies the half-line generator constructed above with the local null-stress charge, and the interval-preserving projective branch transports that identification to the affine-covariant interval weight that vanishes at the two diamond endpoints. On the $t = 0$ slice that weight is exactly $(\ell^2 - r^2)/(2\ell)$. Therefore the null-stress bridge reconstructs the bulk modular charge of the geometric generator as

$$\delta\langle B_C\rangle = \int_{B_\ell} \frac{\ell^2 - r^2}{2\ell}\, \delta\langle T_{00}\rangle\, d^3x + \frac{1}{2\pi}\, \delta\langle E^{(\eta)}_{C,\ell}\rangle.$$

Multiplying by the first-law factor 2π gives the displayed formula for $\delta S_{\text{bulk}}(C)$.

If $\delta\langle T_{00}\rangle$ is approximately constant across B_ℓ, then

$$\int_{B_\ell} \frac{\ell^2 - r^2}{2\ell}\, \delta\langle T_{00}\rangle\, d^3x = \frac{4\pi\ell^4}{15}\, \delta\langle T_{00}\rangle + O(\ell^5 \partial T).$$

The additional hypothesis $\delta\langle E^{(\eta)}_{C,\ell}\rangle = o(\ell^4)$ then yields

$$\delta S_{\text{bulk}}(C) = 2\pi \cdot \frac{4\pi\ell^4}{15}\, \delta\langle T_{00}\rangle + O(\ell^5 \partial T) + o(\ell^4),$$

which is exactly the stated coefficient. No separate EFT small-ball first law has been used: the kernel comes from the geometric cap generator together with the D4 null-stress bridge. □

Theorem 6.21 (Conditional Jacobson-type rest-frame relation from derived fixed-cap generalized-entropy stationarity)**.** *Assume Axioms 3.1–3.4, Assumption 3.6, and the hypotheses of Theorem 6.14 and Lemma 6.20. Then, for every sufficiently small cap centered at p and every admissible fixed-cap MaxEnt variation δ about a realized reference state ω in the locally Lorentzian $d = 4$ scaling regime, if u^a is the future-directed unit tangent of the local diamond rest frame at p, one has*

$$\delta\Big[(G_{ab} + \Lambda g_{ab})u^a u^b\Big] = 8\pi G\, u^a u^b\, \delta\langle T_{ab}\rangle$$

at p. In the adapted rest frame $u^a = (1, 0, 0, 0)$, this is

$$\delta(G_{00} + \Lambda g_{00}) = 8\pi G\, \delta\langle T_{00}\rangle.$$

Proof. Theorem 6.18 gives the fixed-cap generalized-entropy stationarity condition

$$0 = \delta S_{\text{gen}}(C) = \delta S_{\text{bulk}}(C) + \frac{\delta A}{4G}.$$

By Lemma 6.20,

$$\delta S_{\text{bulk}}(C) = \frac{8\pi^2\ell^4}{15}\, u^a u^b\, \delta\langle T_{ab}\rangle + O(\ell^5 \partial T) + o(\ell^4),$$

where in the adapted rest frame the contraction is $\delta\langle T_{00}\rangle$.

For a fixed-volume small geodesic ball in $d = 4$, the standard area-variation identity gives

$$\delta A\big|_{V,\Lambda} = -\frac{4\pi\ell^4}{15}\, \delta\Big[(G_{ab} + \Lambda g_{ab})u^a u^b\Big].$$

Therefore

$$0 = \frac{8\pi^2\ell^4}{15}\, u^a u^b\, \delta\langle T_{ab}\rangle - \frac{\pi\ell^4}{15G}\, \delta\Big[(G_{ab} + \Lambda g_{ab})u^a u^b\Big] + O(\ell^5 \partial T) + o(\ell^4).$$

Dividing by ℓ^4 and taking the small-ball limit removes the $O(\ell^5\partial T) + o(\ell^4)$ terms and yields

$$\delta\Big[(G_{ab} + \Lambda g_{ab})u^a u^b\Big] = 8\pi G\, u^a u^b\, \delta\langle T_{ab}\rangle.$$

This is precisely the desired rest-frame scalar relation. □

Corollary 6.22 (Internal tensor upgrade in the scaling regime)**.** *If the first-variation relation of Theorem 6.21 holds for all local observer four-velocities and all reference states in a connected scaling branch, then*

$$G_{ab} + \Lambda g_{ab} = 8\pi G\, \langle T_{ab}\rangle$$

throughout that branch. The only local ambiguity is the metric term isolated by Lemma 6.15, and that ambiguity is exactly the same branch constant Λ*.*

Proof. Define

$$Y_{ab} := G_{ab} + \Lambda g_{ab} - 8\pi G\, \langle T_{ab}\rangle.$$

At any point p, Theorem 6.21 gives

$$u^a u^b\, \delta Y_{ab} = 0$$

for the unit future-directed four-velocity u^a of the local diamond rest frame. By the Lorentz branch of Theorem 6.3 and overlap consistency across observers through the same point, these local rest frames exhaust all timelike directions. Hence

$$u^a u^b\, \delta Y_{ab} = 0 \qquad \text{for every unit timelike } u^a.$$

Choose local inertial coordinates at p and write

$$u^a = \gamma(1, v^i), \qquad |\vec{v}| < 1, \qquad \gamma = (1 - |\vec{v}|^2)^{-1/2}.$$

Then

$$0 = u^a u^b\, \delta Y_{ab} = \gamma^2\Big(\delta Y_{00} + 2v^i \delta Y_{0i} + v^i v^j \delta Y_{ij}\Big)$$

for every $|\vec{v}| < 1$. The quadratic polynomial in v^i therefore vanishes on an open ball, so all of its coefficients vanish:

$$\delta Y_{00} = 0, \qquad \delta Y_{0i} = 0, \qquad \delta Y_{ij} = 0.$$

Thus

$$\delta Y_{ab} = 0$$

as a full tensor.

Along any path of admissible fixed-cap MaxEnt variations inside the connected scaling branch, Y_{ab} is therefore constant. The null bridge isolated the only purely local freedom as a metric term via Lemma 6.15; fixing that term on one maximally symmetric reference state determines the same branch constant Λ. Equivalently,

$$Y_{ab} = 0,$$

so

$$G_{ab} + \Lambda g_{ab} = 8\pi G\, \langle T_{ab}\rangle$$

throughout the branch. □

6.4 Cosmological-constant / screen-capacity closure

Lemma 6.23 (Vacuum-Energy Blindness)**.** *For any null vector k and any vacuum-energy contribution $T^{\mathrm{vac}}_{ab} = -\rho_{\mathrm{vac}} g_{ab}$,*

$$T^{\mathrm{vac}}_{kk} = 0.$$

Proposition 6.24 (Structural Separation of Λ)**.** *Local null-modular data fix T_{ab} only up to ϕg_{ab}. Therefore Λ is not determined by local overlap consistency and requires a global input.*

Corollary 6.25 (Cosmological Constant from Capacity)**.** *If the global screen capacity is identified with the de Sitter static-patch entropy,*

$$N_{\mathrm{scr}} = S_{\mathrm{dS}},$$

and the standard de Sitter entropy relation

$$S_{\mathrm{dS}} = \frac{A_{\mathrm{dS}}}{4G} = \frac{3\pi}{G\Lambda},$$

then

$$\Lambda = \frac{3\pi}{GN_{\mathrm{scr}}}.$$

Corollary 6.26 (Global screen-capacity closure of the Einstein branch)**.** *Assume the hypotheses of Corollaries 6.22 and 6.25. Then on the same connected scaling branch,*

$$G_{ab} + \frac{3\pi}{GN_{\mathrm{scr}}}\, g_{ab} = 8\pi G \langle T_{ab} \rangle.$$

Thus the local Einstein recovery closes globally once the screen-capacity input N_{scr} is declared.

Proof. Corollary 6.22 gives

$$G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle$$

on the connected scaling branch. Corollary 6.25 identifies

$$\Lambda = \frac{3\pi}{GN_{\mathrm{scr}}}$$

once the global screen-capacity input is supplied. Substituting that value of Λ yields the stated equation. □

Corollary 6.27 (De Sitter static-patch parameter package on the D6 branch)**.** *Assume the hypotheses of Corollary 6.25. Then the same screen-capacity input fixes the coherent static-patch package*

$$S_{\mathrm{dS}} = N_{\mathrm{scr}}, \qquad A_{\mathrm{dS}} = 4GN_{\mathrm{scr}}, \qquad r_{\mathrm{dS}} = \sqrt{\frac{3}{\Lambda}}, \qquad t_\Lambda = \frac{r_{\mathrm{dS}}}{c}.$$

Together with Corollary 6.26, this packages the D6 branch as the global closure of the same Einstein branch plus its derived de Sitter entropy/radius/timescale readout.

Proof. Corollary 6.25 fixes Λ from N_{scr}. The de Sitter entropy-area relation gives

$$S_{\mathrm{dS}} = \frac{A_{\mathrm{dS}}}{4G} = N_{\mathrm{scr}},$$

and the standard static-patch formulas give the stated r_{dS} and t_Λ. □

Remark 6.28 (Capacity normalization)**.** *The entropy capacity and the bare horizon area ratio differ by the factor π. In Planck units,*

$$N_{\text{patch}} = \left(\frac{r_{\text{dS}}}{\ell_P}\right)^2 = \frac{3}{\Lambda \ell_P^2}, \qquad N_{\text{scr}} = \pi N_{\text{patch}} = \frac{3\pi}{\Lambda \ell_P^2}.$$

For $\Lambda \ell_P^2 \simeq 2.85 \times 10^{-122}$, this gives $N_{\text{patch}} \simeq 1.05 \times 10^{122}$ and $N_{\text{scr}} \simeq 3.31 \times 10^{122}$. The D6 branch uses the second quantity because N_{scr} is defined as the de Sitter entropy capacity.

Remark 6.29 (Observed-age benchmark status)**.** *The branch quantity t_Λ is the de Sitter timescale fixed by the same D6 closure. The observed cosmic age t_0 is not an additional D6 theorem output. It is a downstream FLRW comparison quantity once one chooses a cosmological model above the local/global D5→D6 stack. On the flat ΛCDM benchmark,*

$$t_0 = \frac{2}{3H_0\sqrt{\Omega_\Lambda}} \sinh^{-1}\left(\sqrt{\frac{\Omega_\Lambda}{\Omega_m}}\right).$$

Theorem 6.30 (Cosmological-constant / screen-capacity closure stack)**.** *Assume the hypotheses of Corollary 6.22, the external input*

$$N_{\text{scr}} = \log \dim \mathcal{H}_{\text{tot}},$$

the D6 screen-capacity identification

$$N_{\text{scr}} = S_{\text{dS}},$$

the standard de Sitter entropy relation

$$S_{\text{dS}} = \frac{A_{\text{dS}}}{4G} = \frac{3\pi}{G\Lambda},$$

and the standard de Sitter static-patch formulas

$$r_{\text{dS}} = \sqrt{\frac{3}{\Lambda}}, \qquad t_\Lambda = \frac{r_{\text{dS}}}{c}.$$

Then:

(i) *the local null-modular data determine the Einstein branch only modulo a metric term Λg_{ab};*

(ii) *the same branch closes globally as*

$$G_{ab} + \frac{3\pi}{G N_{\text{scr}}} g_{ab} = 8\pi G \langle T_{ab} \rangle;$$

(iii) *the same D6 closure fixes the static-patch package*

$$S_{\text{dS}} = N_{\text{scr}}, \qquad A_{\text{dS}} = 4G N_{\text{scr}}, \qquad r_{\text{dS}} = \sqrt{\frac{3}{\Lambda}}, \qquad t_\Lambda = \frac{r_{\text{dS}}}{c};$$

(iv) *the observed cosmic age t_0 is not an additional theorem output of this stack and is a downstream FLRW benchmark.*

Thus the cosmological-constant package is one local/global theorem stack: the local null-modular branch leaves exactly the null-invisible metric freedom, and the external screen-capacity input closes that same Einstein branch globally.

Proof. Item (i) is Proposition 6.24, whose local ambiguity statement rests on the null-data blindness isolated by Lemma 6.23 together with the D5 Einstein branch. Item (ii) is Corollary 6.26. Item (iii) is Corollary 6.27. Item (iv) is Remark 6.29. □

Theorem 6.30 packages the intended local/global D5→D6 handoff. The imported premises are exactly the external screen-capacity input N_{scr}, the D6 identification $N_{\mathrm{scr}} = S_{\mathrm{dS}}$, the standard de Sitter entropy relation, and the standard static-patch radius/time formulas; none of these are hidden inside the local null-modular reconstruction. The external item is the screen-capacity identification itself: the local theorem stack does not derive $N_{\mathrm{scr}} = S_{\mathrm{dS}}$ from the bare local axioms. The vacuum-energy problem is reorganized by separating local null data from that global capacity closure.

7 Gauge Reconstruction and Standard Model Structure

7.1 Compact gauge reconstruction

At any fixed UV cutoff, edge-center completion equips collars with finitely many sector labels, intertwiners, and fusion rules. These are fixed-cutoff collar data. The local MaxEnt / collar-mixing package established earlier controls only fixed-cutoff recoverability, modular-support localization, and carried error terms on the realized branch. It is logically separate from the question whether transportable edge sectors survive refinement, and it supplies no theorem that the refinement-limit sector category is trivial or nontrivial. A later nontrivial transportable branch is therefore treated here as compatible additional gauge data rather than as something decided by the mixing estimate itself. Whenever compact gauge reconstruction is invoked, the full bosonic refinement ladder of Assumption 3.7 is the explicit input.

On the ordinary or central-defect branch, the transportability referred to here is path-independent movement of charges between patches in the DHR sense. On the central branch the same issue appears on the gluing side as the central loop-coherence obstruction. The paper therefore does not treat vanishing of that obstruction as a second theorem-side premise beyond the supplied transportable sector system; it is the central-branch obstruction bookkeeping for that same transportability condition. The genuinely noncentral fixed-cutoff branch differs: its obstruction is the crossed-module class q_Σ, and that branch is handled by Theorems 5.3–5.6 rather than by the ordinary compact-group theorem below.

Assume therefore that in the EFT regime there is a directed ladder $(\mathsf{Sect}_r, U_{rs}, F_r)$ of transportable bosonic edge-sector packages as in Assumption 3.7, and write

$$\mathsf{Sect}_\infty := \varinjlim_r \mathsf{Sect}_r$$

for the directed colimit retaining the transportable sectors and intertwiners that persist in that system. This is the category on which Doplicher–Roberts / Tannaka reconstruction is applied. The theorem below is neutral about whether Sect_∞ is trivial or nontrivial: if the supplied bosonic refinement ladder of Assumption 3.7 furnished only the tensor unit, the reconstructed compact group would be the trivial group.

Lemma 7.1 (Refinement-stable transportable sector persistence)**.** *Assume the directed transport system in Assumption 3.7, and write $U_{rs} : \mathsf{Sect}_r \to \mathsf{Sect}_s$ for the refinement functor from regulator scale r to a finer scale $s \succeq r$. Let α_{r_0} be a realized transportable edge-sector class on a sufficiently fine collar whose edge-center support is visible on the declared overlap algebra. Suppose there is a cofinal refinement tail T above r_0 and representatives $\alpha_s \in \mathsf{Sect}_s$, $s \in T$, such that $U_{st}\alpha_s \simeq \alpha_t$ for $t \succeq s$ in T, with the same overlap-visible support transported by the corresponding refinement maps.*

Then the tail determines a unique object $[\alpha] \in \mathsf{Sect}_\infty$. A later refinement stage cannot delete, split, or change the transportability status of that realized class while preserving the stated hypotheses and adding no new overlap-visible support.

Proof. The directed colimit identifies two representatives once their images agree at some common finer regulator stage. The cofinal tail T therefore supplies a compatible family $(\alpha_s)_{s\in T}$, and its equivalence class is an object $[\alpha]$ of Sect_∞. If a second cofinal family (β_u) represents the same realized collar class, cofinality gives a common refinement $v \succeq s, u$. Compatibility of the refinement functors and the declared overlap-visible support gives $U_{sv}\alpha_s \simeq U_{uv}\beta_u$, so both families define the same colimit object.

Deletion or loss of transportability at a later stage would mean that no compatible representative exists on a cofinal tail, contradicting the persistence hypothesis rather than producing a different colimit object. A split into inequivalent later classes without new overlap-visible support would assign two incompatible refinement images to the same transported edge-center datum, contradicting functorial compatibility on transportable sector classes. If new overlap-visible support appears, it is a new sector datum, not a silent change of the old persistent class. Thus persistence is unique once the directed transport system is supplied; the lemma does not prove that such a nontrivial system exists. □

Theorem 7.2 (Construction of the refinement-stable bosonic sector category)**.** *Assume the bosonic gauge premises of Assumption 3.7. Then the directed colimit*

$$\mathsf{Sect}_\infty := \varinjlim_r \mathsf{Sect}_r$$

inherits a well-defined semisimple rigid symmetric C^*-tensor structure from the fixed-cutoff collar-sector packages. Its tensor unit is the persistent vacuum sector, its tensor product is induced from collar concatenation, its duals are induced from the fixed-cutoff orientation-reversal / charge-conjugation duals, and its braiding is the induced bosonic symmetry from the $3+1$-dimensional branch. The compatible finite-dimensional multiplicity spaces F_r descend to a faithful bosonic fiber functor

$$\mathcal{F} : \mathsf{Sect}_\infty \to \mathsf{Hilb}_{\mathrm{fd}}.$$

Proof. Because every U_{rs} is a faithful monoidal *-functor and the coherence maps satisfy $U_{su}\circ U_{rs} \cong U_{ru}$, the directed-colimit category exists with objects given by eventually compatible tails and morphisms given by eventual intertwiner classes. The tensor unit on each Sect_r is preserved by refinement, so the vacuum unit determines a unique tensor unit $\mathbf{1}_\infty$ in the colimit.

For persistent objects represented by tails (X_r) and (Y_r), define

$$[X] \otimes [Y] := [X_r \otimes_r Y_r]$$

using any sufficiently fine common stage. This is independent of the chosen stage because the monoidal structure maps of U_{rs} identify $U_{rs}(X_r \otimes_r Y_r)$ with $U_{rs}(X_r) \otimes_s U_{rs}(Y_r)$. The same argument descends associators, unitors, and the bosonic symmetry/braiding maps. Since each Sect_r is symmetric, the induced braiding on Sect_∞ is symmetric as well.

Rigidity descends stagewise. If $X_r^\vee$ is a dual of X_r with evaluation and coevaluation maps

$$\mathrm{ev}_{X_r} : X_r^\vee \otimes_r X_r \to \mathbf{1}_r, \qquad \mathrm{coev}_{X_r} : \mathbf{1}_r \to X_r \otimes_r X_r^\vee,$$

then monoidality of the refinement functors carries these maps to the corresponding duality data for $U_{rs}(X_r)$. Hence dual objects and zig-zag identities pass to the colimit. The C^*-structure and

semisimplicity likewise descend because morphisms are eventual classes of morphisms in semisimple C^*-categories and the refinement functors preserve the *-operation and direct-sum data.

For the fibers, define $\mathcal{F}([X])$ to be the stable finite-dimensional Hilbert space represented by any $F_r(X_r)$. This is well defined because the coherence isomorphisms $F_s \circ U_{rs} \cong F_r$ identify any two sufficiently fine representatives canonically. On morphisms, $\mathcal{F}$ is induced by the eventual intertwiner action on those multiplicity spaces. Monoidality and the bosonic unit condition descend from the corresponding properties of the F_r, and faithfulness descends because each F_r is faithful on its stage. Thus $\mathcal{F}$ is a faithful bosonic fiber functor with finite-dimensional values. □

Theorem 7.3 (Conditional compact gauge reconstruction in the bosonic branch)**.** *Assume Axioms 3.1–3.4 and Assumption 3.7. Let*

$$\mathcal{F} : \mathsf{Sect}_\infty \to \mathsf{Hilb}_{\mathrm{fd}}$$

be the faithful bosonic fiber functor on that refinement-limit sector category in the EFT regime, constructed in Theorem 7.2. Then

$$G := \mathrm{Aut}_\otimes(\mathcal{F})$$

is a compact group and

$$\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G).$$

In particular G is uniquely determined up to isomorphism.

Proof. The theorem begins only once the bosonic refinement ladder of Assumption 3.7 has been assembled into the colimit category and fiber functor of Theorem 7.2; those data are not consequences of the local-Gibbs or Dobrushin-type collar estimates. On the central branch, no second obstruction premise is appended beyond that supplied transportability hypothesis; the same path-independent transport condition is the gluing-side content of the loop-coherence criterion. Fix a small skeleton of Sect_∞, which does not change the tensor category or the fiber functor up to equivalence. For every object X, any monoidal natural automorphism $\eta \in \mathrm{Aut}_\otimes(\mathcal{F})$ has a unitary component $\eta_X \in U(\mathcal{F}(X))$ because $\mathcal{F}$ is a *-functor to finite-dimensional Hilbert spaces. Hence there is an injective map

$$\mathrm{Aut}_\otimes(\mathcal{F}) \hookrightarrow \prod_X U(\mathcal{F}(X)), \qquad \eta \mapsto (\eta_X)_X.$$

For each intertwiner $f : X \to Y$, naturality imposes the closed relation $\mathcal{F}(f)\eta_X = \eta_Y \mathcal{F}(f)$. For each pair X, Y, the monoidal structure isomorphism $J_{X,Y} : \mathcal{F}(X) \otimes \mathcal{F}(Y) \xrightarrow{\sim} \mathcal{F}(X \otimes Y)$ imposes the closed relation

$$\eta_{X\otimes Y} = J_{X,Y}\,(\eta_X \otimes \eta_Y)\,J_{X,Y}^{-1}, \qquad \eta_{\mathbf{1}} = \mathrm{id}_{\mathbb{C}}.$$

Therefore $\mathrm{Aut}_\otimes(\mathcal{F})$ is a closed subgroup of the compact product $\prod_X U(\mathcal{F}(X))$, so $G = \mathrm{Aut}_\otimes(\mathcal{F})$ is compact.

The hypotheses required by Doplicher–Roberts / Tannaka reconstruction are checked explicitly on the constructed pair $(\mathsf{Sect}_\infty, \mathcal{F})$: Theorem 7.2 supplies a rigid symmetric C^*-tensor category with a faithful bosonic fiber functor of finite-dimensional values. The standard reconstruction theorem therefore applies to that pair and yields a symmetric C^*-tensor equivalence

$$\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G),$$

where an object X is sent to the representation $g \mapsto g_X$ on $\mathcal{F}(X)$ and morphisms are sent to the corresponding intertwiners [20, 21]. If the same category were also equivalent to $\mathrm{Rep}(G')$

through another faithful bosonic fiber functor, then composing the two reconstruction equivalences would give a symmetric tensor equivalence $\mathrm{Rep}(G) \simeq \mathrm{Rep}(G')$ compatible with the forgetful fiber functors. Applying the same reconstruction theorem to either side identifies the two automorphism groups of the fiber functor, so $G \cong G'$. Thus the reconstructed compact group is unique up to isomorphism. □

If a fermionic sign object is present, the correct reconstruction statement is super-Tannakian rather than purely Tannakian. This paper does not prove that fermionic/chiral extension or the nontriviality of the realized transportable bosonic ladder; it works only in the bosonic internal-gauge branch once the coherent fixed-cutoff sector packages and their monoidal refinement transport are supplied.

7.2 MAR minimality and the Standard Model quotient

Throughout this subsection, χ_{cpl} means the dimension of the minimal coupled carrier supporting the required weak-type and color-type nonabelian charges on a common block. It is not the abstract minimal faithful representation dimension of the final gauge group.

Definition 7.4 (Weak-type and color-type nonabelian roles)**.** *A weak-type nonabelian charge is a pseudoreal nonabelian doublet role. A color-type nonabelian charge is an intrinsically complex nonabelian role, meaning that its nonabelian factor itself acts by an irreducible representation not equivalent to its conjugate. Abelian twisting of a pseudoreal nonabelian irreducible representation does not count as a color-type role.*

The derivation of the Standard Model quotient separates into five steps:

1. existence of a compact reconstructed group;
2. identification of the minimal nonabelian sector content;
3. identification of the minimal coupled carrier;
4. uniqueness of the connected abelian factor once admitted on that carrier;
5. determination of product gauge structure up to finite quotient, then of the global finite quotient from the realized matter spectrum.

Lemma 7.5 (Minimal nonabelian sector content)**.** *Any admissible low-energy sector must contain both a weak-type pseudoreal nonabelian charge role and an intrinsically complex color-type nonabelian charge role.*

Proof. Light chiral matter requires left-handed multiplets and right-handed singlets to carry inequivalent gauge data. A pseudoreal doublet structure is the minimal way to realize the weak sector without immediately producing vectorlike masses. A genuinely complex nonabelian representation is required to distinguish quarks from antiquarks. A single nonabelian simple factor cannot simultaneously supply both a minimal weak-type role and an intrinsically complex color-type role; abelian twisting does not count toward the color-type requirement. □

Lemma 7.6 (Connected 2D pseudoreal image classification)**.** *Let H be a connected compact group with a faithful irreducible 2-dimensional pseudoreal unitary representation. Then the connected derived subgroup of its image is conjugate to* $\mathrm{SU}(2)$*. Equivalently, the nonabelian factor acting in that role is* $\mathrm{SU}(2)$ *up to finite central quotient.*

Proof sketch. Irreducibility places the image inside U(2) with scalar commutant. Its connected derived subgroup is therefore a connected compact subgroup of SU(2). The connected compact subgroups of SU(2) are either tori or SU(2) itself. The representation is nonabelian and pseudoreal, so the torus case is excluded. Hence the derived subgroup is conjugate to SU(2), with only a finite central kernel left invisible in the representation. □

Lemma 7.7 (Connected 3D intrinsically complex image classification)**.** *Let H be a connected compact group with a faithful irreducible intrinsically complex* 3*-dimensional unitary representation. Then the connected derived subgroup of its image is conjugate to* SU(3)*. Equivalently, the nonabelian factor acting in that role is* SU(3) *up to finite central quotient.*

Proof sketch. Because the representation is irreducible on $\mathbb{C}^3$, the semisimple part of the Lie algebra of the image acts irreducibly on $\mathbb{C}^3$. A product of two nontrivial simple factors would force a tensor-product decomposition of dimension at least 4, so only one simple factor can occur. Among compact simple Lie algebras, the only ones with nontrivial irreducible representations of dimension at most 3 are $\mathfrak{su}(2)$ and $\mathfrak{su}(3)$. The 3-dimensional $\mathfrak{su}(2)$ representation is real, not intrinsically complex, whereas the fundamental $\mathfrak{su}(3)$ representation is intrinsically complex. Therefore the connected derived subgroup is conjugate to SU(3), again up to finite central kernel. □

Lemma 7.8 (Minimal coupled carrier under MAR)**.** *Within the connected positive-dimensional Lie admissible class used by MAR, and under the weak-type / intrinsically complex color-type criteria above, the minimal coupled carrier supporting both roles on a common block has the form*

$$V = \mathbb{C}^3 \otimes \mathbb{C}^2, \qquad \chi_{\mathrm{cpl}} = 6.$$

Proof. By Lemma 7.6, the minimal weak-type nonabelian role is the pseudoreal doublet of SU(2), hence has dimension 2. By Lemma 7.7, the minimal intrinsically complex color-type nonabelian role is the triplet of SU(3), hence has dimension 3.

By definition of χ_{cpl}, the weak-type and color-type roles must act nontrivially on one common irreducible nonabelian block rather than on disconnected direct-sum summands. Commuting irreducible actions on such a common block therefore require dimension at least the product of the minimal weak and color dimensions:

$$\chi_{\mathrm{cpl}} \geq 2 \cdot 3 = 6.$$

Equality is achieved by the tensor-product carrier $\mathbb{C}^3 \otimes \mathbb{C}^2$. The block-diagonal representation $\mathbb{C}^3 \oplus \mathbb{C}^2$ of $S(U(3) \times U(2))$ is faithful of dimension 5, but it is not coupled and hence is not the MAR minimizer. Connected non-semisimple counterexamples of $U(2)$ type are excluded by the explicit color-type criterion rather than being silently ignored. □

Remark 7.9. *The dimension comparisons used in Lemma 7.8 apply to the positive-dimensional connected Lie image carrying the admissible nonabelian charges, equivalently to the identity component of the reconstructed group on that sector. Finite, disconnected, or connected non-semisimple counterexamples exist, but they are outside the admissible class fixed by the explicit weak-type / color-type criteria.*

MAR supplies existence of a connected abelian factor; the next lemma proves only uniqueness and identification on the minimal coupled carrier.

Lemma 7.10 (Uniqueness of the connected abelian factor on the minimal coupled carrier)**.** *Inside* U(6)*, the commutant of* SU(3) × SU(2) *acting on* $\mathbb{C}^3 \otimes \mathbb{C}^2$ *is exactly* U(1)*.*

Proof. The SU(3) action is irreducible on the first tensor factor and trivial on the second. The SU(2) action is irreducible on the second tensor factor and trivial on the first. By Schur's lemma, any operator commuting with both actions is scalar on each irreducible factor. Hence the full commutant is a single U(1). □

Theorem 7.11 (Product Gauge Structure up to Finite Quotient)**.** *Assume Axiom 3.5, the hypotheses of Theorem 7.3 on the ordinary or central-defect realized low-energy branch, the weak-type / color-type MAR premises above, and that the realized connected gauge image acts faithfully on the minimal coupled carrier. Then the realized connected gauge structure has the form*

$$G_{\text{phys}} = \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\Gamma}$$

for some finite central subgroup Γ*.*

Proof. Theorem 7.3 yields some compact group G. Lemma 7.5 identifies the minimal admissible sector content, Lemmas 7.6 and 7.7 identify the connected nonabelian factors realizing the minimal weak-type and color-type roles, and Lemma 7.8 fixes the minimal coupled carrier. The tensor-product structure of that carrier implies commuting weak and color actions, so the connected semisimple part of the realized image is locally $\mathrm{SU}(3) \times \mathrm{SU}(2)$. Axiom 3.5 supplies the existence of one connected abelian charge factor acting nontrivially on the coupled carrier, and Lemma 7.10 shows that any such connected abelian factor is necessarily a single U(1). A compact connected Lie group with Lie algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ is a quotient of $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ by a finite central subgroup. Faithfulness rules out an invisible extra torus, so the only ambiguity is the finite central quotient Γ. □

Theorem 7.12 (Hypercharge lattice on the realized matter package)**.** *Assume gauge group* $\mathrm{SU}(N_c) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y$*, one generation of chiral matter* (Q, u^c, d^c, L, e^c)*, one Higgs doublet* H*, and Yukawa terms*

$$QHu^c, \qquad QH^\dagger d^c, \qquad LH^\dagger e^c.$$

Then anomaly cancellation and Yukawa invariance determine the hypercharge ratios up to an overall $\mathrm{U}(1)_Y$ *normalization:*

$$Y_L = -N_c Y_Q, \quad Y_H = N_c Y_Q, \quad Y_u = -(N_c+1)Y_Q, \quad Y_d = (N_c-1)Y_Q, \quad Y_e = 2N_c Y_Q.$$

Fixing the normalization by $Q = T_3 + Y$ *and* $Q(\nu_L) = 0$ *gives*

$$Y_Q = \frac{1}{2N_c}.$$

For $N_c = 3$ *one recovers the exact Standard Model lattice*

$$Y_Q = \frac{1}{6}, \quad Y_L = -\frac{1}{2}, \quad Y_u = -\frac{2}{3}, \quad Y_d = \frac{1}{3}, \quad Y_e = 1, \quad Y_H = \frac{1}{2}.$$

Proof. Yukawa invariance gives

$$Y_u = -(Y_Q + Y_H), \qquad Y_d = -Y_Q + Y_H, \qquad Y_e = -Y_L + Y_H.$$

The mixed anomalies yield

$$N_c Y_Q + Y_L = 0,$$

and

$$2N_cY_Q + N_cY_u + N_cY_d + 2Y_L + Y_e = 0.$$

Solving the first anomaly gives

$$Y_L = -N_cY_Q.$$

Substituting the Yukawa relations and $Y_L = -N_cY_Q$ into the mixed gravitational anomaly then yields

$$Y_H = N_cY_Q, \qquad Y_u = -(N_c+1)Y_Q, \qquad Y_d = (N_c-1)Y_Q, \qquad Y_e = 2N_cY_Q.$$

With these relations in place, the $\mathrm{SU}(N_c)^2\mathrm{U}(1)$ and $\mathrm{U}(1)^3$ anomalies cancel identically. The normalization of Y_Q is fixed by the electric charge operator $Q = T_3 + Y$ together with $Q(\nu_L) = 0$, which implies

$$Y_L = -\frac{1}{2} = -N_cY_Q, \qquad \text{hence} \qquad Y_Q = \frac{1}{2N_c}.$$

□

Corollary 7.13 (Three Colors on the realized MAR branch)**.** *Under the hypotheses of Theorem 7.11, the realized color-type factor acts in the fundamental triplet on the coupled carrier. Hence the realized quark doublet carries exactly*

$$N_c = 3.$$

Proof. Lemma 7.7 identifies the minimal intrinsically complex color-type role in the connected Lie admissible class as the 3-dimensional fundamental of SU(3). Lemma 7.8 then shows that MAR realizes this role on the coupled block $\mathbb{C}^3 \otimes \mathbb{C}^2$. The color multiplicity of the realized weak doublet Q is therefore fixed by the same D8 carrier to be 3. No later admissibility selector is needed to promote oddness to the specific value 3. □

Corollary 7.14 (Three Generations on the realized MAR branch)**.** *Under the hypotheses of Theorem 7.11, with $N_c = 3$ from Corollary 7.13, the CP-capability and weak-sector UV clauses contained in Axiom 3.5 force*

$$N_g = 3.$$

Proof. On the realized one-Higgs chiral branch, intrinsic CKM CP capability requires

$$\frac{(N_g-1)(N_g-2)}{2} > 0,$$

hence $N_g \geq 3$. For one Higgs doublet, the one-loop weak-sector coefficient is

$$b_2 = \frac{11}{3}C_A - \frac{2}{3}\sum_f T(R_f) - \frac{1}{3}\sum_s T(R_s) = \frac{22}{3} - \frac{N_g(N_c+1)}{3} - \frac{1}{6} = \frac{43}{6} - \frac{N_g(N_c+1)}{3}.$$

Asymptotic freedom therefore implies

$$N_g(N_c+1) < \frac{43}{2}.$$

With $N_c = 3$, this gives

$$4N_g < \frac{43}{2}, \qquad \text{hence} \qquad N_g \leq 5.$$

These are not extra post-MAR selectors: they are the explicit numerical consequences of the CP-capability and weak-sector UV clauses present in Axiom 3.5 on the same realized branch. Once the first three MAR components are fixed by the D8 color/weak structure, the fourth component N_g is minimized on the allowed set $\{3, 4, 5\}$, so $N_g = 3$. □

Remark 7.15 (Witten parity on the realized color branch)**.** *With $N_c = 3$, each generation contributes $N_c + 1 = 4$ left-handed* SU(2) *doublets, so Witten's global anomaly constraint is automatically satisfied generation by generation. In this theorem stack the anomaly is therefore a consistency check on the realized triplet-doublet package, not the step that creates the color count [26].*

Proposition 7.16 (Global quotient from the realized matter spectrum)**.** *If the realized matter spectrum carries the hypercharges of Theorem 7.12 with $N_c = 3$, then the subgroup of* $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ *acting trivially on all realized states is exactly $\mathbb{Z}_6$.*

Proof. Write $q := 6Y \in \mathbb{Z}$ and $\eta := e^{i\pi/3}$, so the U(1) factor acts by η^q on charge-q multiplets. The element

$$g_0 := (\omega_3, -1, \eta)$$

acts trivially on the realized matter and Higgs multiplets:

$$Q: \ \omega_3(-1)\eta = 1, \qquad u^c: \ \omega_3^{-1}\eta^{-4} = 1, \qquad d^c: \ \omega_3^{-1}\eta^2 = 1,$$

$$L: \ (-1)\eta^{-3} = 1, \qquad e^c: \ \eta^6 = 1, \qquad H: \ (-1)\eta^3 = 1.$$

Its powers therefore generate a subgroup of order six acting trivially on all realized states. Conversely, any central element acting trivially on e^c must have U(1) part η^n, and triviality on Q then fixes the SU(3) and SU(2) center factors uniquely. Hence every trivial central element is a power of g_0, so the kernel is exactly $\mathbb{Z}_6$. □

Corollary 7.17 (Standard Model Gauge Group)**.** *Under Theorem 7.11, Theorem 7.12, Corollaries 7.14 and 7.13, and Proposition 7.16,*

$$G_{\text{phys}} = \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}.$$

Proof. Theorem 7.11 yields the product structure up to finite quotient, and Proposition 7.16 fixes that quotient to $\mathbb{Z}_6$ once the realized hypercharge lattice and $N_c = 3$ are in place. □

Corollary 7.18 (Product-Group Consequences)**.** *The adjoint representation of the full connected gauge group contains only*

$$(8, 1, 0) \oplus (1, 3, 0) \oplus (1, 1, 0).$$

Equivalently, the adjoint representation of the derived gauge group contains only

$$(8, 1, 0) \oplus (1, 3, 0).$$

There are no mixed $(3, 2, \pm 5/6)$ gauge bosons. Hence gauge-mediated proton decay is absent. The unbroken gauge symmetries forbid elementary gauge-boson mass terms: the photon is massless, while gluons are massless gauge fields of the unbroken color symmetry.

Remark 7.19. *The same product-group structure removes the usual grand-unification monopole sector. In the present framework, any retained coupling-unification discussion does not proceed by embedding the gauge group into a simple group that is later broken. It proceeds, if at all, only through the separate D10 calibration branch built from shared edge heat-kernel data together with the printed running/matching conventions and extra calibration assumptions. Absence of gauge-mediated proton decay and unification-like running therefore coexist without tension on that branch. Any confinement statement is a separate infrared dynamical issue rather than a corollary of the product-group adjoint argument alone.*

7.3 Fundamental scales

The quantitative D10 branch is a forward quantitative-closure sector, not part of the recovered-core theorem package. Its particle-physics scale variable is

$$P \equiv a_{\text{cell}}/\ell_P^2,$$

together with the realized discrete gauge data of the structural branch, in particular

$$N_c = 3, \qquad \beta_{\text{EW}} = N_c + 1 = 4.$$

The logical direction is therefore

$$P \longmapsto \big(M_U(P), E_{\text{cell}}(P)\big) \longmapsto \alpha_U(P) \longmapsto \big(t_U(P), t_{\text{tr}}(P)\big) \longmapsto \big(t_2(P), t_3(P), v(P)\big) \longmapsto \alpha_i(\mu_*; P).$$

The edge-law input on this lane is split explicitly by claim tier. The fixed-cutoff same-overlap thermal/Casimir theorem is carried on the separate microphysics surface. This compact paper imports that closure only through the declared merge boundary: one fixed-cutoff edge-law package below, then the compact-group / Peter–Weyl lift used in the D10 pixel constraint above it. The open burden is refinement-stable universality/uniqueness of that lift and the scalar-root control of the printed D10 solve, not the existence of the finite-cutoff Casimir branch itself. Measured electroweak quantities are not used to infer the internal transmutation parameters. They enter only at the end as compare-only validation targets for the printed D10 running/matching package. The forward transmutation certificate on the live D10 lane makes that logical order explicit: the source-only basis reconstructs the same $\alpha_U(P)$, hence the same unified diffusion parameter

$$t_U(P) = 4\pi^2 \alpha_U(P),$$

and transmutation exponent

$$t_{\text{tr}}(P) = \frac{2\pi}{(N_c+1)\alpha_U(P)}$$

as the pixel-closure solve itself. Here the paper-side transmutation factor is the same $\beta_{\text{EW}} = N_c + 1$ used below; older overloaded β-ratios are compare-only diagnostic readouts and are not part of the theorem contract.

The scales fixed directly from P are

$$M_U(P) = \frac{E_P}{e^{2\pi}} P^{1/6}, \qquad E_{\text{cell}}(P) = \frac{E_P}{\sqrt{P}}.$$

The factor $e^{-2\pi}$ is the same modular normalization used on the Lorentz/BW side, and $P^{1/6}$ is the cell-area scaling relation carried by the present D10 implementation.

Golden-ratio equilibrium benchmark. The total/bulk/edge hierarchy has one exact self-similar balance point. Writing

$$x(C) := \frac{S_{\text{gen}}(C)}{S_{\text{bulk}}(C)} = 1 + \frac{\langle L_C \rangle}{S_{\text{bulk}}(C)},$$

the exact self-similar balance condition

$$\frac{S_{\text{gen}}(C)}{S_{\text{bulk}}(C)} = \frac{S_{\text{bulk}}(C)}{\langle L_C \rangle}$$

gives

$$x = \frac{1}{x-1}, \qquad x^2 - x - 1 = 0.$$

Hence the unique positive equilibrium point is

$$x = \varphi := \frac{1+\sqrt{5}}{2}.$$

Equivalently, the equilibrium-breaking order parameter

$$A_\varphi(x) := x - 1 - \frac{1}{x}$$

vanishes exactly at $x = \varphi$. In that sense φ is the exact self-similar balance point rather than a numerological comparison constant. The synthesis paper gives the outer/inner closure relation that fixes the realized value of P by matching that detuning to the inner electromagnetic observation scale emitted by the same cell. The role of the equilibrium theorem is to explain why the realized value sits close to φ. Exact equilibrium is too symmetric to support durable records, structure, and dynamics, so the realized branch sits at a small equilibrium-breaking detuning away from it. The technical question for the quantitative branch is the size of that detuning together with the reduced-residual/root-control analysis of the printed solve.

The one-dimensional internal variable solved on the D10 branch is the unified coupling α_U. For a trial value of α_U, define

$$v(\alpha_U, P) := E_{\text{cell}}(P)\, \exp\!\left(-\frac{2\pi}{\beta_{\text{EW}}\alpha_U}\right).$$

Run the one-loop D10 couplings from

$$\alpha_1(M_U) = \alpha_2(M_U) = \alpha_3(M_U) = \alpha_U$$

down to a scale μ using

$$\alpha_i^{-1}(\mu; \alpha_U, P) = \alpha_U^{-1} + \frac{b_i}{2\pi} \log \frac{M_U(P)}{\mu},$$

with the printed D10 one-loop coefficients b_i. Let $\mu_* = \mu_*(\alpha_U, P)$ be the fixed point determined by the tree-level Z-mass relation

$$\mu_* = \frac{v(\alpha_U, P)}{2} \sqrt{g_2(\mu_*)^2 + g_Y(\mu_*)^2}, \qquad \alpha_Y = \frac{3}{5}\,\alpha_1.$$

At that fixed point define

$$t_2(\alpha_U, P) = 4\pi^2 \alpha_2(\mu_*; \alpha_U, P), \qquad t_3(\alpha_U, P) = 4\pi^2 \alpha_3(\mu_*; \alpha_U, P).$$

Let $\bar{\ell}_{\text{SU}(2)}(t)$ and $\bar{\ell}_{\text{SU}(3)}(t)$ denote the nonabelian edge-entropy functions appearing in the D10 pixel constraint. The pixel-closure functional is

$$\mathcal{F}(\alpha_U; P) := \bar{\ell}_{\text{SU}(2)}\big(t_2(\alpha_U, P)\big) + \bar{\ell}_{\text{SU}(3)}\big(t_3(\alpha_U, P)\big) - \frac{P}{4}.$$

The printed D10 package takes $\alpha_U(P)$ to be the branch value selected by solving

$$\mathcal{F}(\alpha_U; P) = 0.$$

Only after that forward solve are the internal transmutation parameters fixed:

$$t_U(P) := 4\pi^2\alpha_U(P), \qquad t_{\rm tr}(P) := \frac{2\pi}{(N_c+1)\alpha_U(P)},$$

and with them the downstream scale data

$$t_2(P) := t_2(\alpha_U(P), P), \qquad t_3(P) := t_3(\alpha_U(P), P), \qquad v(P) := v(\alpha_U(P), P).$$

Thus the branch runs from OPH input P to the internal transmutation data $(t_U(P), t_{\rm tr}(P))$, then to the downstream local scale data $(t_2(P), t_3(P), v(P))$, and only then to the low-energy couplings. It does not use measured $\alpha_i(m_Z)$ to infer those internal t-parameters.

Every quoted D10 electroweak number is downstream of this forward map. In particular, the source-locked running-family anchor $a_0(P) = \alpha_{\rm em}^{-1}(m_Z^2; P)$, the electromagnetic transport family $\alpha_{\rm em}^{-1}(q^2; P)$ and $\sin^2\theta_W(q^2; P)$, the public W/Z pair, and the Thomson endpoint

$$\alpha_{\rm Th}^{-1}(P) = \lim_{q^2\to 0} \alpha_{\rm em}^{-1}(q^2; P)$$

on the Ward-projected $\mathrm{U}(1)_Q$ lane are forward-emitted outputs of the chosen D10 running/matching package. When compared with observation, they serve as validation of that printed implementation on the quantitative-closure branch recorded in the premise ledger. The repository carries a numerical witness for the outer/inner fixed-point closure, while external fine-structure comparisons enter only as explicit compare metadata, together with a local numerical contraction certificate for the implemented map. The theorem-external item is an interval-wide existence/uniqueness or monotonicity proof for the scalar root selected by the printed quantitative package under arbitrary convention changes.

8 Relation to Holography and Existing UV Frameworks

OPH belongs to the holographic family of ideas, but its primitive data are different from those used in asymptotic-boundary constructions. The distinction matters because several of the central claims of this paper, including the treatment of Λ, the handling of factorization, and the emergence of gauge structure, depend on it.

8.1 Static-patch screen versus asymptotic boundary

In asymptotic-boundary holography one starts from a dual theory at infinity and reconstructs the bulk inward. Here one starts from a finite-capacity screen equipped with a net of local patch algebras and reconstructs the effective bulk from overlap consistency. The primitive object is therefore a family of observer patches with nontrivial overlaps, not a single global boundary theory.

This shift has two technical consequences. First, subsystem factorization is treated from the beginning as a gluing problem with centers and sector labels. Second, the physical role of the horizon is local and operational: every observer has direct access only to a patch algebra, and global law is whatever survives reconciliation on overlaps.

8.2 Positive Λ and finite capacity

The D5→D6 cosmological-capacity stack is explicit: the null-modular reconstruction of Section 6 determines the stress tensor only up to a metric term, so local null data do not fix Λ. On the D6 branch the extra global premise is the external screen capacity

$$N_{\rm scr} = \log\dim\mathcal{H}_{\rm tot}$$

together with the screen-capacity identification

$$N_{\mathrm{scr}} = S_{\mathrm{dS}},$$

the standard de Sitter entropy relation

$$S_{\mathrm{dS}} = \frac{A_{\mathrm{dS}}}{4G} = \frac{3\pi}{G\Lambda},$$

On that same branch this yields

$$\Lambda = \frac{3\pi}{GN_{\mathrm{scr}}}$$

and fixes the static-patch package

$$S_{\mathrm{dS}} = N_{\mathrm{scr}}, \qquad r_{\mathrm{dS}} = \sqrt{\frac{3}{\Lambda}}, \qquad t_{\Lambda} = \frac{r_{\mathrm{dS}}}{c},$$

while the observed cosmic age is a downstream FLRW benchmark rather than an additional D6 theorem output. The compact paper is therefore formulated in a de Sitter-first language: the positive cosmological constant is tied to finite capacity rather than to a deformation of a negative-Λ starting point.

8.3 Factorization, gauge structure, and the string sector

The factorization problem of gauge theory and gravity is often treated as a technical nuisance. Here it is part of the architecture. Edge-center completion turns the collar center into a first-class object, and the entropy split

$$S_{\mathrm{gen}} = \langle L_C \rangle + S_{\mathrm{bulk}}$$

is then a structural consequence rather than an added prescription.

In the bosonic EFT branch, compact gauge reconstruction follows only after the explicit coherent refinement ladder of transportable fixed-cutoff collar-sector packages, its bosonic symmetry, and its finite-fiber descent data of Assumption 3.7 are supplied; on the central branch the corresponding loop-coherence criterion is already carried by that supplied transportability. The genuinely noncentral fixed-cutoff branch is handled separately by crossed-module data rather than by the ordinary compact-group theorem. On the realized MAR-admissible branch with the explicit realized one-generation chiral matter plus one-Higgs package, the Standard Model quotient is then selected by minimal admissibility. This differs sharply from approaches in which the gauge group is specified in advance.

The worldsheet relation is similarly reversed. The edge partition function

$$Z_{\mathrm{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)}$$

is the closed partition function obtained after gluing the open-edge weights

$$p_R(t) \propto d_R e^{-tC_2(R)}.$$

Theorem 8.1 (OPH-to-2D-Yang–Mills edge partition theorem)**.** *Assume the compact-group heat-kernel branch supplied by the fixed-cutoff edge-sector law together with the compact-group / Peter–Weyl lift used in the bosonic gauge lane, in the same quadratic-Casimir normalization as the compact heat kernel. Equivalently, assume the open-edge weights satisfy*

$$p_R(t) \propto d_R e^{-tC_2(R)}$$

on a compact gauge group G, and define the closed edge partition function by gluing the two edge boundaries:

$$Z_{\text{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)}.$$

Then:

(i) *the closed edge partition function is exactly the compact-group heat kernel at the identity,*

$$Z_{\text{edge}}(t) = K_t(1);$$

(ii) *the same sum is therefore the standard compact-group two-dimensional Yang–Mills heat-kernel partition function at identity, equivalently the closed-surface heat-kernel partition sum on that branch;*

(iii) *the Chapman–Kolmogorov law for K_t is the corresponding collar-sewing rule for the OPH edge partition.*

Thus the OPH edge-sector partition reorganizes exactly into the two-dimensional Yang–Mills heat-kernel form before any large-N_{edge} continuation is invoked.

Proof. The fixed-cutoff edge-sector theorem gives the heat-kernel weights $d_R e^{-tC_2(R)}$ on the compact-group branch. Gluing the two edge boundaries contributes a second factor of d_R, so the closed partition function is the displayed sum. Peter–Weyl identifies

$$\sum_R d_R \chi_R(g) e^{-tC_2(R)}$$

with the compact-group heat kernel $K_t(g)$, and evaluating at the identity $g = 1$ gives

$$K_t(1) = \sum_R d_R \chi_R(1) e^{-tC_2(R)} = \sum_R d_R^2 e^{-tC_2(R)} = Z_{\text{edge}}(t),$$

since $\chi_R(1) = d_R$. The Chapman–Kolmogorov law for K_t is exactly the semigroup gluing law, so it gives the collar-sewing rule on the same branch. □

Scope boundary. The imported premises are exactly the fixed-cutoff edge heat-kernel / Casimir law, the compact gauge-group / Peter–Weyl lift, the quadratic-Casimir heat-kernel normalization on that lift, and the Peter–Weyl heat-kernel identity. The external content is any large-N_{edge} regime, the Gross–Taylor dictionary on that regime, and every further critical-string ingredient. The theorem above proves the OPH-to-2D-Yang–Mills partition reorganization itself; it does not by itself produce a worldsheet genus expansion.

Theorem 8.2 (Criterion for a controlled large-N_{edge} worldsheet effective description)**.** *Assume Theorem 8.1 and a large-N_{edge} realization of the compact-group heat-kernel branch, with $N_{\text{edge}} \neq N_c = 3$, for which the 't Hooft-style variable*

$$\tau := tN_{\text{edge}}$$

is kept in a compact interval $I \subset (0, \infty)$. Suppose moreover that for every truncation order $G \geq 0$ there exist coefficient functions $F_g : I \to \mathbb{R}$ and constants $C_{G,I}$ such that

$$\log Z_{\text{edge}}\left(\frac{\tau}{N_{\text{edge}}}\right) = \sum_{g=0}^{G} N_{\text{edge}}^{2-2g} F_g(\tau) + R_{G+1}(\tau, N_{\text{edge}})$$

with

$$\left|R_{G+1}(\tau, N_{\text{edge}})\right| \leq C_{G,I} N_{\text{edge}}^{-2G} \qquad (\tau \in I).$$

Then this branch defines a controlled theorem-level worldsheet effective description of the edge dynamics, with control parameters N_{edge}^{-2}, the fixed-τ window I, and the truncation order G.

Proof. Theorem 8.1 puts the edge partition function on the compact-group heat-kernel / two-dimensional Yang–Mills surface. On that surface, the standard Gross–Taylor dictionary reads a genus expansion of the displayed form as a closed-worldsheet rewriting of the free energy. The stated large-N_{edge} premise supplies that expansion on the fixed-τ window I, together with an explicit truncation bound. Therefore the worldsheet rewriting is controlled by the displayed parameters N_{edge}^{-2}, I, and G. The statement is only an effective description on the declared branch: it does not by itself construct a critical worldsheet CFT, derive modular invariance, worldsheet supersymmetry, anomaly cancellation, GSO projection, or full massless-spectrum matching. □

Open blockers. The criterion theorem does not derive the large-N_{edge} regime itself from the compact-gauge chain, and it does not reprove the imported Gross–Taylor large-N worldsheet dictionary for two-dimensional Yang–Mills. The external content is the existence of the large-N_{edge} sequence, the fixed-τ window on which the genus expansion holds, and the uniform remainder control. Any further lift to critical superstring structure additionally requires worldsheet supersymmetry, critical dimension, modular invariance, anomaly cancellation, GSO projection, and full massless-spectrum matching beyond the present continuation theorem.

Feature	Asymptotic-boundary holography	OPH
Primitive data	Boundary theory at infinity	Finite-capacity screen with observer-patch net
Factorization across cuts	Subtle and often indirect	Explicit edge-center completion with central labels
Status of Λ	External background datum	Fixed globally by N_{scr} after local null reconstruction
Gauge group	Usually specified as part of the model	Reconstructed from edge sectors, then fixed on the realized MAR branch
String sector	Fundamental dual description	Controlled large-N_{edge} worldsheet effective description of edge dynamics on the stated branch

9 Claim Tiers and Falsifiability

The recovered core is

$$(D1\text{–}D5) \cup (D7\text{–}D9),$$

namely the relativity chain together with the realized Standard Model structural chain. D6 is the input-dependent global closure of that same Einstein branch, D10 is the integrated quantitative-closure branch, and D12 collects phenomenological continuations.

9.1 Prediction audit by claim tier

Tier	Representative outputs	What would actually falsify the tier
Phase I recovered core (Theorem 3.9; D1–D5, D7–D9)	Confluence of overlap repair, conditional Lorentz kinematics, the conditional Jacobson-type Einstein branch in the stated scaling regime, compact gauge reconstruction in the bosonic branch, the realized Standard Model quotient chain on the explicit realized matter package, the exact hypercharge lattice on that realized matter package, the realized color triplet $N_c = 3$, the generation count $N_g = 3$, and the product-group consequence of no gauge-mediated proton decay	A mathematical failure in the derivation chain, or data that require a different realized gauge quotient, different realized hypercharges, a different color or generation count on the admissible branch, or gauge-mediated proton decay
Phase II input-dependent corollary (D6)	$\Lambda = \frac{3\pi}{GN_{\mathrm{scr}}}$, $S_{\mathrm{dS}} = N_{\mathrm{scr}}$, $r_{\mathrm{dS}} = \sqrt{3/\Lambda}$, and $t_\Lambda = r_{\mathrm{dS}}/c$ once the screen-capacity identification is made, while any retained capacity-level neutrino side estimate on that input is legacy bookkeeping rather than the live neutrino theorem lane; the observed cosmic age is a downstream FLRW benchmark	Failure of the screen-capacity identification or of the specific global capacity relation. Such a failure would not by itself erase the recovered core
Phase II quantitative-closure branch (D10)	forward transmutation data $\alpha_U(P)$, $t_U(P)$, $t_{\mathrm{tr}}(P)$, pixel-closure gauge-coupling consistency, $\alpha_i(m_Z)$, the source-locked anchor $a_0(P) = \alpha_{\mathrm{em}}^{-1}(m_Z^2; P)$, the electroweak transport family $(W, Z, \alpha_{\mathrm{em}}^{-1}(q^2), \sin^2\theta_W(q^2), v)$, and the Thomson endpoint $\alpha_{\mathrm{Th}}^{-1}(P)$ on the Ward-projected $\mathrm{U}(1)_Q$ lane	Failure of the integrated quantitative closure once P and the printed running/matching conventions are imposed
Phase III phenomenological continuations (D12 and beyond)	Flavor ansätze, charged-lepton continuation ansätze beyond exact centered readback, texture branches, the weighted-cycle / Majorana-holonomy neutrino branch above the legacy D6 side estimate, further neutrino mass/mixing refinements, dark-sector response laws, heuristic baryogenesis continuations, proton-spin bookkeeping, proton-lifetime estimates beyond the gauge-channel exclusion, black-hole spectroscopy templates, controlled large-N_{edge} string/worldsheet effective descriptions, conjectural critical-superstring extensions, and other downstream phenomenology	Retraction of the corresponding continuation only. These branches are not part of the recovered core and their failure is not, by itself, a falsification of relativity-plus-Standard-Model recovery

Phase I is the recovered-core tier, Phase II is adjacent but non-core, and Phase III contains

phenomenological continuations. Interpretive strange-loop discussions lie outside these tiers: the paper does not define an OPH closure map on an invariant observer-supporting sector or prove uniqueness or stability for such a map.

9.2 Recovered-core discriminators

Once the claim boundary is made explicit, the sharpest discriminators are the ones that do not depend on auxiliary flavor, dark-sector proposals, heuristic baryogenesis continuations, or string ansätze.

9.2.1 Realized Standard Model gauge structure

On the realized MAR-admissible branch, this paper fixes

$$G_{\text{phys}} = \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_g = 3, \qquad N_c = 3,$$

with the exact one-generation hypercharge lattice. Here the count statements are internal to the same realized branch: the D8 minimal coupled-carrier theorem stack fixes the color triplet and hence $N_c = 3$; CKM phase counting and weak-sector asymptotic freedom then bound N_g; MAR fixes the smallest realized value $N_g = 3$; and Witten parity remains only a consistency check on the resulting triplet-doublet package.

$$Y_Q = \frac{1}{6}, \qquad Y_L = -\frac{1}{2}, \qquad Y_u = -\frac{2}{3}, \qquad Y_d = \frac{1}{3}, \qquad Y_e = 1, \qquad Y_H = \frac{1}{2}.$$

Evidence that the realized low-energy gauge structure or realized hypercharges differ from these values would directly contradict the recovered core.

9.2.2 Product-group corollary

Because the realized gauge structure is a product group up to the finite central quotient fixed above, there are no mixed $(3, 2, \pm 5/6)$ gauge bosons. Hence

$$\tau_p^{(\text{gauge})} = \infty$$

for gauge-mediated proton decay. Observation of proton decay specifically attributable to gauge-boson exchange would therefore falsify the realized product-group branch.

9.2.3 Relativity branch

The relativity claim is conditional and sharp about its scope: if the stated geometric-modular and null-bridge premises are realized in the intended scaling regime, and if the same realized cap-label-preserving MaxEnt family satisfies the derived fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap MaxEnt variations on that branch, then this paper predicts Lorentz kinematics on the screen and a Jacobson-type Einstein branch. What is being tested here is the existence of that scaling regime and its premises, not a fixed-cutoff matrix-algebra identity.

9.3 What does *not* count as a contradiction of the recovered core

The following do *not* falsify Theorem 3.9 if they fail in their present form:

1. the uniform $\mathbb{Z}_6$ center-label ensemble and the associated $\varepsilon = 1/6$ flavor ansatz;
2. charged-lepton continuation ansätze beyond the exact centered-readback / common-shift frontier, texture exponents, or legacy capacity-level neutrino side estimates;
3. modular-anomaly dark-sector response ansätze, including MOND/RAR-style scaling claims;
4. heuristic baryogenesis continuations based on extra suppression-counting assumptions; a baryogenesis theorem requires a concrete out-of-equilibrium mechanism, derived CP-odd source terms, justified defect/sphaleron counting, washout control, and a freeze-out computation of the final asymmetry;
5. proton-spin ansätze or proton-lifetime estimates beyond the structural exclusion of gauge-mediated decay;
6. discrete-horizon spectroscopy templates;
7. controlled large-N_{edge} string/worldsheet effective descriptions.

These are all post-Phase-I, non-core branches. Some are Phase-II input-dependent estimates, while the rest are Phase-III continuations. None is part of this paper's recovered relativity-plus-Standard-Model core.

10 Common Objections and Clarifications

10.1 Is the use of P circular?

The pixel area P is fixed by the outer/inner closure condition in the synthesis paper. That condition matches the outer pixel detuning to the inner electromagnetic observation scale emitted by the same cell. Within the printed quantitative implementation, that P first fixes $M_U(P)$ and $E_{\text{cell}}(P)$, then a one-dimensional pixel-closure solve fixes $\alpha_U(P)$, equivalently the internal transmutation data $t_U(P)$ and $t_{\text{tr}}(P)$, and only then do $t_2(P)$, $t_3(P)$, $v(P)$, and the running electroweak outputs appear. Quantities algebraically entangled with that quantitative branch is on the Phase-II implementation surface: they are forward-emitted there, and comparison with observation checks that printed implementation rather than enlarging the recovered-core claim set.

Operationally, changing P moves the quantitative family

$$(\alpha_U, \alpha_i(m_Z), a_0, \alpha_{\text{em}}^{-1}(q^2), \sin^2\theta_W(q^2))$$

through that forward solve. It does not alter the parameter-free structural outputs such as the gauge quotient or hypercharge lattice. Downstream matter-sector continuations, including charged-lepton continuation ansätze beyond the exact centered-readback / common-shift frontier, are outside this SM/GR derivation paper's recovered-core theorem package.

10.2 Does a fixed UV cell size break Lorentz invariance?

The UV regulator is placed on the screen algebra, not on an emergent bulk spacetime lattice. Lorentz kinematics arise from the continuum modular action on caps, summarized in Corollary 6.6, rather than from microscopic invariance of a bulk discretization.

This is conceptually similar to other continuum limits in statistical mechanics and lattice field theory. A regulator may break a symmetry at finite cutoff while the infrared fixed point restores it exactly. Here the restoration is stronger than a generic RG expectation because the modular-flow theorem identifies the symmetry group itself:

$$\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1).$$

Residual Lorentz violation would therefore have to arise from a failure of the regulator premises or from a failure of the modular-flow theorem, not merely from the existence of a UV cell size. The UV-side burden on the Lorentz branch is therefore the explicit scaling-limit plus geometric-subnet extraction-and-rigidity package under **T2**, not the mere existence of a finite cell size.

10.3 Why use type-I algebras if continuum QFT uses type III?

Type I is a regulator statement. At every finite cutoff the patch and collar algebras are finite-dimensional matrix algebras, so entropy, recovery maps, and

$$K = -\log \rho$$

are literal finite-dimensional objects. The Lorentz branch is *not* a claim that the continuum observer algebra stays in that class. The refinement picture is

$$\text{finite type-I regulator net} \longrightarrow \text{scaling-limit observer net } \mathcal{A}_\infty,$$

and the limit can leave the regulator class. Axiom 3.3 controls the realized state-side branch across refinement; it does *not* by itself prove that the refinement limit is type I or that the emitted scaling-limit pair lies in the BW / canonical cap phase with standard geometric modular action.

On the geometric-subnet branch of Theorem 6.3, the scaling-limit cap algebras may be non-type-I and, in the continuum-QFT case of interest, are expected to be type III. Then there is generally no cap density matrix $\rho_C \in \mathcal{A}_\infty^{\mathrm{geo}}(C)$ and no bounded operator $K_C = -\log \rho_C$ inside the limit algebra. The correct continuum object is the modular automorphism group of the pair $(\mathcal{A}_\infty^{\mathrm{geo}}(C), \omega_\infty^{\mathrm{geo},C})$, which on that branch acts geometrically and typically outerly.

There is therefore no contradiction between using type-I regulators and aiming at a type-III continuum. The finite regulator provides the collar decomposition, recovery control, and carried errors. What the fixed-cutoff MaxEnt package adds is control of the realized state-side branch through one common finite-dimensional multiplier family under refinement. It does *not* prove the algebraic type of the continuum limit or that the realized branch lands in the BW / canonical cap phase with standard geometric modular action. The corresponding shorthand in this paper means only the case in which the realized refinement branch emits a scaling-limit geometric cap pair with the standard geometric modular automorphism group. In the generic non-type-I case that is an outer geometric modular action, not an inner density-matrix identity.

The clean internal extension route is a two-step chain, and the first object is the pressure point. First, one needs a realized scaling-limit observer cap-pair extraction on the geometric subnet from transported marginals. The theorem contract for that object is explicit: the realized transported cap-local system packages the cap-local test family on $\mathcal{A}_n^{\mathrm{geo}}(C_n)$, the projectively compatible transported marginal family, and the asymptotic transport-equivalence certificate. The emitted witness

is a derived vanishing carried-collar schedule on fixed local collars, explicitly decomposed into the constructive-recovery remainder and the faithful modular-defect term, namely the convergence to zero of the carried quantity

$$r_{\mathrm{FR}}(\varepsilon_{n,m,\delta}) + 4\lambda_*^{-1}\,\delta^{\mathrm{M}}_{A_{n,m,\delta}:B_{n,m,\delta}:D_{n,m,\delta}}(\varepsilon_{n,m,\delta})$$

after transport to each fixed local collar model. Only after that can local weak-$*$ extraction and GNS gluing emit the realized scaling-limit geometric cap pair.

Second, one needs ordered cut-pair rigidity on that realized limit. That stage would collapse the residual cap-preserving conformal freedom to the unique BW hyperbolic subgroup, after which the modular KMS condition fixes the 2π normalization. Ordered cut-pair rigidity is symbolic until the realized scaling-limit geometric cap pair exists. Until those two objects are internalized, every stronger vacuum or canonical-cap identification is explicitly theorem-external shorthand for the geometric-subnet branch of Theorem 6.3, not a derived theorem.

This is analogous to ordinary lattice field theory: one computes at finite regulator in a type-I algebra and then asks which continuum algebraic phase the scaling limit realizes. The additional structural point here is that the realized state-side branch is controlled under refinement. The open problem is to identify which scaling-limit observer algebra/state pair the realized branch produces and then to prove BW rigidity on that pair.

10.4 UV branch and scaling-limit scope

At fixed cutoff, Axiom 3.3 yields a quasi-local finite-range interacting branch, and Propositions 4.8 and 4.12 show that the physical normal form is unique only modulo quotient-level OPH-stable equivalence, while the terminal expectation functionals on the declared fixed-point / quotient-local physical algebras are schedule-independent on that carrier even when microscopic representatives differ by gauge labels globally or by sector labels on one regional quotient-local glued state. The invariant fixed by the axiom language is the class $[\mathfrak{U}]_{\mathrm{OPH}}$ of the physical branch modulo implementation hiding and inert ancillary stabilization, not a unique microscopic representative. The genuinely noncentral topological case is also closed at fixed cutoff by the higher-gauge package of Theorems 5.3–5.6 and Proposition 5.7, so the fixed-cutoff topological UV surface is closed on the ordinary, central-defect, and crossed-module branches alike. Beyond fixed cutoff, the continuum modular/geometric lift is the geometric-subnet extraction-plus-rigidity problem under **T2**; whenever the bosonic compact-gauge lane is invoked at refinement limit, the remaining external burden is exactly the transportable directed sector system together with its $3+1$-dimensional symmetric braiding and faithful bosonic fiber functor from Assumption 3.7. The consensus paper also classifies that D1 lane explicitly: on each fixed finite patch net, normal-form computation is finite-state and decidable with the Lyapunov step bound, the approximate-stability results are only the collar-local splice and record controls, and no universality theorem for a uniform family of growing patch nets and repair laws is part of the emitted theorem surface.

10.5 Why is charge quantization possible without a simple GUT?

The framework uses the global quotient and anomaly structure rather than a simple-group embedding. Once the realized matter content is fixed, the subgroup acting trivially on all states is exactly $\mathbb{Z}_6$, and the anomaly equations force the sixth-integer hypercharge lattice. Integer electric charge for color singlets is then a property of the realized quotient structure. A simple unified gauge group is not required.

11 Discussion

The framework is strongest where the outputs are discrete or structurally rigid:

1. the Lorentz branch on the extracted prime geometric cap pair once the theorem-local BW lift objects are supplied;
2. the conditional scaling-limit Einstein branch under the derived fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap MaxEnt variations and the null modular bridge together with the bounded-interval projective branch;
3. the Standard Model gauge quotient chain on the explicit realized matter package;
4. exact hypercharges on the realized one-generation matter package with one Higgs doublet;
5. the realized color triplet $N_c = 3$ and generation count $N_g = 3$ on the realized MAR-admissible branch;
6. the compact two-input quantitative layer, kept explicit as a secondary layer rather than as part of Phase I.

The local MaxEnt branch gives a finite-range interacting fixed-cutoff dynamics, Propositions 4.8 and 4.12 fix uniqueness of the UV branch only modulo OPH-stable equivalence, and Theorems 5.3–5.6 with Proposition 5.7 close the genuinely noncentral topological branch at fixed cutoff. The open items on the UV side are the continuum BW/geometric lift on the extracted prime geometric subnet under **T2** and, on the bosonic gauge lane, derivation of the refinement-limit transportable sector system together with its $3+1$-dimensional symmetric braiding and faithful bosonic fiber functor from Assumption 3.7. The smallest internal extension route to the BW/geometric lift is the two-step chain of a scaling-limit cap-pair extraction theorem that extracts $(\mathcal{A}_\infty^{\mathrm{geo}}(C), \omega_\infty^{\mathrm{geo},C})$ as local weak-$*$ limits of transported cap marginals on $\mathcal{A}_n^{\mathrm{geo}}(C_n)$ with vanishing carried collar errors, followed by an ordered null cut-pair rigidity theorem that collapses the residual cap-preserving conformal freedom on those extracted cap pairs to the unique BW hyperbolic subgroup. Beneath the derived carried-collar witness, the sharper raw datum is a transported fixed-local-collar Markov/faithfulness package: collarwise conditional mutual information vanishing together with an eventual common floor on the finite modular-transport family on each fixed local collar model. Within that lower datum, the single missing clause is the modular-transport common floor, which feeds the faithful modular-defect term. No separate exact-Markov-reference faithfulness datum is missing: on one fixed collar model, the comparison marginals inherit the same eventual floor once the exact-Markov modulus goes to zero. The corpus does not derive a refinement-uniform common floor on the finite modular-transport marginals; this single clause is external to the emitted theorem chain. That two-step chain and its transported fixed-local-collar datum are not part of the proved theorem surface stated here.

Shared excitation dictionary and flavor theorem boundaries. The D10 quantitative-closure branch gives integrated gauge-coupling closure on the displayed carrier. That carrier closes its own exact W/Z chart, and the promoted target-free source-only repair theorem emits the public electroweak quintet on the Phase II quantitative-closure branch; the freeze-once coherent repair surface serves only as compare-only validation. On that frozen authoritative repair surface the canonical W/Z reference pair is hit exactly, but it is a compare-only sidecar below the target-free theorem. The separate D6 cosmological-parameter package is an input-dependent corollary. On

the flavor side the manuscript treats one shared OPH excitation dictionary as the only proof-facing family datum; lane-specific integers are not introduced independently. For each realized same-label refinement arrow e on the three-generation bundle, let ω_e be the same-label overlap holonomy scalar, let

$$d_e := 1 - \omega_e$$

be the derived edge defect, let g_e be the same-label gap scalar, and define

$$q_e := \sqrt{g_e d_e}, \qquad \eta_e := \log q_e - \frac{1}{3}\sum_f \log q_f, \qquad \mu_e := \frac{e^{\eta_e}}{\frac{1}{3}\sum_f e^{\eta_f}}.$$

These are the common descendants of overlap holonomy and edge-defect data. They exist once the same-label gap and overlap witnesses are fixed, and they are the shared family-side input used by the charged, quark, and neutrino continuations.

To include the charge-side bookkeeping explicitly, let s_e denote the realized sector-charge step carried by the same-label arrow e. For any lane-specific readout γ on the realized same-label transport graph, let

$$m_{\gamma,e} \in \mathbb{Z}_{\geq 0}$$

be the multiplicity with which γ traverses the elementary same-label excitation arrow e. Equivalently, along the realized transport path one has

$$s_\gamma = \sum_e m_{\gamma,e} s_e.$$

The associated excitation weight and excitation action are

$$A_\gamma := -\sum_e m_{\gamma,e} \log q_e = -\log\Big(\prod_e q_e^{m_{\gamma,e}}\Big), \qquad Y_\gamma^{\mathrm{exc}} := \prod_e q_e^{m_{\gamma,e}}.$$

So overlap holonomy, edge defects, gap data, and sector-charge transport determine the suppression law before any base choice is made. What earlier D12 language called a "texture exponent" or "defect count" is therefore not an extra primitive beyond the structural branch. The primitive flavor-side integers are the transport multiplicities $m_{\gamma,e}$. A one-number exponent in "units of $\log 6$" is only the compressed summary

$$n_\gamma^{(6)} := A_\gamma / \log 6,$$

and it becomes a literal defect count only after the additional uniform sixfold center-label collapse compatible with the realized

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}$$

quotient. Without that extra collapse, the derived OPH data are the multiplicity vector $m_{\gamma,\bullet}$ and the excitation action A_γ, not an independently postulated flavor integer.

The imported realized-branch premise here is the D9 result that the same realized branch carries the color triplet $N_c = 3$ and the generation count $N_g = 3$. This shared dictionary feeds the downstream particle continuations. Their lane-specific theorem objects, exact witnesses, and quantitative closures are developed in Ref. [2]. On the compact-paper surface, the dictionary serves as the common family-side input beneath those continuations. Old base-6 exponents are compare-only compressions of excitation action rather than primitive flavor data.

Exact non-hadron output lane. For reader-facing exact hits, the non-hadron surface is explicit and lane-split. The structural carrier chain

$$\text{Axioms 1–5} \longrightarrow D7\text{–}D9 \longrightarrow (m_\gamma, m_g, m_{\text{grav}}) = (0,0,0)$$

is theorem-grade structural exactness. The electroweak quantitative-closure chain

$$\text{Axioms 1–5} \longrightarrow D7\text{–}D10$$

has an exact frozen authoritative repair sidecar

$$(M_W, M_Z) = (80.377,\ 91.18797809193725)\,\text{GeV}$$

on the frozen authoritative repair surface, but that exact pair is compare-only beneath the target-free source-only theorem. The D11 Higgs/top branch closes as one source-only split theorem on the declared D10/D11 quantitative surface. On the emitted D10 repair tuple $(\eta_{\text{source}}, \beta_{EW}, \lambda_{EW}, \tau_{2,\text{tree}}^{\text{exact}}, \delta n_{\text{tree}}^{\text{exact}})$, define

$$\rho_{HT} = \log\!\big(1 + \tau_{2,\text{tree}}^{\text{exact}}\big),$$

$$R_T = -\tau_{2,\text{tree}}^{\text{exact}}\eta_{\text{source}}^2 + \Big(1 + \frac{\beta_{EW}}{28}\Big)\eta_{\text{source}}^6 + \frac{\eta_{\text{source}}^8}{14} + \frac{\eta_{\text{source}}^9}{27},$$

$$R_H = \eta_{\text{source}}^5 - \frac{3}{25}\eta_{\text{source}}^6 + \frac{\lambda_{EW}\eta_{\text{source}}^6}{18} + \frac{\eta_{\text{source}}^8}{2\beta_{EW}}.$$

Then the forward split coordinates are

$$\pi_y = \frac{\eta_{\text{source}} + \left(\frac{3}{2} + \frac{\beta_{EW}}{4}\right)\rho_{HT} + R_T}{\sqrt{\pi}}, \qquad \pi_\lambda = \frac{\eta_{\text{source}} - \left(\frac{4}{3} - \frac{\beta_{EW}}{54}\right)\rho_{HT} + R_H}{\sqrt{\pi}},$$

and the declared D11 Jacobian reads out

$$\delta y_t(\mu_t) = \pi_y\, y_t^{\text{core}}(\mu_t), \qquad \delta\lambda(\mu_t) = -\frac{16}{9}\pi_\lambda\, \lambda^{\text{core}}(\mu_t).$$

On that same declared surface,

$$m_H = 125.1995304097179\ \text{GeV}, \qquad m_t^{D11} = 172.3523553288312\ \text{GeV}.$$

At the precision quoted by the PDG Higgs listing, the Higgs row lands on the 2025 Higgs average. The same surface emits a companion top coordinate. The exact public running-top row is carried by the selected-class quark theorem and uses the PDG 2025 cross-section entry. The bridge to the auxiliary direct-top PDG row is closed as a current-corpus codomain no-go; the auxiliary row remains compare-only until a source-side extraction-response kernel is emitted. The old one-scalar seed

$$\sigma_{D11,\text{HT}} = \frac{\alpha_U \cos(2\theta_{W0})}{\sqrt{\pi}}$$

is on disk only as the lower-rank fixed-ray companion branch with $\pi_y = \pi_\lambda = \sigma_{D11,\text{HT}}$. The exact inverse slice on the same D11 Jacobian is compare-only and does not define the theorem surface. The same D11 Jacobian also has a compare-only exact inverse slice at the canonical Higgs/top reference pair

$$(m_H, m_t) = (125.1995304097179,\ 172.3523553288312)\,\text{GeV}.$$

Detailed particle-spectrum continuation statements are carried by Ref. [2]. That companion paper develops the quark selected-class theorem, the charged-lepton determinant-line boundary, the weighted-cycle neutrino theorem surface, and the hadron execution boundary. The hadron backend tasks #153/#157 are closed as out-of-scope/computationally blocked, not solved; the compact paper therefore promotes no hadron masses and reopens that branch only after a working OPH hadron backend emits production spectral data and systematics. These particle-continuation statements do not belong to the compact-paper claim surface.

Local unification surface and exact-release frontier. The local quantitative bridge can be stated without inflating the recovered-core claim. On the bosonic side, the declared pixel input P fixes the D10/D11 trunk

$$P \longmapsto \alpha_U(P) \longmapsto \big(t_U(P), t_{\mathrm{tr}}(P)\big) \longmapsto v(P),$$

after which the D10 source basis emits the public electroweak pair and the D11 source-only split theorem with

$$\rho_{HT} = \log\!\big(1 + \tau_{2,\mathrm{tree}}^{\mathrm{exact}}\big), \qquad (\pi_y, \pi_\lambda) \longmapsto (m_t, m_H)$$

by the declared D11 Jacobian, again with no inverse readback of the internal transmutation data from measured low-energy couplings. emits the Higgs/top branch. On the gravity side, the same D10 pixel law packages

$$\bar{\ell}_{\mathrm{SU}(2)}(t_{2,\mathrm{run}}) \;+\; \bar{\ell}_{\mathrm{SU}(3)}(t_{3,\mathrm{run}}) = P/4, \qquad G = \frac{a_{\mathrm{cell}}}{4\,\bar{\ell}(t)}.$$

The exact local release burden consists of two theorem-sized objects: one strict classical-regime clause above the BW/Lorentz support package and one target-free D10 chart-identity theorem for exact W/Z. The familiar-unit readout is not a third theorem-sized burden. It is the explicit display package attached to the theorem-side data a_{cell}, c, and $\bar{\ell}_{\mathrm{shared}}$. On the gravity side, Proposition 5.18 gives

$$\bar{\ell}_{\mathrm{shared}} = \bar{\ell}_{\mathrm{SU}(2)}(t_{2,\mathrm{run}}) \;+\; \bar{\ell}_{\mathrm{SU}(3)}(t_{3,\mathrm{run}}) = P/4,$$

and Proposition 5.19 gives the local SI readout

$$G_{\mathrm{SI}} = \frac{c^3 a_{\mathrm{cell}}}{\hbar P}$$

relative to the declared microscopic datum a_{cell}. The same proposition fixes the full familiar-unit package:

$$L_{\mathrm{loc}} = \sqrt{a_{\mathrm{cell}}}\,\widehat{L}(P), \qquad t_{\mathrm{loc}} = \frac{\sqrt{a_{\mathrm{cell}}}}{c}\,\widehat{T}(P),$$

$$E_{\mathrm{loc}} = \frac{\hbar c}{\sqrt{a_{\mathrm{cell}}}}\,\widehat{E}(P), \qquad \Theta_{\mathrm{loc}} = \frac{\hbar c}{k_B\sqrt{a_{\mathrm{cell}}}}\,\widehat{\Theta}(P),$$

with dimensionless $\widehat{L}, \widehat{T}, \widehat{E}, \widehat{\Theta}$. Meters and seconds are therefore read from the single local ruler $\sqrt{a_{\mathrm{cell}}}$ together with the structural Lorentz output c, while GeV and Kelvin are downstream familiar-unit displays of the inverse local ruler through $\hbar$ and k_B. On that declared extension surface the local release values are

$$c = 299792458\,\mathrm{m/s}, \qquad G = 6.674299995910528 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}},$$

$$(M_W, M_Z, M_H) = (80.377,\ 91.18797809193725,\ 125.1995304097179)\,\mathrm{GeV}.$$

The displayed c row is structural from the Lorentz branch rather than P-fitted, and the displayed G row is an exact emitted branch value relative to the stated a_{cell} datum rather than a literal zero-difference identity against the rounded benchmark 6.6743×10^{-11}.

Particle-spectrum continuation boundaries beyond the compact SM/GR ledger are stated in Ref. [2].

The paper does not claim complete closure of all low-energy observables.[1]

12 Conclusion

Observer-Patch Holography starts from overlap consistency of observer patches and yields, within one framework, several effective structures usually treated separately in modern fundamental physics. This SM/GR derivation paper supports a unique schedule-independent normal form for overlap repair and hence a unique physical UV branch only modulo boundary redundancy, implementation hiding, and inert ancillary stabilization, a conditional Lorentz branch on the extracted prime geometric cap pair, a conditional Jacobson-type Einstein branch under the derived fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap MaxEnt variations and the null modular bridge together with the bounded-interval projective branch, the input-dependent global closure of that same gravity lane once the screen-capacity input is supplied, the Standard Model gauge quotient with exact hypercharges on the realized matter package on the realized MAR one-Higgs branch in the bosonic internal-gauge sector, the realized color triplet $N_c = 3$ and generation count $N_g = 3$ on that branch, and a separate integrated quantitative-closure branch. Phenomenological continuations are discussed separately from the recovered core.

The framework therefore presents general relativity and the Standard Model as effective descriptions arising once the stated scaling-limit, geometric-subnet, and categorical premises are realized. The local quantitative claim surface is equally explicit: the same declared pixel input P controls the electroweak/Higgs mass trunk and the classical-regime gravity coupling, the invariant causal speed c is the structural Lorentz output, and the familiar-unit package reads meters, seconds, GeV, and Kelvin from the single local ruler $\sqrt{a_{\text{cell}}}$ with $\hbar$ and k_B used only as display conventions. Open items are sharper quantitative control, proof that the required coherent transportable bosonic refinement ladder is emitted on a realized nontrivial branch rather than merely supplied as gauge data, a fuller fermionic gauge branch, explicit microscopic realizations that emit the prime geometric cap pair and support ordered cut-pair rigidity, and proof of the theorem-side local exact-release objects on the live corpus.

A Supplemented Gravity and Structural Gauge Details

This appendix carries the gravity-side and structural gauge items that support the compact paper's technical surface. They sharpen the compact paper's Lorentz, Einstein, cosmological-constant, and product-group claim surfaces without widening the recovered core beyond its stated premises.

[1] Questions outside this theorem package do not alter the specific claim set established here. Even a separate habitat theorem for OPH state-and-law data would not enlarge this SM/GR derivation paper's Phase-I scope unless the actual closure map, invariant admissible sector, and stability hypotheses were also proved at the same tier.

A.1 Lorentz and Einstein Bridge

Theorem A.1 (Conformal group isomorphism)**.** *The orientation-preserving conformal group of S^2 is isomorphic to the connected Lorentz group:*

$$\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1).$$

Theorem A.2 (Geometric modular flow on caps on the extracted prime geometric subnet)**.** *Assume the OPH axioms, the derived fixed-cutoff collar package, the controlled scaling-limit scope clause **T2**, and the theorem-local realized geometric cap-pair extraction plus ordered cut-pair rigidity hypotheses on the extracted prime geometric subnet. Then the scaling-limit modular automorphism group of the cap pair is geometric conformal dilation on that subnet, with the standard 2π normalization. If the emitted scaling-limit cap algebra is type I, this may be written as $K_C = 2\pi B_C$; in the generic continuum case the theorem is the automorphism statement on a non-type-I cap algebra with outer geometric modular action.*

Definition A.3 (Conditional BW-branch observer-relative time reading)**.** *On the branch satisfying the hypotheses of the preceding geometric modular-flow theorem, OPH uses the modular automorphism parameter t of the extracted cap pair as that cap observer's relative time parameter. This is a declared BW-branch reading of the derived geometric modular flow; it is not an additional proof that arbitrary operational clocks, global time, or the full problem of time have been derived from the axioms.*

BW-side UV scaffold. The UV burden on this branch is not a separate cap-isotropy or Euclidean-regularity selector. The fixed-cutoff cap algebras are type-I regulators, but the scaling-limit observer algebra may leave that class, and MaxEnt alone does not select the BW / canonical cap phase. The UV burden is the realized transported geometric cap-local system together with the carried-collar schedule derived from the transported fixed-local-collar Markov/faithfulness datum on each fixed local collar model, followed by ordered cut-pair rigidity on the emitted scaling-limit geometric cap pair.

Null-strip completion status. On the D4 side, the fixed-cutoff strip package is more specific than the generic inherited-strip summary. The null cuts first transfer the same cut-center data as the spatial collar branch, which fixes the central sector-pair decomposition of the strip algebra. The stronger left/right tensor decomposition used by the null modular bridge then requires the extra inherited strip-split condition on the multiplicity spaces, together with the exact-or-controlled Markov hypotheses on one fixed inherited strip model. On that same fixed strip model, the renormalized half-line family is endpoint-Lipschitz, hence defines the weak tail generator, and on the scaling-limit geometric-cap branch the half-line blow-up net carries the derived half-sided modular pair. Borchers–Wiesbrock then supplies the positive null-translation generator on its Stone domain together with the affine half-line modular relation $K_a(\Omega) = K_0(\Omega) - 2\pi a P_\Omega$, and the same half-line family fixes the generator/charge identification internally. Bounded-interval formulas are downstream on the separate interval-preserving projective branch.

Lemma A.4 (Null data ambiguity)**.** *If a symmetric tensor X_{ab} satisfies*

$$X_{ab}k^a k^b = 0$$

for every null vector k, then $X_{ab} = \phi\, g_{ab}$ for some scalar ϕ.

Corollary A.5 (Null modular data determine Einstein only up to the metric term)**.** *Null modular data determine T_{ab} only up to ϕg_{ab}. Consequently the Einstein equation is fixed locally only up to Λg_{ab}.*

Theorem A.6 (Conditional Jacobson-type rest-frame relation)**.** *In the local Lorentzian scaling regime, once the realized cap-label-preserving MaxEnt family satisfies the derived fixed-cap generalized-entropy stationarity theorem for admissible fixed-cap variations, the half-line generator/charge identification of the null bridge, the bounded-interval transport input used in the local Lorentzian regime, and the internal small-ball bridge yield the rest-frame first-variation relation that drives the compact paper's local Einstein branch; on the same scaling branch, if that rest-frame relation holds for all local observer four-velocities and all reference states, the compact paper upgrades it to the full tensor equation by the explicit local quadratic-polarization argument of Corollary 6.22.*

Theorem A.7 (Newton coupling from edge entropy density)**.** *The gravitational coupling is fixed by the edge entropy density:*

$$G = \frac{a_{\text{cell}}}{4\bar{\ell}(t)}.$$

A.2 Cosmological-Constant / Screen-Capacity Closure

Lemma A.8 (Vacuum energy blindness)**.** *For any null vector k,*

$$T^{\text{vac}}_{kk} = T^{\text{vac}}_{ab} k^a k^b = -\rho_{\text{vac}}\, g_{ab} k^a k^b = 0.$$

Vacuum-energy contributions therefore lie in the kernel of the null map $T_{ab} \mapsto T_{kk}$.

Proposition A.9 (Structural separation)**.** *Within the structural split used here:*

1. *local modular/null data fix T_{ab} only up to ϕg_{ab}; and*
2. *Λ is fixed by global screen capacity, not by local null data.*

So the large vacuum-energy bookkeeping of EFT is not itself the local quantity determining curvature on this branch.

Theorem A.10 (No local Λ prediction)**.** *Within the null-modular reconstruction used here:*

1. *all T_{kk} data are unchanged under $T_{ab} \to T_{ab} + \phi g_{ab}$;*
2. *therefore Λ cannot be fixed by local overlap consistency alone; and*
3. *Λ requires the global screen-capacity input that closes the same Einstein branch beyond the local null data.*

Corollary A.11 (Benchmark cosmological readout on the D6 branch)**.** *On the same D6 branch as Corollaries 6.25–6.27, inserting the observed value $\Lambda \approx 1.09 \times 10^{-52}\,\text{m}^{-2}$ gives*

$$r_{\text{dS}} = \sqrt{\frac{3}{\Lambda}} \approx 1.66 \times 10^{26}\ \text{m}, \qquad t_\Lambda = \frac{r_{\text{dS}}}{c} \approx 17.5\ \text{Gyr},$$

$$N_{\text{patch}} = \left(\frac{r_{\text{dS}}}{\ell_P}\right)^2 \approx 1.05 \times 10^{122}, \qquad N_{\text{scr}} = S_{\text{dS}} = \pi N_{\text{patch}} \approx 3.31 \times 10^{122},$$

$$\Lambda \ell_P^2 \approx 2.85 \times 10^{-122}.$$

Proof. Immediate from Corollary 6.27. □

Observed-age benchmark status. The branch quantity t_Λ is the de Sitter static-patch timescale. The usual cosmic age t_0 is a compare-only FLRW benchmark rather than an additional D6 theorem output. On the flat ΛCDM benchmark

$$t_0 = \frac{2}{3H_0\sqrt{\Omega_\Lambda}} \sinh^{-1}\left(\sqrt{\frac{\Omega_\Lambda}{\Omega_m}}\right)$$

one gets the standard comparison value $t_0 \approx 13.8\,\mathrm{Gyr}$ at $H_0 \approx 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $\Omega_\Lambda \approx 0.685$.

Theorem A.12 (D5–D6 cosmological-capacity closure stack)**.** *Assume the compact paper's local Einstein branch, the external input*

$$N_{\mathrm{scr}} = \log \dim \mathcal{H}_{\mathrm{tot}},$$

the D6 screen-capacity identification

$$N_{\mathrm{scr}} = S_{\mathrm{dS}},$$

the standard de Sitter entropy relation

$$S_{\mathrm{dS}} = \frac{A_{\mathrm{dS}}}{4G} = \frac{3\pi}{G\Lambda},$$

and the standard static-patch formulas

$$r_{\mathrm{dS}} = \sqrt{\frac{3}{\Lambda}}, \qquad t_\Lambda = \frac{r_{\mathrm{dS}}}{c}.$$

Then the cosmological-constant package is one local/global theorem stack: local null data fix the Einstein branch only modulo Λg_{ab}, the same branch closes globally as

$$G_{ab} + \frac{3\pi}{GN_{\mathrm{scr}}}\, g_{ab} = 8\pi G\, \langle T_{ab}\rangle,$$

the same D6 closure fixes

$$S_{\mathrm{dS}} = N_{\mathrm{scr}}, \qquad A_{\mathrm{dS}} = 4GN_{\mathrm{scr}}, \qquad r_{\mathrm{dS}} = \sqrt{\frac{3}{\Lambda}}, \qquad t_\Lambda = \frac{r_{\mathrm{dS}}}{c},$$

and the observed cosmic age is a downstream FLRW benchmark rather than an additional theorem output.

Scope boundary. The imported premises are exactly the D5 local Einstein branch, the external input N_{scr}, the D6 identification $N_{\mathrm{scr}} = S_{\mathrm{dS}}$, the standard de Sitter entropy relation, and the standard static-patch formulas. The external item is the screen-capacity identification itself; it is not derived from the bare local null-data route.

Separate pixel-side benchmarks. The companion pixel numbers

$$a_{\mathrm{cell}} \approx 1.63\,\ell_P^2, \qquad \ell_{\mathrm{UV}} = \sqrt{a_{\mathrm{cell}}} \approx 1.28\,\ell_P, \qquad \bar{\ell} \approx 0.408$$

belong to the separate pixel-closure/local-readout package rather than to the D6 cosmological-parameter corollary itself.

A.3 Product-Group Structural Corollaries

Theorem A.13 (Factorization equivalence)**.**

Factorizing edge weights $\Longleftrightarrow$ *Additive boundary Laplacian* $\Longleftrightarrow$ *Product gauge group.*

If

$$H_\partial = H_\partial^{(1)} + H_\partial^{(2)} + H_\partial^{(3)} \qquad \textit{with} \qquad [H_\partial^{(i)}, H_\partial^{(j)}] = 0,$$

then

$$p(R_1, R_2, R_3) \propto \prod_{i=1}^{3} d_{R_i} e^{-t_i C_2(R_i)}.$$

Applied to the assumed transportable refinement-directed edge-sector colimit on the realized branch, together with the rigid symmetric C^*-tensor and faithful bosonic fiber-functor premises of the compact-gauge theorem, Tannaka–Krein reconstruction first yields some compact group G. Under the MAR admissibility package, and once the admissible class includes one connected abelian charge factor, the selected connected Lie realization is $SU(3) \times SU(2) \times U(1)$ up to finite quotient.

Corollary A.14 (No leptoquarks, hence no gauge-mediated proton decay)**.** *With product gauge group, the adjoint representation of the full connected gauge group is*

$$(8, 1, 0) \oplus (1, 3, 0) \oplus (1, 1, 0),$$

equivalently the derived nonabelian adjoint is

$$(8, 1, 0) \oplus (1, 3, 0).$$

There are therefore no gauge generators in mixed representations $(3, 2, \pm 5/6)$, i.e. no simple-GUT X, Y bosons on the realized branch. The structural claim is exactly

$$\tau_p^{(\text{gauge})} = \infty,$$

meaning that the gauge-boson proton-decay channel is absent. Scalar-mediated or higher-dimensional baryon-violating operators are outside this structural corollary.

Proton-spin continuation benchmark. On the realized D8–D9 branch the color factor is SU(3), so the structural OPH input for proton-spin bookkeeping is

$$C_F = \frac{4}{3}, \qquad C_A = 3.$$

If one adds the QCD-dependent continuation ansatz that quark/gluon spin sharing equilibrates according to these Casimirs, then

$$\Delta\Sigma \approx \frac{C_F}{C_F + C_A} = \frac{4}{13} \approx 0.308.$$

Against a representative lattice benchmark $\Delta\Sigma \approx 0.286$, this lands within about 8%, but it is only a deferred continuation benchmark. A first-principles OPH derivation of proton spin would require the nonperturbative light-quark/hadron completion plus an explicit map from OPH data to renormalized proton matrix elements.

A.4 Controlled Worldsheet Effective Description

Exact OPH-to-2D-Yang–Mills bridge. On the compact-group heat-kernel branch, Theorem 8.1 proves the exact identity

$$Z_{\text{edge}}(t) = K_t(1)$$

by gluing the open-edge weights $p_R(t) \propto d_R e^{-tC_2(R)}$ into the closed partition sum

$$Z_{\text{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)}.$$

This is the precise theorem-level sense in which the OPH edge-sector partition reorganizes into the two-dimensional Yang–Mills heat-kernel surface, and the Chapman–Kolmogorov law supplies the matching collar-sewing rule.

Large-N_{edge} boundary. The large-N_{edge} worldsheet interpretation is carried only when one fixes a distinct large-N_{edge} sequence, with $N_{\text{edge}} \neq N_c = 3$, a fixed-$\tau$ window for

$$\tau = tN_{\text{edge}},$$

and the uniform genus-remainder control of Theorem 8.2. On that branch the edge free energy satisfies the compact paper's theorem-level criterion for a controlled genus expansion, so the standard Gross–Taylor rewriting becomes a controlled worldsheet effective description of edge dynamics rather than a loose analogy.

External items. The exact bridge theorem does not derive the compact-group heat-kernel branch itself from bare local axioms, and the continuation theorem does not derive the large-N_{edge} regime itself, nor does it rederive the imported Gross–Taylor large-N worldsheet dictionary for two-dimensional Yang–Mills, nor does it identify a critical worldsheet CFT. Worldsheet supersymmetry, critical dimension, modular invariance, anomaly cancellation, GSO projection, and full massless-spectrum matching is separate extra premises beyond the compact recovered-core chain.

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